Introduction to the Finite Element Method

Theory, Implementation, and MATLAB Examples

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Dedicated to all students learning structural mechanics.

Preface

This book provides a concise introduction to the Finite Element Method (FEM) for structural and mechanical engineering applications. It is intended for senior undergraduate and graduate students who have a background in solid mechanics and linear algebra.

Each chapter introduces new element types, derives the governing equations from first principles, and concludes with a worked example. MATLAB code listings in Chapter A demonstrate practical implementation of the key algorithms.

The topics covered are:

- **Chapter 2** — Weighted residual methods for an axially loaded bar
- **Chapter 3** — One-dimensional bar and rod elements
- **Chapter 4** — Two-dimensional truss elements
- **Chapter 5** — Euler-Bernoulli beam elements
- **Chapter 6** — Two-dimensional frame elements
- **Chapter 7** — Two-dimensional continuum elements (CST and isoparametric quad)

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Introduction to the Finite Element Method

1.1 Purpose of the Finite Element Method

Many engineering problems are governed by differential equations together with boundary conditions. For simple geometries and idealized loading, these equations may have closed-form analytical solutions. In practical design, however, the geometry, material behavior, loading, and supports are often too complex for an exact solution.

The Finite Element Method (FEM) is a systematic numerical method for replacing a continuous problem by a finite set of algebraic equations. Instead of solving for an unknown field at infinitely many points, FEM solves for a finite number of unknowns at selected points called nodes. The solution between nodes is then interpolated by simple functions.

FEM is widely used in structural, mechanical, civil, aerospace, biomedical, and multiphysics engineering because it combines physical modeling with a practical computational procedure.

1.2 Historical Background

The modern finite element method developed in the 1950s from matrix structural analysis and the need to solve complex aerospace structures. Turner, Clough, Martin, and Topp [5] published an important early paper in 1956 describing the stiffness approach for plane stress analysis. The term *finite element* was introduced by Clough in 1960.

Since then, FEM has grown into one of the central tools of computational engineering. It is now implemented in commercial and open-source software for linear and nonlinear statics, heat transfer, dynamics, fluid flow, and coupled field problems.

1.3 Basic Idea

The central idea of FEM is discretization. A domain is divided into small, simple parts called **elements**. Elements meet at **nodes**, where the primary unknowns are stored. Within each element, the unknown field is approximated using interpolation functions called **shape functions**.

For a structural displacement problem, the final algebraic system commonly has the form

$$\boldsymbol{K}\,\boldsymbol{u} = \boldsymbol{f} \tag{1.1}$$

where $\boldsymbol{K}$ is the global stiffness matrix, $\boldsymbol{u}$ is the vector of unknown nodal displacements, and $\boldsymbol{f}$ is the global load vector.

Although Equation (1.1) looks compact, it represents the combined effect of element behavior, material properties, geometry, loading, and boundary conditions. Much of FEM is the careful construction of this system.

1.4 General FEM Procedure

A typical finite element analysis follows these steps:

1. **Define the physical problem:** Identify the domain, material properties, loads, and boundary conditions.

2. **Choose the element type:** Select elements appropriate for the physics and expected behavior of the solution.

3. **Discretize the domain:** Divide the domain into a mesh of finite elements connected at nodes.

4. **Form element equations:** Derive the element stiffness matrix and element load vector.

5. **Assemble the global system:** Combine all element equations into the global matrix equation.

6. **Apply boundary conditions:** Enforce prescribed values and include natural loading terms.

7. **Solve the algebraic system:** Compute the unknown nodal quantities.

8. **Post-process the results:** Recover derived quantities such as strains, stresses, reactions, or internal forces.

1.5 Role of Approximation

FEM is an approximate method. The accuracy of the result depends on the element formulation, mesh density, interpolation order, numerical integration, and the quality of the physical model. A refined mesh or higher-order interpolation can improve accuracy, but it also increases computational cost.

Good finite element modeling therefore requires engineering judgment. The goal is not only to obtain numbers, but to obtain numbers that are meaningful for the problem being studied.

1.6 Organization of This Book

This book begins with weighted residual methods because they provide a clear mathematical bridge from differential equations to finite element equations. It then develops one-dimensional elements, truss and beam elements, and two-dimensional continuum elements.

Each chapter emphasizes the connection between theory, element formulation, and implementation. MATLAB examples in the appendix show how the main algorithms can be programmed directly.

1.7 Notation Used in This Book

Throughout this book, we use the following conventions:

Symbol	Meaning	Example
$\boldsymbol{u}$	Column vector (bold italic)	displacement vector
$\boldsymbol{K}$	Matrix (bold italic)	stiffness matrix
$(\cdot)^{\mathsf{T}}$	Transpose	$\boldsymbol{K}^{\mathsf{T}}$
$(\cdot)^{-1}$	Inverse	$\boldsymbol{K}^{-1}$
d	Differential operator	$\mathrm{d}x,\ \mathrm{d}V$
δ	Virtual quantity	δu, δW

Chapter 2

Weighted Residual Methods

2.1 Motivation

The finite element method can be introduced from several points of view: direct stiffness, energy methods, virtual work, or weighted residual methods. The weighted residual viewpoint is useful because it shows how an approximate solution is obtained when the differential equation cannot be satisfied exactly at every point of the domain.

Consider a trial solution $\tilde{u}(x)$ that satisfies the essential boundary conditions but is not the exact solution. After substitution into the governing differential equation, a nonzero residual remains:

$$R(x) = \mathcal{L}\tilde{u}(x) - f(x) \tag{2.1}$$

where $\mathcal{L}$ is the differential operator and $f(x)$ is the loading term. Weighted residual methods determine the unknown coefficients in $\tilde{u}$ by requiring the residual to vanish in an averaged or weighted sense:

$$\int_\Omega w_i(x)R(x)\,\mathrm{d}\Omega = 0, \qquad i = 1, 2, \ldots, n \tag{2.2}$$

where $w_i(x)$ are weighting functions. Different choices of w_i lead to different methods.

2.2 Axially Loaded Bar

Consider the prismatic member in Figure 2.1, fixed at $x = 0$ and free at $x = L$. Since the load acts only along the axis of the member, the structural model is a bar or rod rather than a bending beam. The bar is subjected to a distributed axial load

$$p(x) = p_0\left(\frac{x}{L}\right)^2 \tag{2.3}$$

with constant E and A.

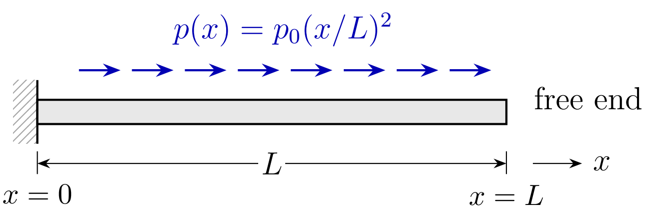

Figure 2.1: Fixed-free bar subjected to a nonuniform axial load.

2.2.1 Equilibrium of a Differential Element

To derive the governing equation, isolate a differential element of length $\mathrm{d}x$ as shown in Figure 2.2. Let $N(x)$ denote the internal axial force resultant. The sign convention is that $p(x)$ is positive in the positive x-direction and N is positive in tension.

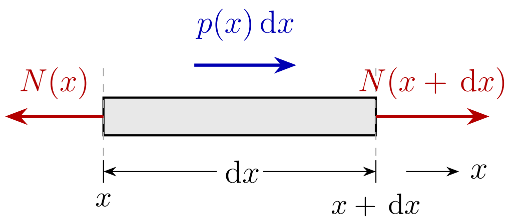

Figure 2.2: Free-body diagram of a differential bar element.

Force equilibrium in the axial direction gives

$$-N(x) + N(x+\mathrm{d}x) + p(x)\,\mathrm{d}x = 0. \tag{2.4}$$

Using the Taylor expansion

$$N(x+\mathrm{d}x) = N(x) + \frac{\mathrm{d}N}{\mathrm{d}x}\,\mathrm{d}x$$

and neglecting higher-order terms, Equation (2.4) becomes

$$\frac{\mathrm{d}N}{\mathrm{d}x} + p(x) = 0. \tag{2.5}$$

2.2.2 Constitutive and Kinematic Relations

The axial strain is related to the displacement field by

$$\varepsilon(x) = \frac{\mathrm{d}u}{\mathrm{d}x}. \tag{2.6}$$

For a linear elastic material,

$$\sigma(x) = E\varepsilon(x). \tag{2.7}$$

The axial force resultant is

$$N(x) = \sigma(x)A = EA\frac{\mathrm{d}u}{\mathrm{d}x}. \tag{2.8}$$

Substituting Equation (2.8) into Equation (2.5) gives the differential equation of an axially loaded bar:

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(EA\frac{\mathrm{d}u}{\mathrm{d}x}\right) + p(x) = 0. \tag{2.9}$$

Equivalently,

$$-\frac{\mathrm{d}}{\mathrm{d}x}\left(EA\frac{\mathrm{d}u}{\mathrm{d}x}\right) = p(x). \tag{2.10}$$

For constant E and A, and for the load in Equation (2.3), the governing equation becomes

$$-EA\frac{\mathrm{d}^2u}{\mathrm{d}x^2} = p_0\left(\frac{x}{L}\right)^2, \qquad 0 < x < L \tag{2.11}$$
$$u(0) = 0, \qquad EAu'(L) = 0.$$

The first boundary condition is essential because it prescribes displacement. The second is natural because it prescribes axial force. If an external axial force F were applied at $x = L$, the natural boundary condition would be $EAu'(L) = F$.

For this introductory comparison, use the one-term trial function

$$\tilde{u}(x) = a\left[\left(\frac{x}{L}\right)^2 - 2\left(\frac{x}{L}\right)\right] \tag{2.12}$$

where a is the unknown coefficient. This function satisfies both boundary conditions:

$$\tilde{u}(0) = 0, \qquad EA\tilde{u}'(L) = 0.$$

It is not flexible enough to satisfy the differential equation exactly, so a residual remains.

Let

$$\xi = \frac{x}{L}, \qquad 0 \le \xi \le 1.$$

Substituting Equation (2.12) into Equation (2.11) gives

$$R(\xi) = -EA\frac{\mathrm{d}^2\tilde{u}}{\mathrm{d}x^2} - p_0\xi^2 = -\frac{2EA}{L^2}a - p_0\xi^2. \tag{2.13}$$

The three methods below differ only in how they choose a from this residual.

2.3 Collocation Method

The collocation method requires the residual to vanish at selected points in the domain. Since the approximation contains one unknown coefficient, one collocation point is needed. Choosing the midpoint $\xi = 1/2$ gives

$$R\left(\frac{1}{2}\right) = 0. \tag{2.14}$$

Using Equation (2.13),

$$-\frac{2EA}{L^2}a - \frac{p_0}{4} = 0 \qquad \Rightarrow \qquad a_{\mathrm{col}} = -\frac{p_0L^2}{8EA}. \tag{2.15}$$

The approximate displacement is therefore

$$\tilde{u}_{\mathrm{col}}(x) = -\frac{p_0L^2}{8EA}\left(\xi^2 - 2\xi\right). \tag{2.16}$$

2.4 Subdomain Method

The subdomain method requires the residual to have zero average over each subdomain. With one unknown coefficient, the whole bar is taken as one subdomain:

$$\int_0^L R(x)\,\mathrm{d}x = 0. \tag{2.17}$$

Equivalently,

$$L\int_0^1 \left(-\frac{2EA}{L^2}a - p_0\xi^2\right)\mathrm{d}\xi = 0. \tag{2.18}$$

Thus,

$$-\frac{2EA}{L^2}a - \frac{p_0}{3} = 0 \qquad \Rightarrow \qquad a_{\text{sub}} = -\frac{p_0L^2}{6EA}. \tag{2.19}$$

2.5 Galerkin Method

In the Galerkin method, the weighting function is chosen from the same family as the trial function. Here,

$$w(\xi) = \xi^2 - 2\xi. \tag{2.20}$$

The weighted residual statement is

$$\int_0^L w(x)R(x)\,\mathrm{d}x = 0. \tag{2.21}$$

Substituting Equations (2.13) and (2.20) and changing to ξ gives

$$L\int_0^1 (\xi^2 - 2\xi)\left(-\frac{2EA}{L^2}a - p_0\xi^2\right)\mathrm{d}\xi = 0. \tag{2.22}$$

The required integrals are

$$\int_0^1 (\xi^2 - 2\xi)\,\mathrm{d}\xi = -\frac{2}{3}, \qquad \int_0^1 (\xi^2 - 2\xi)\xi^2\,\mathrm{d}\xi = -\frac{3}{10}.$$

Therefore,

$$\frac{4EA}{3L^2}a + \frac{3p_0}{10} = 0 \qquad \Rightarrow \qquad a_{\text{gal}} = -\frac{9p_0L^2}{40EA}. \tag{2.23}$$

Key Result

The Galerkin method is the weighted residual method that leads directly to the standard finite element formulation when piecewise polynomial trial functions are used and the governing equation is written in weak form.

2.6 Comparison with the Exact Solution

For this example, the exact solution is obtained by integrating Equation (2.11) twice and applying the boundary conditions:

$$u(x) = \frac{p_0L^2}{EA}\left(\frac{\xi}{3} - \frac{\xi^4}{12}\right). \tag{2.24}$$

Table 2.1: Comparison of one-term weighted residual approximations.

Method	Coefficient a	Tip displacement $\tilde{u}(L)$
Collocation	$-\dfrac{p_0L^2}{8EA}$	$\dfrac{p_0L^2}{8EA}$
Subdomain	$-\dfrac{p_0L^2}{6EA}$	$\dfrac{p_0L^2}{6EA}$
Galerkin	$-\dfrac{9p_0L^2}{40EA}$	$\dfrac{9p_0L^2}{40EA}$
Exact	–	$\dfrac{p_0L^2}{4EA}$

At the free end,

$$u(L) = \frac{p_0L^2}{4EA}. \tag{2.25}$$

Since $\tilde{u}(L) = -a$, the three approximations give the tip displacements listed in Table 2.1.

The collocation method makes the residual zero at one selected point, but the residual may still be large elsewhere. The subdomain method balances the total residual over the domain. The Galerkin method weights the residual with the same function used to approximate the displacement, which gives the best result among these one-term approximations for this problem.

2.7 MATLAB Implementation and Results

The following MATLAB script evaluates the coefficients, prints a numerical tip-displacement comparison, and generates the normalized displacement curves shown in Figure 2.3.

Figure 2.3 compares the three weighted residual approximations with the exact solution. The displacement is normalized as $uEA/(p_0L^2)$, so the plot shows the method error independently of the chosen numerical values of E, A, p_0, and L.

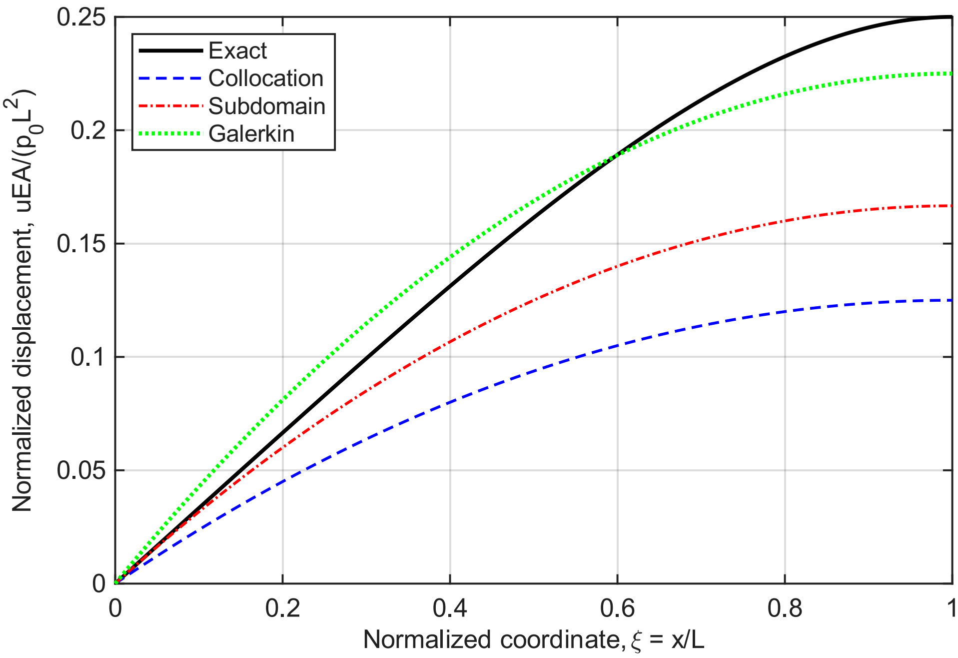

Figure 2.3: MATLAB comparison of collocation, subdomain, and Galerkin one-term approximations against the exact solution.

```
% weighted_residual_comparison.m
% One-term weighted residual approximation for an axially
   loaded bar

clear; clc; close all;

%% Problem parameters
L   = 1.0;          % bar length [m]
E   = 210e9;        % Young's modulus [Pa]
A   = 1.0e-4;       % cross-sectional area [m^2]
p0  = 1.0e4;        % load intensity scale [N/m]
EA  = E*A;          % axial stiffness [N]

%% Method coefficients for u_tilde = a*(xi^2 - 2*xi)
a_col = -p0*L^2/(8*EA);         % collocation at xi = 1/2
a_sub = -p0*L^2/(6*EA);         % one subdomain over [0,L]
a_gal = -9*p0*L^2/(40*EA);      % Galerkin weighting

methodNames = {'Collocation'; 'Subdomain'; 'Galerkin'};
aValues = [a_col; a_sub; a_gal];
tipValues = -aValues;           % xi=1 gives u_tilde(L) = -a

%% Exact tip displacement and percent error
uTipExact = p0*L^2/(4*EA);
percentError = abs(tipValues - uTipExact)/uTipExact * 100;

fprintf('Weighted residual comparison for axial bar\n');
fprintf('L = %.3f m, EA = %.4e N, p0 = %.4e N/m\n\n', L, EA, p0
   );
```

```
fprintf('%-14s %14s %18s %12s\n', ...
        'Method', 'a [m]', 'u_tilde(L) [m]', 'Error [%]');
fprintf('%s\n', repmat('-', 1, 62));

for i = 1:numel(methodNames)
    fprintf('%-14s %14.6e %18.6e %12.3f\n', ...
        methodNames{i}, aValues(i), tipValues(i), percentError(i));
end
fprintf('%-14s %14s %18.6e %12.3f\n', 'Exact', '--', uTipExact, 0);

%% Normalized displacement curves
xi = linspace(0, 1, 400);
phi = xi.^2 - 2*xi;

u_col_norm = -(1/8)  * phi;
u_sub_norm = -(1/6)  * phi;
u_gal_norm = -(9/40) * phi;
u_exact_norm = xi/3 - xi.^4/12;

figure('Name','Weighted Residual Comparison','Color','w');
plot(xi, u_exact_norm, 'k-',  'LineWidth', 1.8); hold on;
plot(xi, u_col_norm,   'b--', 'LineWidth', 1.3);
plot(xi, u_sub_norm,   'r-.', 'LineWidth', 1.3);
plot(xi, u_gal_norm,   'g:',  'LineWidth', 1.9);
grid on; box on;
xlabel('Normalized coordinate, \xi = x/L');
ylabel('Normalized displacement, uEA/(p_0L^2)');
legend('Exact', 'Collocation', 'Subdomain', 'Galerkin', ...
       'Location', 'northwest');
```

Listing 2.1: MATLAB comparison of weighted residual methods

2.8 Connection to FEM

The finite element method uses the Galerkin idea locally on each element. The unknown displacement field is approximated by shape functions,

$$u^e(x) = \boldsymbol{N}(x)\boldsymbol{u}^e, \tag{2.26}$$

and the weighting functions are chosen as the same shape functions. After integration by parts, the weak form for the axial bar becomes

$$\int_0^L EA \frac{\mathrm{d}w}{\mathrm{d}x}\frac{\mathrm{d}u}{\mathrm{d}x}\,\mathrm{d}x = \int_0^L wp\,\mathrm{d}x + w(L)F. \tag{2.27}$$

This is the starting point for the bar element stiffness matrix derived in Chapter 3. In other words, the familiar finite element equations are not separate from weighted residual methods; they are the Galerkin weighted residual method applied element by element.

Chapter 3

One-Dimensional Bar and Rod Elements

3.1 Direct Formulation of the Bar Element

The simplest finite element is the two-node bar (rod) element, suitable for axially loaded members. Consider an element of length L, cross-sectional area A, and Young's modulus E, with nodes 1 and 2 having axial displacements u_1 and u_2 [4, 2].

Rather than starting from a variational principle, we derive the element stiffness directly from *mechanics of materials*: equilibrium of axial forces combined with Hooke's law. The principle of virtual work, derived later in this chapter (Section 3.8), reproduces the same stiffness matrix and provides the variational foundation for the entire finite element method.

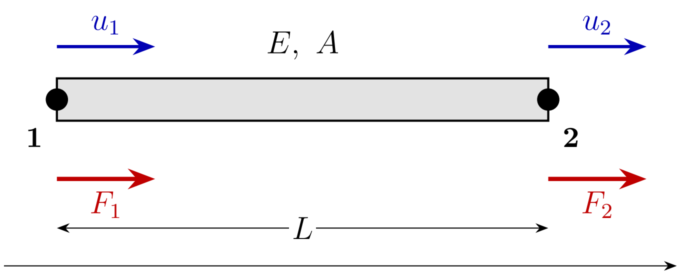

Figure 3.1: Two-node bar element: nodal displacements u_1, u_2 (blue, above) and external nodal forces F_1, F_2 (red, below), with cross-section A, modulus E, length L.

3.1.1 Element Kinematics

For small deformations, the axial elongation of the element is the difference of nodal displacements:

$$\Delta L = u_2 - u_1. \tag{3.1}$$

Assuming the strain is uniform along the element (consistent with linear displacement variation between nodes),

$$\varepsilon = \frac{\Delta L}{L} = \frac{u_2 - u_1}{L}. \tag{3.2}$$

Hooke's law and the axial force resultant $N = \sigma A$ give

$$\sigma = E\varepsilon = \frac{E(u_2 - u_1)}{L}, \qquad N = \frac{EA(u_2 - u_1)}{L}. \tag{3.3}$$

3.1.2 Stiffness Matrix from Equilibrium

Taking $N > 0$ in tension, the external nodal forces required to produce the displacements u_1, u_2 are

$$F_1 = -N = \frac{EA}{L}(u_1 - u_2), \tag{3.4}$$
$$F_2 = +N = \frac{EA}{L}(u_2 - u_1). \tag{3.5}$$

Collecting these in matrix form,

$$\begin{bmatrix} F_1 \\ F_2 \end{bmatrix} = \frac{EA}{L}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}\begin{bmatrix} u_1 \\ u_2 \end{bmatrix} \iff \boldsymbol{k}^e\,\boldsymbol{u}^e = \boldsymbol{f}^e. \tag{3.6}$$

The element stiffness matrix is therefore

$$\boldsymbol{k}^e = \frac{EA}{L}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}. \tag{3.7}$$

Key Result

$$\boldsymbol{k}^e = \frac{EA}{L}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

3.1.3 Unit Displacement Interpretation

The columns of $\boldsymbol{k}^e$ have a clear physical meaning. Set $u_1 = 1$ and $u_2 = 0$: the bar shortens, the strain is $\varepsilon = -1/L$, and the internal force is $N = -EA/L$ (compression). Equilibrium then requires the external forces $F_1 = +EA/L$ at node 1 and $F_2 = -EA/L$ at node 2 — exactly the first column of $\boldsymbol{k}^e$. The second column follows from $u_1 = 0,\ u_2 = 1$ by an identical argument.

Remark 3.1. The stiffness matrix is symmetric (Maxwell's reciprocal theorem) and singular because rows sum to zero: a uniform translation $u_1 = u_2 = c$ produces no strain and no force, reflecting the rigid-body mode of the unconstrained element.

3.2 Displacement Field Approximation

The direct formulation has used only nodal displacements. To analyze the response *between* nodes — for postprocessing strains and stresses, or for representing distributed loading — we need an approximate displacement field $u(x)$.

3.2.1 Linear Shape Functions

The displacement field consistent with the constant-strain assumption used in Equation (3.2) is linear in x:

$$u(x) = N_1(x)\,u_1 + N_2(x)\,u_2 = \boldsymbol{N}\,\boldsymbol{u}^e \tag{3.8}$$

with the linear Lagrange shape functions

$$N_1(x) = 1 - \frac{x}{L}, \qquad N_2(x) = \frac{x}{L}, \qquad x \in [0, L]. \tag{3.9}$$

Remark 3.2. Shape functions satisfy the *partition of unity*, $N_1 + N_2 = 1$, and the *Kronecker delta property*, $N_i(x_j) = \delta_{ij}$.

The strain operator follows by differentiation,

$$\varepsilon(x) = \frac{\mathrm{d}u}{\mathrm{d}x} = \boldsymbol{B}\,\boldsymbol{u}^e, \qquad \boldsymbol{B} = \frac{1}{L}\begin{bmatrix} -1 & 1 \end{bmatrix}, \tag{3.10}$$

which recovers $\varepsilon = (u_2 - u_1)/L$, in agreement with Equation (3.2).

3.2.2 Enrichment by a Particular Solution

The strong form of the loaded bar, derived in Chapter 2, is

$$-EA\,\frac{\mathrm{d}^2 u}{\mathrm{d}x^2} = p(x), \qquad 0 < x < L. \tag{3.11}$$

The linear interpolation Equation (3.8) solves the *homogeneous* equation $u'' = 0$ exactly, but cannot represent the inhomogeneous response when $p(x) \neq 0$.

To recover the exact response within the element, augment the linear field with a *particular solution* $u_p(x)$:

$$u(x) = N_1(x)\,u_1 + N_2(x)\,u_2 + u_p(x), \tag{3.12}$$

where $u_p(x)$ satisfies the inhomogeneous equation with homogeneous endpoint conditions:

$$-EA\,u_p''(x) = p(x), \qquad u_p(0) = 0, \qquad u_p(L) = 0. \tag{3.13}$$

The boundary conditions on u_p ensure that the linear part still recovers the nodal displacements at the endpoints, $u(0) = u_1$ and $u(L) = u_2$.

Key Result

Enriched trial function:

$$u(x) = N_1(x)\,u_1 + N_2(x)\,u_2 + u_p(x).$$

The first two terms carry the nodal displacements. The particular solution $u_p(x)$ carries the local response to the distributed load — solved once analytically per load case and added to every element under that loading.

3.2.3 Worked Example: Uniform Distributed Load

For a uniform distributed axial load $p(x) = p_0$, integrating Equation (3.13) twice and applying $u_p(0) = u_p(L) = 0$ gives

$$u_p(x) = \frac{p_0\, x(L-x)}{2EA}. \tag{3.14}$$

This is a parabolic profile peaking at midspan with $u_p(L/2) = p_0 L^2/(8EA)$. Substituting into Equation (3.12) gives the complete element displacement field for a uniformly loaded bar:

$$u(x) = \left(1 - \frac{x}{L}\right) u_1 + \frac{x}{L}\, u_2 + \frac{p_0\, x(L-x)}{2EA}. \tag{3.15}$$

Differentiating once gives the axial strain

$$\varepsilon(x) = \frac{u_2 - u_1}{L} + \frac{p_0(L-2x)}{2EA}, \tag{3.16}$$

and the internal axial force

$$N(x) = EA\,\varepsilon(x) = \frac{EA(u_2 - u_1)}{L} + \frac{p_0(L-2x)}{2}. \tag{3.17}$$

3.3 Element Load Vector

External forces at the element ends follow from equilibrium with the internal axial force in Equation (3.17): $F_1 = -N(0)$ and $F_2 = +N(L)$. Substituting,

$$F_1 = \frac{EA}{L}(u_1 - u_2) - \frac{p_0 L}{2}, \tag{3.18}$$
$$F_2 = \frac{EA}{L}(u_2 - u_1) - \frac{p_0 L}{2}. \tag{3.19}$$

Rearranging into the standard finite element form,

$$\boldsymbol{k}^e\, \boldsymbol{u}^e = \boldsymbol{f}^e, \qquad \boldsymbol{f}^e = \frac{p_0 L}{2}\begin{bmatrix}1\\1\end{bmatrix}. \tag{3.20}$$

Half of the total distributed load is placed at each node, recovering the familiar consistent load vector. Note that the same vector is obtained from the variational expression $\boldsymbol{f}^e = \int_0^L \boldsymbol{N}^\mathsf{T} p\, \mathrm{d}x$ used in Chapter 2 and in the PVW derivation of Section 3.8 — direct mechanics, weak form, and PVW all agree.

Remark 3.3. For constant EA and uniform p_0, the enriched element with $u_p(x)$ is *nodally exact*: even a single element gives the exact analytical displacement and stress at the nodes. Refining the mesh only improves the resolution of the smooth interior fields, not the nodal accuracy.

3.4 Global Assembly

Consider a bar discretized into n elements with $n+1$ nodes. Each element connects local nodes 1–2 to global nodes i-$(i+1)$. The global stiffness matrix $\boldsymbol{K} \in \mathbb{R}^{(n+1)\times(n+1)}$ is assembled by the *direct stiffness method*:

$$\boldsymbol{K} = \sum_{e=1}^{n} \boldsymbol{L}_e^\mathsf{T} \boldsymbol{k}^e \boldsymbol{L}_e \tag{3.21}$$

where $\boldsymbol{L}_e$ is the Boolean connectivity matrix that maps local to global DOFs.

In practice, assembly is performed by adding element stiffness entries at the appropriate global positions:

$$K_{ij} \mathrel{+}= k^e_{rs}, \quad \text{where } i = \text{global}(r),\ j = \text{global}(s) \tag{3.22}$$

3.5 Boundary Conditions

3.5.1 Homogeneous Essential BC (fixed node)

Zero out the row and column corresponding to the prescribed DOF, and set the diagonal to 1:

$$K_{ii} = 1, \quad K_{ij} = K_{ji} = 0\ \forall j \neq i, \quad f_i = 0 \tag{3.23}$$

3.5.2 Penalty Method

Add a large number $\alpha \gg \max(K_{ii})$ to the diagonal:

$$K_{ii} \mathrel{+}= \alpha, \quad f_i \mathrel{+}= \alpha \bar{u}_i \tag{3.24}$$

where $\bar{u}_i$ is the prescribed displacement.

3.6 Worked Example: Three-Element Bar

Example 3.1 (Three-element bar under tip load)**.** Consider a stepped bar of total length $L = 0.9$ m fixed at $x = 0$, with a point load $F = 10{,}000$ N at $x = L$. Three equal elements, $EA = 70 \times 10^6$ N.

Element stiffness:

$$\boldsymbol{k}^e = \frac{EA}{L/3} = \frac{70 \times 10^6}{0.3} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} = 2.\overline{3} \times 10^8 \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

Graphical assembly of the global stiffness matrix. The three identical 2×2 element matrices are added into the 4×4 global matrix at the rows and columns of their nodes (Figure 3.2): element 1 occupies DOFs 1–2, element 2 DOFs 2–3, element 3 DOFs 3–4. The two interior diagonals (DOFs 2 and 3) each receive contributions from two elements, so their global values equal $1 + 1 = 2$.

The resulting global system (4 DOFs, node 1 fixed; no distributed loads, so $\boldsymbol{f}$ contains only the concentrated tip force) is

$$\boldsymbol{K} = 2.\overline{3} \times 10^8 \begin{bmatrix} 1 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 1 \end{bmatrix}, \qquad \boldsymbol{f} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 10000 \end{bmatrix}.$$

Applying BC $u_1 = 0$ (strike row/column 1):

$$\boldsymbol{u} = \boldsymbol{K}^{-1}\boldsymbol{f} \implies u_4 = \frac{FL}{EA} = 1.286 \times 10^{-4} \text{ m}$$

See Section A.1 in the Appendix for the MATLAB implementation.

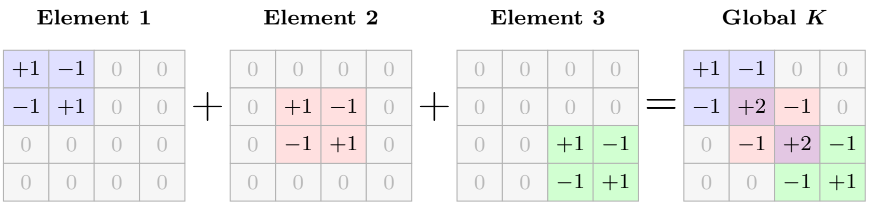

Figure 3.2: Graphical assembly of the global stiffness matrix (in units of $EA/L_e = 2.\overline{3} \times 10^8$ N/m). Element 1 (blue) at DOFs 1–2, element 2 (red) at DOFs 2–3, element 3 (green) at DOFs 3–4. The violet diagonals at DOFs 2 and 3 receive contributions from two elements each.

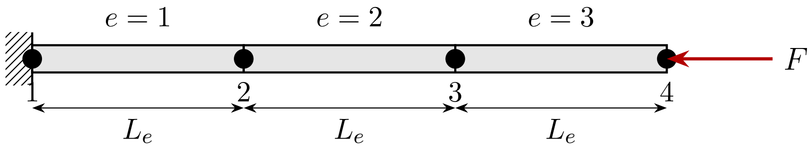

Figure 3.3: Three-element bar discretization: fixed at node 1, tip load F at node 4.

3.7 Worked Example: Stepped Bar with Combined Loading

Example 3.2 (Stepped axial bar with distributed and concentrated loads)**.** A cantilever stepped bar comprises two prismatic segments. Element 1 has length $L_1 = 500$ mm and area $A = 225$ mm^2; element 2 has length $L_2 = 400$ mm and area $2A = 450$ mm^2. The Young's modulus is $E = 210{,}000$ MPa. A uniform distributed axial load $p_1 = -4$ N/mm acts on element 1 (negative = directed toward the fixed support), and concentrated forces of $+6$ kN at node 2 and -15 kN at node 3 are applied (Figure 3.4).

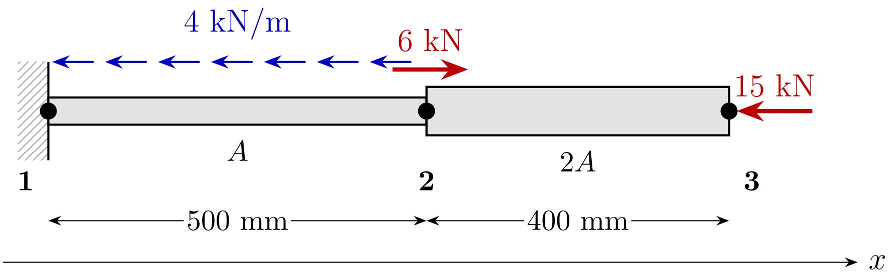

Figure 3.4: Stepped bar problem: distributed axial load (blue) on element 1 and concentrated forces (red) at nodes 2 and 3.

Discretization. Two linear bar elements connect three nodes located at $x_1 = 0$, $x_2 = 500$, $x_3 = 900$ mm. Each node carries one axial DOF, giving $n_{\text{dof}} = 3$.

Node coordinates		**Connectivity**		
Node	x [mm]	Element	Nodes	DOFs
1	0	1	1, 2	1, 2
2	500	2	2, 3	2, 3
3	900			

Element matrices. Using the direct stiffness from Equation (3.7),

$$\boldsymbol{k}^{(1)} = \frac{EA}{L_1}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} = 94{,}500\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}\frac{\mathrm{N}}{\mathrm{mm}}, \quad \boldsymbol{k}^{(2)} = \frac{E(2A)}{L_2}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} = 236{,}250\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}\frac{\mathrm{N}}{\mathrm{mm}}.$$

The consistent distributed-load vector for element 1, from Equation (3.20), is

$$\boldsymbol{f}_p^{(1)} = \tfrac{p_1 L_1}{2}\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -1000 \\ -1000 \end{bmatrix}\mathrm{N}, \qquad \boldsymbol{f}_p^{(2)} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}\mathrm{N}.$$

Graphical assembly of the global stiffness matrix. Each 2×2 element matrix is added into the 3×3 global matrix at the rows and columns corresponding to its nodes (Figure 3.5): element 1 occupies the top-left block (rows/columns 1 and 2), element 2 the bottom-right block (rows/columns 2 and 3). The shared entry at row/column 2 receives contributions from both elements.

Element 1 contribution

94500	−94500	0
−94500	94500	0
0	0	0

+

Element 2 contribution

0	0	0
0	236250	−236250
0	−236250	236250

=

Global $\boldsymbol{K}$

94500	−94500	0
−94500	330750	−236250
0	−236250	236250

Figure 3.5: Graphical assembly of the global stiffness matrix. Blue cells receive contributions from element 1, red cells from element 2, and the violet cell at row/column 2 receives contributions from both, giving $94{,}500 + 236{,}250 = 330{,}750$.

Graphical assembly of the load vector. The same row-mapping rule applies to the load vector: each element's distributed-load contribution $\boldsymbol{f}_p^{(e)}$ is placed into the global vector at the rows corresponding to its nodes (Figure 3.6). Element 1 contributes to DOFs 1 and 2; element 2 to DOFs 2 and 3; DOF 2 receives both, identically to the matrix case.

Element 1 contribution		Element 2 contribution		Distributed $\boldsymbol{f}_p$
−1000		0		−1000
−1000	+	0	=	−1000
0		0		0

Figure 3.6: Graphical assembly of the distributed-load vector. Element 1's 2×1 contribution lands at DOFs 1 and 2 (blue); element 2's at DOFs 2 and 3 (red, all zero here because $p_2 = 0$); the violet entry at DOF 2 receives contributions from both elements ($-1000 + 0 = -1000$). The colour scheme is identical to Figure 3.5.

The concentrated forces $\boldsymbol{f}_c$, applied directly at their respective nodes, are then added on top of $\boldsymbol{f}_p$ to give the total load vector:

$$\boldsymbol{f}_p = \begin{bmatrix} -1000 \\ -1000 \\ 0 \end{bmatrix}, \quad \boldsymbol{f}_c = \begin{bmatrix} 0 \\ 6000 \\ -15000 \end{bmatrix}, \quad \boldsymbol{f} = \boldsymbol{f}_p + \boldsymbol{f}_c = \begin{bmatrix} -1000 \\ 5000 \\ -15000 \end{bmatrix} \text{ N.}$$

Boundary conditions and reduced system. Node 1 is fixed, $u_1 = 0$. The free-DOF set is $\{2, 3\}$. Striking the first row and column,

$$\underbrace{\begin{bmatrix} 330{,}750 & -236{,}250 \\ -236{,}250 & 236{,}250 \end{bmatrix}}_{\boldsymbol{K}_{ff}} \begin{bmatrix} u_2 \\ u_3 \end{bmatrix} = \underbrace{\begin{bmatrix} 5{,}000 \\ -15{,}000 \end{bmatrix}}_{\boldsymbol{f}_f}.$$

Solving,

$$u_2 = -\tfrac{20}{189} \approx -0.1058 \text{ mm}, \qquad u_3 = -\tfrac{32}{189} \approx -0.1693 \text{ mm}.$$

Reaction at the fixed support. The reaction follows from the first row of the unreduced system:

$$R_1 = \boldsymbol{K}_{1,*}\,\boldsymbol{u} - f_1 = (-94{,}500)(-\tfrac{20}{189}) - (-1000) = +11{,}000 \text{ N},$$

which balances the resultant of all applied axial forces: $-(-2000) - 6000 - (-15000) = +11{,}000$ N.

Internal displacement and force fields. Inside each element the displacement follows the enriched trial Equation (3.12), evaluated with the nodal solution and the particular solution Equation (3.14). Differentiating yields $N(x)$ from Equation (3.17). The resulting curves (Figures 3.7 and 3.8) show the parabolic curvature in element 1 caused by $p_1 \neq 0$, the linear field in element 2, and the +6 kN jump in $N(x)$ at $x = 500$ mm corresponding to the concentrated load applied there.

See Section A.2 in the Appendix for the full MATLAB implementation that produced these plots.

3.8 Principle of Virtual Work for the Axial Bar

The direct formulation of Section 3.1 and the worked examples above gave the bar element stiffness and load vector from physical equilibrium. The principle of virtual work (PVW) provides an alternative, energy-based statement of equilibrium that reproduces the same results and generalizes immediately to non-uniform loading, variable cross-section, and higher-order elements. PVW is the variational foundation on which the entire finite element method rests.

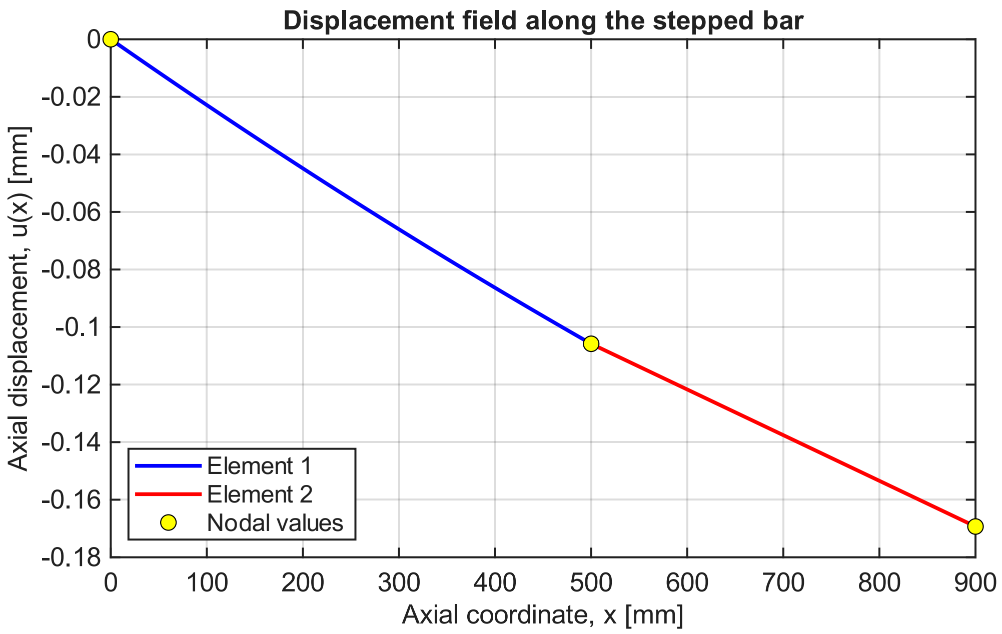

Figure 3.7: Displacement field $u(x)$ along the stepped bar.

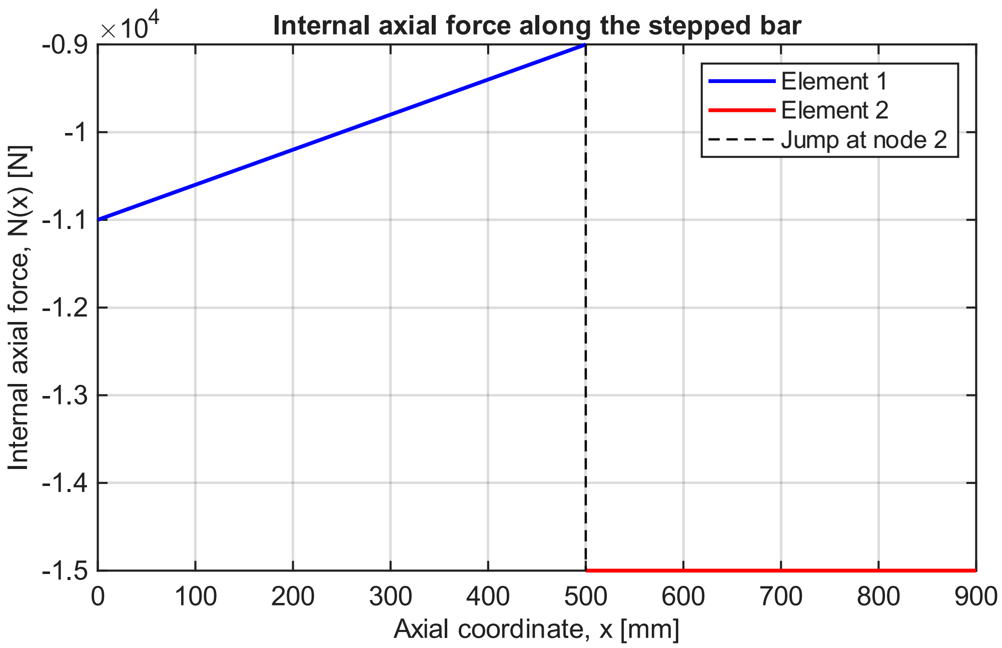

Figure 3.8: Internal axial force $N(x)$ along the stepped bar. The discontinuity at $x = 500$ mm corresponds to the +6 kN concentrated force applied at node 2.

3.8.1 Statement of the Principle

Key Result

Principle of Virtual Work: For a bar in static equilibrium, and for any *kinematically admissible* virtual displacement field $\delta u(x)$,

$$\delta W_{\text{int}} = \delta W_{\text{ext}}, \tag{3.25}$$

with internal and external virtual work

$$\delta W_{\text{int}} = \int_0^L N(x)\,\delta\varepsilon(x)\;\mathrm{d}x = \int_0^L EA\,\frac{\mathrm{d}u}{\mathrm{d}x}\,\frac{\mathrm{d}\delta u}{\mathrm{d}x}\;\mathrm{d}x,$$

$$\delta W_{\text{ext}} = \int_0^L p(x)\,\delta u(x)\;\mathrm{d}x + \sum_i F_i\,\delta u(x_i).$$

A virtual displacement is *kinematically admissible* if it satisfies the essential boundary conditions: $\delta u = 0$ wherever u is prescribed.

The virtual displacement $\delta u(x)$ is an arbitrary, infinitesimal perturbation. It need not satisfy equilibrium or any natural boundary conditions — its sole role is to test the equilibrium of the actual displacement field $u(x)$.

3.8.2 Derivation from Equilibrium

Begin with the local equilibrium equation derived in Chapter 2:

$$\frac{\mathrm{d}N}{\mathrm{d}x} + p(x) = 0, \qquad 0 < x < L. \tag{3.26}$$

Multiply both sides by an arbitrary virtual displacement $\delta u(x)$ and integrate over the bar:

$$\int_0^L \left(\frac{\mathrm{d}N}{\mathrm{d}x} + p\right)\delta u\;\mathrm{d}x = 0. \tag{3.27}$$

Integration by parts on the first term gives

$$\int_0^L \frac{\mathrm{d}N}{\mathrm{d}x}\,\delta u\;\mathrm{d}x = \Big[N(x)\,\delta u(x)\Big]_0^L - \int_0^L N(x)\,\frac{\mathrm{d}\delta u}{\mathrm{d}x}\;\mathrm{d}x. \tag{3.28}$$

Substituting back into Equation (3.27) and rearranging,

$$\int_0^L N\,\frac{\mathrm{d}\delta u}{\mathrm{d}x}\;\mathrm{d}x = \Big[N\,\delta u\Big]_0^L + \int_0^L p\,\delta u\;\mathrm{d}x. \tag{3.29}$$

The boundary term vanishes wherever the displacement is prescribed (kinematic admissibility forces $\delta u = 0$ there) and reduces to $F_i\,\delta u(x_i)$ at any free end with a concentrated force F_i (because N at that end equals the applied force). Substituting the constitutive law $N = EA\,\mathrm{d}u/\mathrm{d}x$ and $\delta\varepsilon = \mathrm{d}\delta u/\mathrm{d}x$ on the left, equation Equation (3.29) becomes

$$\int_0^L EA\,\frac{\mathrm{d}u}{\mathrm{d}x}\,\frac{\mathrm{d}\delta u}{\mathrm{d}x}\;\mathrm{d}x \;=\; \int_0^L p\,\delta u\;\mathrm{d}x \;+\; \sum_i F_i\,\delta u(x_i), \tag{3.30}$$

which is exactly the PVW statement Equation (3.25).

Remark 3.4. Equation Equation (3.30) is the *weak form* of the bar equilibrium equation. The strong form $-EA\,u'' = p$ requires twice-differentiable trial functions; the weak form requires only first derivatives. This relaxation makes the weak form well-suited to piecewise-polynomial finite element approximations whose derivatives may be discontinuous between elements.

3.8.3 Recovery of the Bar Stiffness Matrix

To reproduce the element stiffness derived in Section 3.1, substitute the linear interpolation

$$u(x) = N_1(x)\,u_1 + N_2(x)\,u_2, \qquad \delta u(x) = N_1(x)\,\delta u_1 + N_2(x)\,\delta u_2, \tag{3.31}$$

with strain-displacement matrix $\boldsymbol{B} = \frac{1}{L}[\,-1,\;\;1\,]$, into the left-hand side of Equation (3.30):

$$\delta \boldsymbol{u}^{e\mathsf{T}} \left[\int_0^L EA\,\boldsymbol{B}^\mathsf{T}\boldsymbol{B}\;\mathrm{d}x\right]\boldsymbol{u}^e = \delta \boldsymbol{u}^{e\mathsf{T}}\,\boldsymbol{k}^e\,\boldsymbol{u}^e,$$

giving

$$\boldsymbol{k}^e = \int_0^L EA\,\boldsymbol{B}^\mathsf{T}\boldsymbol{B}\;\mathrm{d}x = \frac{EA}{L}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix},$$

identical to Equation (3.7). The right-hand side likewise produces the consistent load vector $\boldsymbol{f}^e = \int_0^L \boldsymbol{N}^\mathsf{T} p\;\mathrm{d}x + (\text{nodal forces})$. Direct mechanics, weak form, and PVW all give the same element equations.

3.8.4 Worked Example: Verification of PVW on the Stepped Bar

Example 3.3 (Numerical verification of $\delta W_{\mathrm{int}} = \delta W_{\mathrm{ext}}$)**.** To illustrate that PVW holds for a real loaded bar in equilibrium, we return to the stepped bar of Example 3.2 (already solved exactly). Recall the actual nodal solution

$$u_1 = 0, \quad u_2 = -\tfrac{20}{189} \approx -0.10582 \text{ mm}, \quad u_3 = -\tfrac{32}{189} \approx -0.16931 \text{ mm},$$

the support reaction $R_1 = +11{,}000$ N, and the internal axial force per element from Equation (3.17),

$$N^{(1)}(x_1) = \tfrac{EA_1(u_2-u_1)}{L_1} + \tfrac{p_1(L_1-2x_1)}{2} = -10{,}000 + (-2)(500 - 2x_1) = -11{,}000 + 4x_1 \;\;\text{[N]},$$
$$N^{(2)}(x_2) = \tfrac{EA_2(u_3-u_2)}{L_2} = -15{,}000 \;\;\text{[N]} \quad \text{(constant)},$$

where $x_1 \in [0, 500]$ and $x_2 \in [0, 400]$ are local element coordinates.

We now choose two different kinematically admissible virtual displacements $\delta u(x)$ (each must satisfy $\delta u(0) = 0$) and verify the equality $\delta W_{\mathrm{int}} = \delta W_{\mathrm{ext}}$ by direct calculation of both sides.

Case A: $(\delta u_1, \delta u_2, \delta u_3) = (0, 1, 0)$ **"tent".** Piecewise linear with the apex at node 2:

$$\delta u^{(1)}(x_1) = \frac{x_1}{L_1} = \frac{x_1}{500}, \qquad \delta u^{(2)}(x_2) = 1 - \frac{x_2}{L_2} = 1 - \frac{x_2}{400}.$$

The corresponding virtual strains are constant on each element:

$$\delta\varepsilon^{(1)} = +\tfrac{1}{500}, \qquad \delta\varepsilon^{(2)} = -\tfrac{1}{400}.$$

Internal virtual work. For an element with constant EA and uniform p, $\int_0^L N(x)\,\mathrm{d}x = EA(u_b - u_a)$ (the distributed-load contribution integrates to zero). Hence

$$\begin{aligned}
\delta W_{\text{int}}^{(1)} &= \delta\varepsilon^{(1)}\int_0^{L_1} N^{(1)}\,\mathrm{d}x_1 = \tfrac{1}{500}\,EA_1(u_2 - u_1) = \tfrac{1}{500}\,(-5{,}000{,}000) = -10{,}000\ \mathrm{N\,mm},\\
\delta W_{\text{int}}^{(2)} &= \delta\varepsilon^{(2)}\int_0^{L_2} N^{(2)}\,\mathrm{d}x_2 = -\tfrac{1}{400}\,EA_2(u_3 - u_2) = -\tfrac{1}{400}\,(-6{,}000{,}000) = +15{,}000\ \mathrm{N\,mm},\\
\delta W_{\text{int}} &= -10{,}000 + 15{,}000 = +5{,}000\ \mathrm{N\,mm}.
\end{aligned}$$

External virtual work. The reaction at node 1 does no virtual work because $\delta u_1 = 0$.

$$\begin{aligned}
\int_0^{L_1} p_1\,\delta u^{(1)}\,\mathrm{d}x_1 &= \int_0^{500}(-4)\left(\tfrac{x_1}{500}\right)\mathrm{d}x_1 = -1{,}000\ \mathrm{N\,mm},\\
F_2\,\delta u_2 + F_3\,\delta u_3 &= (6{,}000)(1) + (-15{,}000)(0) = +6{,}000\ \mathrm{N\,mm},\\
\delta W_{\text{ext}} &= -1{,}000 + 6{,}000 = +5{,}000\ \mathrm{N\,mm}.
\end{aligned}$$

Result: $\delta W_{\text{int}} = \delta W_{\text{ext}} = +5{,}000$ N mm. ✓

Case B: $(\delta u_1, \delta u_2, \delta u_3) = (0, 1, 1)$ **"ramp".** Linear in element 1, constant in element 2:

$$\delta u^{(1)} = \frac{x_1}{500}, \qquad \delta u^{(2)} = 1, \qquad \delta\varepsilon^{(1)} = \tfrac{1}{500}, \qquad \delta\varepsilon^{(2)} = 0.$$

$$\begin{aligned}
\delta W_{\text{int}} &= \tfrac{1}{500}(-5{,}000{,}000) + 0 = -10{,}000\ \mathrm{N\,mm},\\
\delta W_{\text{ext}} &= \underbrace{-1{,}000}_{\text{distributed load}} + \underbrace{(6{,}000)(1) + (-15{,}000)(1)}_{\text{concentrated forces}} = -1{,}000 - 9{,}000 = -10{,}000\ \mathrm{N\,mm}.
\end{aligned}$$

Result: $\delta W_{\text{int}} = \delta W_{\text{ext}} = -10{,}000$ N mm. ✓

The two virtual displacements above coincide with the basis used in the FEM discretization (piecewise linear, one unit value at a free DOF, zero elsewhere). Choosing δu as the i-th basis function reproduces exactly the i-th row of the reduced FEM equations $\boldsymbol{K}_{ff}\boldsymbol{u}_f = \boldsymbol{f}_f$. Indeed, +5,000 and −10,000 are the assembled effective forces at DOF 2 and DOF 2+3 for these two perturbations.

Beyond piecewise linear. PVW holds for any kinematically admissible virtual displacement, not just those built from the FEM basis. The MATLAB script Section A.3 also evaluates a smooth quadratic $\delta u(x) = (x/900)^2$ and reports

$$\delta W_{\text{int}} = \delta W_{\text{ext}} \approx -13{,}353.91\ \mathrm{N\,mm} \qquad (|\Delta| < 10^{-11}\ \mathrm{N\,mm}),$$

confirming that the work balance is independent of the chosen test field.

Remark 3.5. This verification is more than a numerical check — it is the defining property of the equilibrium configuration. Of all kinematically admissible displacement fields, the actual $u(x)$ is the unique one for which the work balance holds for *every* admissible perturbation. The finite element method exploits this property: by enforcing PVW with the basis functions of the chosen mesh as test fields, it produces the discrete equations $\boldsymbol{K}\boldsymbol{u} = \boldsymbol{f}$ whose solution is the best approximation to the true equilibrium within that finite-dimensional space.

3.9 Galerkin Finite Element Formulation of the Bar

Remark 3.6. Chapter 2 applied the Galerkin method with a single global polynomial – one free parameter, one weight function – to the continuous bar. The formulation developed here uses the finite element shape functions N_1, N_2 of Equation (3.9) as both trial and test functions on each element. With piecewise-linear shape functions and one test function per node, this is the Galerkin principle applied at the element level: the method through which every finite element stiffness integral is derived in practice.

3.9.1 From Weighted Residuals to Shape Functions

Start from the strong form of the loaded bar (Equation (3.11)):

$$-EA\,\frac{\mathrm{d}^2u}{\mathrm{d}x^2} = p(x), \qquad 0 < x < L.$$

Substitute the finite element approximation $u^h(x) = \boldsymbol{N}(x)\,\boldsymbol{u}^e$ from Equation (3.8). Because N_1 and N_2 are linear in x, their second derivative vanishes identically, so the residual is

$$R(x) = -EA\,\frac{\mathrm{d}^2u^h}{\mathrm{d}x^2} - p(x) = -p(x) \neq 0. \tag{3.32}$$

The linear approximation cannot satisfy the ODE pointwise when $p(x) \neq 0$; it satisfies it instead in an *average* sense. The **Galerkin statement** requires the residual to be orthogonal to each shape function:

$$\int_0^L N_i(x)\,R(x)\,\mathrm{d}x = 0, \qquad i = 1,\,2. \tag{3.33}$$

This choice – test functions identical to the trial functions – is called the *Bubnov–Galerkin* method and is standard for self-adjoint problems such as linear elasticity.

3.9.2 Galerkin Weak Form for a Single Element

Substituting Equation (3.32) into Equation (3.33) and expanding:

$$\int_0^L N_i\left(-EA\,\frac{\mathrm{d}^2u^h}{\mathrm{d}x^2} - p\right)\,\mathrm{d}x = 0.$$

Integration by parts on the second-derivative term (the same step as Equation (3.28)) gives

$$\int_0^L EA\,\frac{\mathrm{d}N_i}{\mathrm{d}x}\,\frac{\mathrm{d}u^h}{\mathrm{d}x}\,\mathrm{d}x \;-\; \left[N_i\,EA\,\frac{\mathrm{d}u^h}{\mathrm{d}x}\right]_0^L \;=\; \int_0^L N_i\,p\,\mathrm{d}x. \tag{3.34}$$

The boundary term vanishes at nodes where the displacement is prescribed (kinematic admissibility requires $N_i = 0$ there), and reduces to the applied concentrated force F at a free end via the natural boundary condition $EA\,\mathrm{d}u/\mathrm{d}x = F$. The result is the **Galerkin weak form**:

$$\int_0^L EA\,\frac{\mathrm{d}N_i}{\mathrm{d}x}\,\frac{\mathrm{d}u^h}{\mathrm{d}x}\,\mathrm{d}x \;=\; \int_0^L N_i\,p\,\mathrm{d}x \;+\; F_i^{\mathrm{natural}}. \tag{3.35}$$

3.9.3 Element Stiffness Matrix and Load Vector

Writing both rows ($i = 1$ and $i = 2$) simultaneously in matrix form,

$$\underbrace{\left[\int_0^L EA\,\boldsymbol{B}^{\mathsf{T}}\boldsymbol{B}\;\mathrm{d}x\right]}_{\boldsymbol{k}^e}\boldsymbol{u}^e \;=\; \underbrace{\int_0^L \boldsymbol{N}^{\mathsf{T}} p\;\mathrm{d}x}_{\boldsymbol{f}^e},$$

where $\boldsymbol{B} = \mathrm{d}\boldsymbol{N}/\,\mathrm{d}x = \frac{1}{L}[\,-1,\;1\,]$ from Equation (3.10). Evaluating the integrals:

$$\boldsymbol{k}^e = \int_0^L EA\,\boldsymbol{B}^{\mathsf{T}}\boldsymbol{B}\;\mathrm{d}x = \frac{EA}{L}\begin{bmatrix}1 & -1\\ -1 & 1\end{bmatrix}, \tag{3.36}$$

$$\boldsymbol{f}^e = \int_0^L \boldsymbol{N}^{\mathsf{T}} p_0\;\mathrm{d}x = p_0\int_0^L \begin{bmatrix}1 - x/L\\ x/L\end{bmatrix}\mathrm{d}x = \frac{p_0 L}{2}\begin{bmatrix}1\\ 1\end{bmatrix}, \tag{3.37}$$

in exact agreement with Equation (3.7) and Equation (3.20).

Key Result

The Galerkin finite element formulation yields

$$\boldsymbol{k}^e = \int_0^L EA\,\boldsymbol{B}^{\mathsf{T}}\boldsymbol{B}\;\mathrm{d}x = \frac{EA}{L}\begin{bmatrix}1 & -1\\ -1 & 1\end{bmatrix}, \qquad \boldsymbol{f}^e = \int_0^L \boldsymbol{N}^{\mathsf{T}} p\;\mathrm{d}x = \frac{p_0 L}{2}\begin{bmatrix}1\\ 1\end{bmatrix},$$

identical to Equation (3.7) and Equation (3.20).

3.10 Equivalence of Galerkin and Virtual Work Formulations

The Galerkin weak form Equation (3.35) and the PVW equation Equation (3.30) are the same integral statement. To see this, write them side by side:

$$\begin{aligned}
\text{PVW:}\quad & \int_0^L EA\,\frac{\mathrm{d}u}{\mathrm{d}x}\,\frac{\mathrm{d}\delta u}{\mathrm{d}x}\;\mathrm{d}x = \int_0^L p\,\delta u\;\mathrm{d}x + \sum_i F_i\,\delta u(x_i),\\
\text{Galerkin FE:}\quad & \int_0^L EA\,\frac{\mathrm{d}u^h}{\mathrm{d}x}\,\frac{\mathrm{d}w_i}{\mathrm{d}x}\;\mathrm{d}x = \int_0^L p\,w_i\;\mathrm{d}x + \sum_i F_i\,w_i(x_i).
\end{aligned} \tag{3.38}$$

Setting $w_i = \delta u$, the two expressions are identical: both require the integral to vanish for every kinematically admissible test field, and both produce the same discrete system $\boldsymbol{k}^e\,\boldsymbol{u}^e = \boldsymbol{f}^e$.

Remark 3.7. PVW speaks the language of mechanics: δu is an infinitesimal virtual displacement, and the equation expresses the balance of virtual internal and external work. Galerkin speaks the language of weighted residuals: $w_i = N_i$ is a test function chosen to make the residual orthogonal to the approximation space. They are the same mathematical object. In the finite element context, both methods mandate choosing test functions from the same polynomial space as the trial – which is why they produce identical element matrices.

3.11 Principle of Minimum Potential Energy

3.11.1 Total Potential Energy of the Axial Bar

Definition 3.1. The **strain energy** stored in a linearly elastic axially loaded bar is

$$U = \frac{1}{2}\int_0^L EA\left(\frac{\mathrm{d}u}{\mathrm{d}x}\right)^2 \mathrm{d}x. \tag{3.39}$$

The **total potential energy** is

$$\Pi[u] = U - W_{\mathrm{ext}}, \qquad W_{\mathrm{ext}} = \int_0^L p(x)\,u(x)\;\mathrm{d}x + \sum_i F_i\,u(x_i). \tag{3.40}$$

The strain energy $U \geq 0$ is always non-negative. The total potential energy Π is a functional of the displacement field $u(x)$: each admissible u gives a scalar value of Π, and the equilibrium field is singled out by a stationarity condition.

3.11.2 Stationarity Condition and the Weak Form

Key Result

Principle of Minimum Potential Energy (PMPE): Of all kinematically admissible displacement fields, the one that satisfies equilibrium makes the total potential energy stationary:

$$\delta\Pi = 0 \tag{3.41}$$

for every admissible variation δu. For a stable elastic equilibrium, Π is in fact a global minimum.

Taking the first variation of Equation (3.40),

$$\delta\Pi = \int_0^L EA\,\frac{\mathrm{d}u}{\mathrm{d}x}\,\frac{\mathrm{d}\delta u}{\mathrm{d}x}\;\mathrm{d}x \;-\; \int_0^L p\,\delta u\;\mathrm{d}x \;-\; \sum_i F_i\,\delta u(x_i) \;=\; 0. \tag{3.42}$$

Recognizing the first integral as δW_{int} and the remaining terms as δW_{ext}, this is precisely the PVW statement Equation (3.25):

$$\delta\Pi = 0 \iff \delta W_{\mathrm{int}} = \delta W_{\mathrm{ext}}. \tag{3.43}$$

Remark 3.8. PMPE is a special case of PVW valid for *conservative* systems, where the material is linearly elastic and the external loads are independent of the deformation. For the axially loaded bar with fixed applied loads, the two principles are exactly equivalent. PVW is the more general statement: it extends to non-conservative loading (e.g. follower loads) for which no potential energy exists.

3.11.3 Element Stiffness Matrix and Load Vector from PMPE

Substitute the finite element trial $u = \boldsymbol{N}\,\boldsymbol{u}^e$, $\mathrm{d}u/\,\mathrm{d}x = \boldsymbol{B}\,\boldsymbol{u}^e$ into the element contributions to Equation (3.40). The strain energy becomes

$$U^e = \frac{1}{2}\int_0^L EA\,\boldsymbol{u}^{e\mathsf{T}}\boldsymbol{B}^{\mathsf{T}}\boldsymbol{B}\,\boldsymbol{u}^e\;\mathrm{d}x = \frac{1}{2}\,\boldsymbol{u}^{e\mathsf{T}}\underbrace{\left[\int_0^L EA\,\boldsymbol{B}^{\mathsf{T}}\boldsymbol{B}\;\mathrm{d}x\right]}_{\boldsymbol{k}^e}\boldsymbol{u}^e = \frac{1}{2}\,\boldsymbol{u}^{e\mathsf{T}}\boldsymbol{k}^e\,\boldsymbol{u}^e,$$

and the work of external loads (uniform p_0):

$$W^e_{\text{ext}} = \int_0^L (\boldsymbol{N}\,\boldsymbol{u}^e)\,p_0\,\mathrm{d}x = \boldsymbol{u}^{e\mathsf{T}} \underbrace{\left[\int_0^L \boldsymbol{N}^\mathsf{T} p_0\,\mathrm{d}x\right]}_{\boldsymbol{f}^e} = \boldsymbol{u}^{e\mathsf{T}} \boldsymbol{f}^e.$$

The element potential energy is therefore a quadratic function of the nodal displacements:

$$\Pi^e = \frac{1}{2}\boldsymbol{u}^{e\mathsf{T}}\boldsymbol{k}^e\,\boldsymbol{u}^e - \boldsymbol{u}^{e\mathsf{T}}\boldsymbol{f}^e. \tag{3.44}$$

Differentiating with respect to $\boldsymbol{u}^e$ and setting the result to zero,

$$\frac{\partial \Pi^e}{\partial \boldsymbol{u}^e} = \boldsymbol{k}^e\,\boldsymbol{u}^e - \boldsymbol{f}^e = \boldsymbol{0} \quad\Longrightarrow\quad \boldsymbol{k}^e\,\boldsymbol{u}^e = \boldsymbol{f}^e. \tag{3.45}$$

The integrals evaluate to Equation (3.36) and Equation (3.37), so $\boldsymbol{k}^e$ and $\boldsymbol{f}^e$ are again identical to Equation (3.7) and Equation (3.20).

Key Result

Minimizing the element potential energy yields

$$\boldsymbol{k}^e = \int_0^L EA\,\boldsymbol{B}^\mathsf{T}\boldsymbol{B}\,\mathrm{d}x = \frac{EA}{L}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}.$$

The stiffness matrix is the Hessian of the strain energy with respect to the nodal displacements.

3.11.4 Summary: Four Equivalent Paths to the Same Element Matrices

Four independent derivations all produce the same element stiffness matrix $\boldsymbol{k}^e$ (Equation (3.7)) and load vector $\boldsymbol{f}^e$ (Equation (3.20)).

Table 3.1: Four derivation paths for the bar element matrices.

Method	**Starting point**	**Path to $\boldsymbol{k}^e$**	**Load vector $\boldsymbol{f}^e$**
Direct Equilibrium	Hooke's law + nodal force balance	$N = EA\varepsilon$; collect into matrix form	Enriched trial + end reactions (Section 3.2)
PVW	$\delta W_{\text{int}} = \delta W_{\text{ext}}$ (Equation (3.25))	Substitute $u = \boldsymbol{N}\boldsymbol{u}^e$, $\delta u = \boldsymbol{N}\delta\boldsymbol{u}^e$	RHS of weak form (Section 3.8)
Galerkin FE	$\int N_i R\,\mathrm{d}x = 0$, $w_i = N_i$ (Equation (3.33))	Integration by parts + $\boldsymbol{B}$-matrix (Section 3.9)	RHS of weak form (Equation (3.37))
PMPE	$\delta\Pi = 0$, $\Pi = U - W_{\text{ext}}$ (Equation (3.40))	Substitute FE trial into U^e; stationarity $\partial\Pi^e/\partial\boldsymbol{u}^e = \boldsymbol{0}$	$\partial W^e_{\text{ext}}/\partial\boldsymbol{u}^e$ (Section 3.11.3)

The choice of method is a matter of convenience and generalizability. Direct equilibrium is intuitive for one-dimensional elements. PVW and Galerkin FE extend unchanged to two- and three-dimensional elements (Chapter 7); PMPE extends to any conservative system. Numerically, all four paths lead to the same linear system $\boldsymbol{K}\,\boldsymbol{u} = \boldsymbol{f}$, as confirmed by the MATLAB scripts Section A.3 and Section A.2.

3.12 Natural Coordinates and Isoparametric Mapping

The element formulations developed so far were written directly in the physical coordinate $x \in [0, L]$. For higher-order elements, distorted geometries, and two- or three-dimensional continua, it is far more convenient to work in a fixed *natural* (or *parametric*) coordinate $\xi \in [-1, 1]$, the same for every element regardless of its size or position. The mapping between ξ and x is built from the same shape functions used to interpolate the displacement — the defining property of the *isoparametric* concept. We now re-derive the bar element matrices in this framework and recover the familiar results Equation (3.7) and Equation (3.20).

3.12.1 The Isoparametric Mapping

Define the linear coordinate transformation

$$x(\xi) = N_1(\xi)\,x_1 + N_2(\xi)\,x_2 = \frac{1-\xi}{2}\,x_1 + \frac{1+\xi}{2}\,x_2, \qquad \xi \in [-1, 1], \tag{3.46}$$

which sends $\xi = -1 \mapsto x = x_1$ and $\xi = +1 \mapsto x = x_2$. The shape functions in natural coordinates are

$$N_1(\xi) = \frac{1-\xi}{2}, \qquad N_2(\xi) = \frac{1+\xi}{2}, \tag{3.47}$$

and they satisfy $N_i(\xi_j) = \delta_{ij}$ at the parent nodes $\xi_1 = -1$, $\xi_2 = +1$, together with the partition of unity $N_1(\xi) + N_2(\xi) = 1$.

The displacement is interpolated with the *same* two functions,

$$u(\xi) = N_1(\xi)\,u_1 + N_2(\xi)\,u_2 = \boldsymbol{N}(\xi)\,\boldsymbol{u}^e, \tag{3.48}$$

which is what makes the formulation isoparametric: a single set of $\{N_i(\xi)\}$ describes both the geometry and the field.

3.12.2 Jacobian of the Mapping

The Jacobian of Equation (3.46) measures how a unit length in the parent domain stretches into the physical domain:

$$J(\xi) = \frac{\mathrm{d}x}{\mathrm{d}\xi} = \sum_{i=1}^{2} \frac{\mathrm{d}N_i}{\mathrm{d}\xi}\,x_i = -\tfrac{1}{2}\,x_1 + \tfrac{1}{2}\,x_2 = \frac{x_2 - x_1}{2} = \frac{L}{2}. \tag{3.49}$$

For the two-node linear bar the Jacobian is *constant* and equal to half the element length. The differential transforms as $\mathrm{d}x = J\,\mathrm{d}\xi = (L/2)\,\mathrm{d}\xi$, and the inverse Jacobian needed for the chain rule is $J^{-1} = 2/L$.

Remark 3.9. $J(\xi)$ being constant is a special feature of the linear 1D element on a uniform mesh. For curved sides (Q4 in Chapter 7) or higher-order shape functions, J varies over the element and the integrals below can no longer be evaluated by inspection.

3.12.3 Strain-Displacement Matrix in Natural Coordinates

The strain $\varepsilon = \mathrm{d}u/\,\mathrm{d}x$ requires a derivative with respect to x, while u is now a function of ξ. The chain rule supplies the missing factor:

$$\varepsilon = \frac{\mathrm{d}u}{\mathrm{d}x} = \frac{\mathrm{d}u}{\mathrm{d}\xi}\frac{\mathrm{d}\xi}{\mathrm{d}x} = J^{-1}\frac{\mathrm{d}u}{\mathrm{d}\xi} = \frac{2}{L}\frac{\mathrm{d}u}{\mathrm{d}\xi}. \tag{3.50}$$

Substituting Equation (3.48), $\mathrm{d}u/\,\mathrm{d}\xi = -\tfrac{1}{2}\,u_1 + \tfrac{1}{2}\,u_2$, gives

$$\varepsilon(\xi) = \boldsymbol{B}(\xi)\,\boldsymbol{u}^e, \qquad \boldsymbol{B}(\xi) = J^{-1}\frac{\mathrm{d}\boldsymbol{N}}{\mathrm{d}\xi} = \frac{2}{L}\,\tfrac{1}{2}\begin{bmatrix}-1 & 1\end{bmatrix} = \frac{1}{L}\begin{bmatrix}-1 & 1\end{bmatrix}, \tag{3.51}$$

which is identical to Equation (3.10). Note that $\boldsymbol{B}$ is again constant in ξ: a linear displacement field produces a constant strain, exactly as in the physical-coordinate derivation.

3.12.4 Element Stiffness Matrix via Natural Coordinates

The element stiffness derived from PVW or Galerkin (Sections 3.8 and 3.9) is

$$\boldsymbol{k}^e = \int_0^L \boldsymbol{B}^\mathsf{T}\,EA\,\boldsymbol{B}\;\mathrm{d}x. \tag{3.52}$$

Changing variables to ξ via Equation (3.46), the limits become ± 1 and $\mathrm{d}x = J\;\mathrm{d}\xi$:

$$\boldsymbol{k}^e = \int_{-1}^{+1} \boldsymbol{B}(\xi)^\mathsf{T}\,EA\,\boldsymbol{B}(\xi)\,J(\xi)\;\mathrm{d}\xi. \tag{3.53}$$

With $\boldsymbol{B}^\mathsf{T}\boldsymbol{B} = (1/L)^2\,[-1\;\;1]^\mathsf{T}[-1\;\;1] = (1/L^2)\,\begin{smallmatrix}1 & -1\\ -1 & 1\end{smallmatrix}$ and $J = L/2$, the integrand is constant:

$$\boldsymbol{B}^\mathsf{T} EA\,\boldsymbol{B}\,J = \frac{EA}{L^2}\begin{bmatrix}1 & -1\\ -1 & 1\end{bmatrix}\frac{L}{2} = \frac{EA}{2L}\begin{bmatrix}1 & -1\\ -1 & 1\end{bmatrix}. \tag{3.54}$$

Carrying out the trivial integration $\int_{-1}^{+1}\,\mathrm{d}\xi = 2$,

$$\boldsymbol{k}^e = \frac{EA}{2L}\begin{bmatrix}1 & -1\\ -1 & 1\end{bmatrix}\cdot 2 = \frac{EA}{L}\begin{bmatrix}1 & -1\\ -1 & 1\end{bmatrix}, \tag{3.55}$$

recovering Equation (3.7) exactly.

3.12.5 Consistent Load Vector via Natural Coordinates

For a uniform distributed axial load $p(x) = p_0$, the variational load vector is $\boldsymbol{f}^e = \int_0^L \boldsymbol{N}^\mathsf{T} p\;\mathrm{d}x$. The same change of variables produces

$$\boldsymbol{f}^e = \int_{-1}^{+1} \boldsymbol{N}(\xi)^\mathsf{T}\,p_0\,J\;\mathrm{d}\xi = p_0\,\frac{L}{2}\int_{-1}^{+1}\begin{bmatrix}(1-\xi)/2\\ (1+\xi)/2\end{bmatrix}\mathrm{d}\xi. \tag{3.56}$$

Each scalar integral equals one,

$$\int_{-1}^{+1}\frac{1-\xi}{2}\;\mathrm{d}\xi = 1, \qquad \int_{-1}^{+1}\frac{1+\xi}{2}\;\mathrm{d}\xi = 1,$$

so

$$\boldsymbol{f}^e = \frac{p_0 L}{2}\begin{bmatrix}1\\ 1\end{bmatrix}, \tag{3.57}$$

in agreement with Equation (3.20): half of the total distributed load is lumped at each node, regardless of which coordinate system is used to evaluate the integral.

3.12.6 Numerical Integration: A Preview of Gauss Quadrature

For the bar both integrands above are simple enough to evaluate by hand, but for higher-order or curved elements they generally cannot be. The standard remedy is *Gauss–Legendre quadrature*, which approximates

$$\int_{-1}^{+1} g(\xi)\, \mathrm{d}\xi \approx \sum_{p=1}^{n_g} w_p\, g(\xi_p), \tag{3.58}$$

with the points ξ_p and weights w_p chosen so that polynomials up to degree $2n_g - 1$ are integrated *exactly*. For the linear bar: the integrand of $\boldsymbol{k}^e$ is degree 0 in ξ, and the integrand of $\boldsymbol{f}^e$ is degree 1, so a single Gauss point ($n_g = 1$) at $\xi_1 = 0$ with weight $w_1 = 2$ is already exact.

For the bilinear quadrilateral introduced in Chapter 7 the Jacobian is no longer constant, $\boldsymbol{B}$ depends on (ξ, η), and a 2×2 Gauss rule is required. The structure of the calculation is nevertheless identical to Equation (3.53): shape functions and their natural derivatives, a Jacobian, an inverse Jacobian for the chain rule, and a sum (or integral) over the parent domain. The 1D derivation in this section is a complete dress rehearsal for that more general case.

3.13 Three-Node Quadratic Rod Element

The two-node bar of Section 3.12 carries a linear displacement field and therefore a constant strain — adequate for a uniform p_0 on a constant cross-section, but unable to represent the parabolic displacement that arises whenever the distributed load varies with x or EA varies along the element. Adding a third node at the interior of the element produces a *quadratic* (parabolic) bar element with a linear strain field, a stiffness matrix whose entries can no longer be evaluated by inspection, and a consistent load vector that distributes a uniform p_0 in the familiar Simpson-rule pattern 1:4:1. This is the simplest non-trivial setting in which to introduce *Gauss–Legendre numerical integration*, the workhorse of every isoparametric finite-element code.

3.13.1 Quadratic Lagrange Shape Functions

Place three nodes at the parent points $\xi_1 = -1$, $\xi_2 = 0$, $\xi_3 = +1$ and demand that each shape function $N_i(\xi)$ be a quadratic polynomial satisfying $N_i(\xi_j) = \delta_{ij}$. The unique solution is the Lagrange triple

$$N_1(\xi) = \tfrac{1}{2}\xi(\xi - 1), \qquad N_2(\xi) = 1 - \xi^2, \qquad N_3(\xi) = \tfrac{1}{2}\xi(\xi + 1), \tag{3.59}$$

which collectively form a partition of unity, $N_1 + N_2 + N_3 \equiv 1$. Figure 3.9 plots the three functions on the parent interval; each rises to unity at its own node and vanishes at the other two.

The natural-coordinate derivatives are linear in ξ:

$$\frac{\mathrm{d}N_1}{\mathrm{d}\xi} = \xi - \tfrac{1}{2}, \qquad \frac{\mathrm{d}N_2}{\mathrm{d}\xi} = -2\xi, \qquad \frac{\mathrm{d}N_3}{\mathrm{d}\xi} = \xi + \tfrac{1}{2}, \tag{3.60}$$

and they sum to zero, as required for a partition of unity.

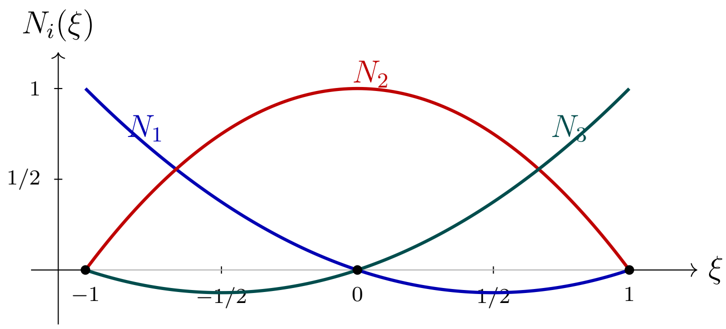

Figure 3.9: Quadratic Lagrange shape functions on the parent domain $\xi \in [-1, 1]$. Each N_i equals 1 at its own node and 0 at the others; the three sum identically to one. The end-node functions N_1 and N_3 dip slightly below zero between $\xi = \pm 1/2$ and the opposite end, reaching a minimum of $-1/8$.

3.13.2 Geometry, Jacobian, and Strain

The same three functions interpolate both the geometry and the displacement (the isoparametric postulate),

$$x(\xi) = \sum_{i=1}^{3} N_i(\xi)\, x_i, \qquad u(\xi) = \sum_{i=1}^{3} N_i(\xi)\, u_i = \boldsymbol{N}(\xi)\, \boldsymbol{u}^e, \tag{3.61}$$

with $\boldsymbol{u}^e = [u_1, u_2, u_3]^\mathsf{T}$. Locating the interior node at the geometric midpoint, $x_1 = 0$, $x_2 = L/2$, $x_3 = L$, the quadratic terms in $x(\xi)$ cancel exactly:

$$x(\xi) = \tfrac{L}{2}(1 - \xi^2) + L\, \tfrac{1}{2}\xi(\xi + 1) = \tfrac{L}{2}(1 + \xi),$$

so the Jacobian is the same constant as for the linear bar,

$$J = \frac{\mathrm{d}x}{\mathrm{d}\xi} = \frac{L}{2}, \qquad J^{-1} = \frac{2}{L}. \tag{3.62}$$

For an interior node displaced from the midpoint, $J(\xi)$ becomes a linear function of ξ and the integrals below acquire an additional ξ-dependence. The 2D analogue, where the Jacobian varies through the element regardless of node placement, is the Q4 of Chapter 7.

The strain matrix follows from the chain rule $\boldsymbol{B}(\xi) = J^{-1}\, \mathrm{d}\boldsymbol{N}/\, \mathrm{d}\xi$:

$$\boldsymbol{B}(\xi) = \frac{1}{L}\begin{bmatrix} 2\xi - 1 & -4\xi & 2\xi + 1 \end{bmatrix}. \tag{3.63}$$

$\boldsymbol{B}$ is now *linear* in ξ, so the strain $\varepsilon(\xi) = \boldsymbol{B}(\xi)\, \boldsymbol{u}^e$ varies along the element. At the three nodal points, $\varepsilon(-1) = (-3u_1 + 4u_2 - u_3)/L$, $\varepsilon(0) = (-u_1 + u_3)/L$ (the midpoint average), and $\varepsilon(+1) = (u_1 - 4u_2 + 3u_3)/L$ — a clear improvement over the constant strain of the linear bar.

3.13.3 Gauss–Legendre Quadrature

The element stiffness now requires an integral that is no longer constant in ξ:

$$\boldsymbol{k}^e = \int_{-1}^{+1} \boldsymbol{B}(\xi)^\mathsf{T}\, EA\, \boldsymbol{B}(\xi)\, J\, \mathrm{d}\xi, \tag{3.64}$$

and each entry of the integrand $\boldsymbol{B}^{\mathsf{T}}\boldsymbol{B}$ is a quadratic polynomial in ξ. Although these particular integrals can be evaluated by hand using $\int_{-1}^{1} \xi^2 \, \mathrm{d}\xi = 2/3$, the standard practice in finite-element codes — and the *only* option once one moves to higher-order, distorted, or two-dimensional elements — is to replace the integral by a discrete sum:

$$\int_{-1}^{+1} g(\xi)\, \mathrm{d}\xi \;\approx\; \sum_{p=1}^{n_g} w_p\, g(\xi_p). \tag{3.65}$$

The *Gauss–Legendre* rule with n_g points selects $\{\xi_p, w_p\}$ so that any polynomial of degree $\leq 2n_g - 1$ is integrated *exactly*. Equivalently, n_g Gauss points integrate exactly an integrand whose entries are polynomials of degree $\leq 2n_g - 1$ — the highest accuracy any rule with n_g function evaluations can achieve. The first three rules are listed in Table 3.2.

Table 3.2: Gauss–Legendre points and weights on $[-1, 1]$, and the highest polynomial degree they integrate exactly.

n_g	Points ξ_p	Weights w_p	Exact for degree $\leq$
1	0	2	1
2	$\pm 1/\sqrt{3} \approx \pm 0.5774$	$1,\ 1$	3
3	$0,\ \pm\sqrt{3/5} \approx \pm 0.7746$	$8/9,\ 5/9,\ 5/9$	5

3.13.4 Element Stiffness Matrix by Two-Point Gauss

Substituting Equation (3.63) and $J = L/2$,

$$\boldsymbol{k}^e = \frac{EA}{2L} \int_{-1}^{+1} \begin{bmatrix} 2\xi - 1 \\ -4\xi \\ 2\xi + 1 \end{bmatrix} \begin{bmatrix} 2\xi - 1 & -4\xi & 2\xi + 1 \end{bmatrix} \mathrm{d}\xi. \tag{3.66}$$

Each entry of the outer product is a polynomial of degree 2 in ξ. The smallest Gauss rule that integrates degree 2 polynomials exactly is $n_g = 2$ (Table 3.2); its points and weights are $\xi_{1,2} = \mp 1/\sqrt{3}$ and $w_1 = w_2 = 1$. Writing $g \equiv 1/\sqrt{3}$ and evaluating the row $\boldsymbol{B}(\xi)\, L$ at the two Gauss points,

$$\boldsymbol{B}(-g)\, L = \begin{bmatrix} -2g - 1 & 4g & -2g + 1 \end{bmatrix}, \quad \boldsymbol{B}(+g)\, L = \begin{bmatrix} 2g - 1 & -4g & 2g + 1 \end{bmatrix}.$$

Forming the symmetric matrix $\boldsymbol{B}(\xi_p)^{\mathsf{T}}\boldsymbol{B}(\xi_p)$ at each point and summing,

$$\boldsymbol{k}^e = \frac{EA}{2L} \sum_{p=1}^{2} w_p \Big[\boldsymbol{B}(\xi_p)\, L\Big]^{\mathsf{T}} \Big[\boldsymbol{B}(\xi_p)\, L\Big], \tag{3.67}$$

where the explicit factor L^2 has been moved outside to keep the inner matrices dimensionless. Each entry contains terms in $g^2 = 1/3$ and linear g-terms that cancel between $\xi = -g$ and $\xi = +g$. For example, the $(1,1)$ entry evaluates to $(-2g - 1)^2 + (2g - 1)^2 = 8g^2 + 2 = 14/3$. Carrying out the analogous algebra for all six independent entries gives

$$\sum_{p=1}^{2} \Big[\boldsymbol{B}(\xi_p)\, L\Big]^{\mathsf{T}} \Big[\boldsymbol{B}(\xi_p)\, L\Big] = \frac{1}{3} \begin{bmatrix} 14 & -16 & 2 \\ -16 & 32 & -16 \\ 2 & -16 & 14 \end{bmatrix}. \tag{3.68}$$

Multiplying through by $EA/(2L)$ and clearing the common factor 2 in every entry yields the standard quadratic-bar stiffness:

Key Result

$$\boldsymbol{k}^e = \frac{EA}{3L}\begin{bmatrix} 7 & -8 & 1 \\ -8 & 16 & -8 \\ 1 & -8 & 7 \end{bmatrix}. \tag{3.69}$$

The 2-point rule is exact here because the integrand is quadratic and $2n_g - 1 = 3 \geq 2$; analytical integration gives exactly the same matrix. The MATLAB script Section A.1 can be modified by replacing the linear shape functions with Equation (3.59), looping over two Gauss points, and accumulating the contributions $w_p\boldsymbol{B}(\xi_p)^{\mathsf{T}}\boldsymbol{D}\boldsymbol{B}(\xi_p)\,J$ to recover Equation (3.69) numerically.

3.13.5 Consistent Load Vector

For a uniform distributed axial load $p(x) = p_0$, the consistent load vector follows from the same change of variables:

$$\boldsymbol{f}^e = \int_{-1}^{+1} \boldsymbol{N}(\xi)^{\mathsf{T}}\, p_0\, J \,\mathrm{d}\xi = p_0\,\frac{L}{2}\int_{-1}^{+1}\begin{bmatrix} \frac{1}{2}\xi(\xi-1) \\ 1-\xi^2 \\ \frac{1}{2}\xi(\xi+1) \end{bmatrix}\mathrm{d}\xi. \tag{3.70}$$

Each scalar integrand is a quadratic in ξ, integrated exactly by the same 2-point Gauss rule (or by direct use of $\int_{-1}^{1}\xi^2\,\mathrm{d}\xi = 2/3$):

$$\int_{-1}^{+1}\tfrac{1}{2}\xi(\xi\mp 1)\,\mathrm{d}\xi = \tfrac{1}{3}, \qquad \int_{-1}^{+1}(1-\xi^2)\,\mathrm{d}\xi = \tfrac{4}{3}.$$

Hence

$$\boldsymbol{f}^e = \frac{p_0 L}{6}\begin{bmatrix} 1 \\ 4 \\ 1 \end{bmatrix}. \tag{3.71}$$

Total force, p_0L, is recovered as $\frac{1}{6}p_0L(1+4+1)$, and the weighting 1:4:1 is exactly Simpson's rule for evaluating the integral of p_0 from 0 to L at three equally-spaced sample points — a numerical coincidence that reflects the underlying quadratic interpolation on both sides.

3.13.6 Why Two Gauss Points: Spurious Modes from Under-Integration

It is tempting to use a single Gauss point for $\boldsymbol{k}^e$ to save work — after all, one evaluation of $\boldsymbol{B}$ is faster than two. The 1-point rule is, however, only exact for polynomials of degree ≤ 1 (Table 3.2), and our integrand is degree 2. The error has a striking consequence.

Evaluating $\boldsymbol{B}$ at the single Gauss point $\xi_1 = 0$, $w_1 = 2$:

$$\boldsymbol{B}(0) = \frac{1}{L}\begin{bmatrix} -1 & 0 & +1 \end{bmatrix}.$$

The mid-node entry vanishes identically. Substituting into Equation (3.65),

$$\boldsymbol{k}^e_{\text{1-pt}} = \frac{EA}{2L} \cdot 2 \cdot \begin{bmatrix} -1 \\ 0 \\ +1 \end{bmatrix} \begin{bmatrix} -1 & 0 & +1 \end{bmatrix} = \frac{EA}{L} \begin{bmatrix} 1 & 0 & -1 \\ 0 & 0 & 0 \\ -1 & 0 & 1 \end{bmatrix}. \tag{3.72}$$

This matrix has rank one. Its null space contains the legitimate rigid-body translation $\boldsymbol{u}^e = [1, 1, 1]^\mathsf{T}$ *and* the spurious vector $\boldsymbol{u}^e = [0, 1, 0]^\mathsf{T}$: the mid-node may translate freely while the end nodes remain fixed, with no internal strain energy generated by the under-integrated stiffness. This is the 1D analogue of the famous *hourglass mode* that plagues under-integrated elements in 2D and 3D.

Remark 3.10. The 1-point rule misses precisely the energy associated with the linearly varying part of $\varepsilon(\xi)$: it sees only the average strain $\varepsilon(0) = (u_3 - u_1)/L$, to which the mid-node displacement contributes nothing. Two Gauss points sample the strain field at $\xi = \pm 1/\sqrt{3}$, where the mid-node *does* affect ε, and the rank deficiency disappears. The simple rule of thumb that emerges is: for an element with shape functions of polynomial order k, use $n_g = k$ Gauss points to integrate $\boldsymbol{k}^e$ exactly on an undistorted geometry — here $k = 2$ requires $n_g = 2$.

3.13.7 Worked Example: Cantilever Bar Under Uniform Load

Example 3.4 (Single quadratic element captures the exact solution)**. Problem.** A uniform bar of length L and axial stiffness EA is fixed at $x = 0$ and free at $x = L$. A constant distributed axial load $p(x) = p_0$ acts along its length (Figure 3.10). Find the nodal displacements, the strain at each node, and the support reaction using a *single* three-node quadratic element. Compare the result with the analytical solution.

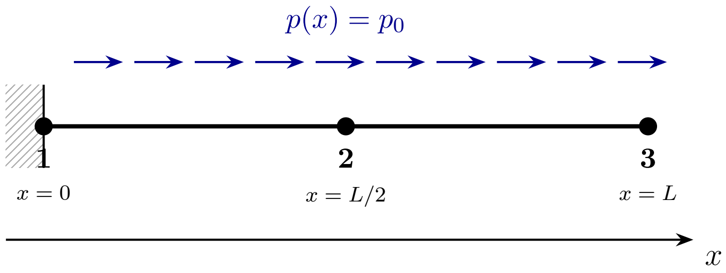

Figure 3.10: Cantilever bar under uniform axial load p_0, modelled by a single 3-node quadratic element. Nodes 1, 2, 3 lie at $x = 0,\ L/2,\ L$.

Step 1: Element matrices. With the interior node placed at the geometric midpoint, the element stiffness and consistent load vector are taken directly from Equations (3.69) and (3.71):

$$\boldsymbol{k}^e = \frac{EA}{3L} \begin{bmatrix} 7 & -8 & 1 \\ -8 & 16 & -8 \\ 1 & -8 & 7 \end{bmatrix}, \qquad \boldsymbol{f}^e = \frac{p_0 L}{6} \begin{bmatrix} 1 \\ 4 \\ 1 \end{bmatrix}.$$

With a single element, $\boldsymbol{K} = \boldsymbol{k}^e$ and $\boldsymbol{f} = \boldsymbol{f}^e$.

Step 2: Boundary conditions and reduced system. The clamp enforces $u_1 = 0$. Eliminating the first row and column of the global system leaves the two free DOFs

$\{u_2, u_3\}$ governed by

$$\frac{EA}{3L}\begin{bmatrix}16 & -8\\ -8 & 7\end{bmatrix}\begin{bmatrix}u_2\\ u_3\end{bmatrix} = \frac{p_0 L}{6}\begin{bmatrix}4\\ 1\end{bmatrix}.$$

Multiplying both sides by $3L/(EA)$ collects the dimensional groups on the right-hand side:

$$\begin{bmatrix}16 & -8\\ -8 & 7\end{bmatrix}\begin{bmatrix}u_2\\ u_3\end{bmatrix} = \frac{p_0 L^2}{2\,EA}\begin{bmatrix}4\\ 1\end{bmatrix}.$$

Step 3: Solve. The 2×2 system has determinant $16\cdot 7 - 64 = 48$ and inverse $\frac{1}{48}\left[\begin{smallmatrix}7 & 8\\ 8 & 16\end{smallmatrix}\right]$. Hence

$$\begin{bmatrix}u_2\\ u_3\end{bmatrix} = \frac{1}{48}\begin{bmatrix}7 & 8\\ 8 & 16\end{bmatrix}\frac{p_0 L^2}{2\,EA}\begin{bmatrix}4\\ 1\end{bmatrix} = \frac{p_0 L^2}{96\,EA}\begin{bmatrix}36\\ 48\end{bmatrix},$$

so

$$u_1 = 0, \qquad u_2 = \frac{3\,p_0 L^2}{8\,EA}, \qquad u_3 = \frac{p_0 L^2}{2\,EA}. \tag{3.73}$$

Step 4: Strain recovery. Evaluating $\boldsymbol{B}(\xi) = (1/L)[2\xi - 1,\ -4\xi,\ 2\xi + 1]$ (Equation (3.63)) at the three nodes ($\xi = -1, 0, +1$) and applying $\varepsilon = \boldsymbol{B}\,\boldsymbol{u}^e$:

$$\begin{aligned}
\boldsymbol{B}(-1) &= \tfrac{1}{L}\begin{bmatrix}-3 & 4 & -1\end{bmatrix}, & \varepsilon_1 &= \frac{4\,u_2 - u_3}{L} = \frac{p_0 L}{EA},\\
\boldsymbol{B}(0) &= \tfrac{1}{L}\begin{bmatrix}-1 & 0 & +1\end{bmatrix}, & \varepsilon_2 &= \frac{u_3}{L} = \frac{p_0 L}{2\,EA},\\
\boldsymbol{B}(+1) &= \tfrac{1}{L}\begin{bmatrix}+1 & -4 & +3\end{bmatrix}, & \varepsilon_3 &= \frac{-4\,u_2 + 3\,u_3}{L} = 0.
\end{aligned}$$

The strain decreases linearly from $p_0 L/EA$ at the clamp to zero at the free end — exactly what the equilibrium equation $-EAu'' = p_0$ predicts.

Step 5: Reaction at the fixed support. The reaction is the residual of the first row of the unmodified element equation,

$$R_1 = \frac{EA}{3L}\Big(7\,u_1 - 8\,u_2 + u_3\Big) - \frac{p_0 L}{6} = \frac{EA}{3L}\left(-\frac{3\,p_0 L^2}{EA} + \frac{p_0 L^2}{2\,EA}\right) - \frac{p_0 L}{6} = -\,p_0 L.$$

The minus sign indicates a leftward pull at the clamp. Global equilibrium is satisfied: the applied total $\int_0^L p_0 \,\mathrm{d}x = p_0 L$ is balanced exactly by $-R_1$.

Step 6: Comparison with the exact solution. Solving the strong form $-EA\,u''(x) = p_0$ with $u(0) = 0$ and $EA\,u'(L) = 0$ gives

$$u_{\text{exact}}(x) = \frac{p_0}{EA}\Big(L\,x - \tfrac{1}{2}x^2\Big), \qquad \varepsilon_{\text{exact}}(x) = \frac{p_0}{EA}(L - x). \tag{3.74}$$

Sampling at the three nodes,

$$u_{\text{exact}}(0) = 0, \qquad u_{\text{exact}}(L/2) = \frac{3\,p_0 L^2}{8\,EA}, \qquad u_{\text{exact}}(L) = \frac{p_0 L^2}{2\,EA},$$

which agrees *exactly* with the FE displacements Equation (3.73). The same is true of the strain field at the nodes. Figure 3.11 overlays the FE displacement field on the analytical solution; the two curves coincide and the three nodal values fall on the analytical line. The MATLAB implementation that reproduces every number in this example is given as Section A.4.

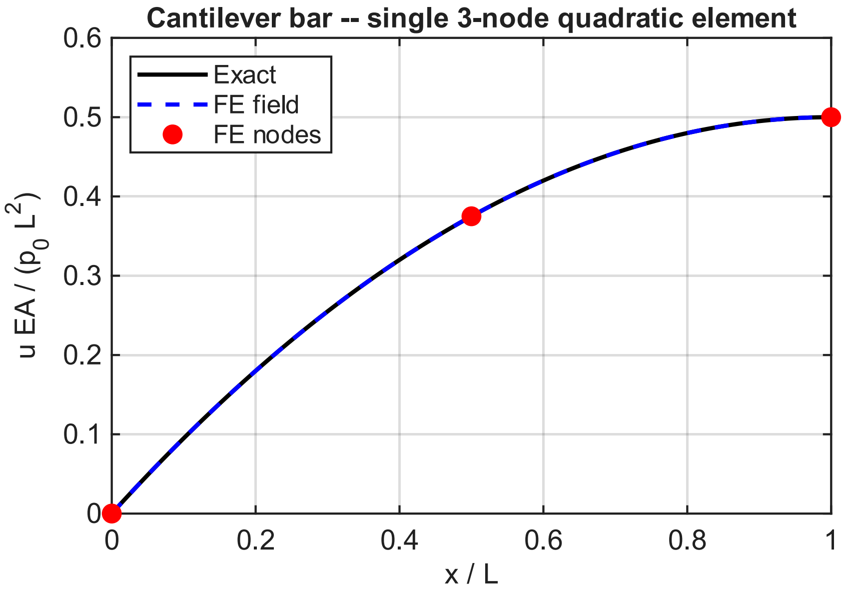

Figure 3.11: Cantilever bar under uniform axial load, single 3-node quadratic element. The FE displacement field (blue dashed) overlies the analytical solution (black solid); red markers show the three FE nodal values. Generated by Section A.4.

Remark 3.11. The single 3-node element recovers the analytical solution *everywhere on the bar*, not only at the nodes, because $u_{\text{exact}}(x)$ is itself a polynomial of degree 2 — exactly the polynomial space spanned by the three element shape functions Equation (3.59). Refining the mesh would change nothing.

For loadings whose exact solution lies outside the element's polynomial space — a linearly varying $p(x) = p_0\, x/L$ produces a cubic $u_{\text{exact}}(x)$, for example — the 3-node element remains *nodally* exact (a structural property of 1D constant-coefficient Galerkin problems) but is only an interior approximation to the true field. Capturing such cases more accurately requires either mesh refinement or higher-order shape functions.

Chapter 4

2D Truss Element

4.1 2D Truss Element

A truss element can carry axial loads only. In a 2D structure, each node has two translational DOFs: u (horizontal) and v (vertical). The element is inclined at angle θ with respect to the global x-axis [2].

4.1.1 Local Stiffness Matrix

In the local coordinate system aligned with the element axis, the stiffness matrix is identical to the 1D bar element Equation (3.7), expanded to account for the transverse DOF (which has zero stiffness):

$$\bar{\boldsymbol{k}}^e = \frac{EA}{L}\begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \tag{4.1}$$

where the DOF ordering is $[\bar{u}_1,\ \bar{v}_1,\ \bar{u}_2,\ \bar{v}_2]$.

4.1.2 Coordinate Transformation

The transformation matrix $\boldsymbol{T}$ rotates from global (x, y) to local $(\bar{x}, \bar{y})$ coordinates. With $c = \cos\theta,\ s = \sin\theta$:

$$\boldsymbol{T} = \begin{bmatrix} c & s & 0 & 0 \\ -s & c & 0 & 0 \\ 0 & 0 & c & s \\ 0 & 0 & -s & c \end{bmatrix} \tag{4.2}$$

4.1.3 Global Element Stiffness

The element stiffness in the global frame is:

$$\boldsymbol{k}^e = \boldsymbol{T}^\top \bar{\boldsymbol{k}}^e \boldsymbol{T} = \frac{EA}{L}\begin{bmatrix} c^2 & cs & -c^2 & -cs \\ cs & s^2 & -cs & -s^2 \\ -c^2 & -cs & c^2 & cs \\ -cs & -s^2 & cs & s^2 \end{bmatrix} \tag{4.3}$$

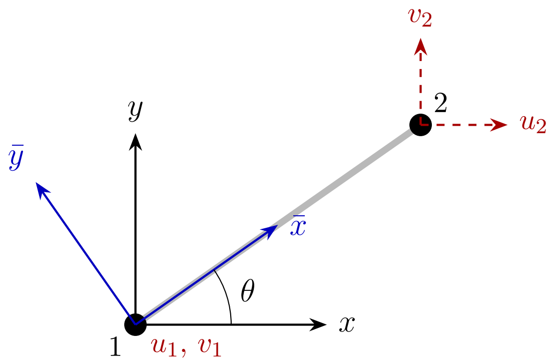

Figure 4.1: 2D truss element inclined at angle θ: global axes (x, y) and local axes $(\bar{x}, \bar{y})$.

> **Key Result**
>
> **Angle convention:** θ is measured counter-clockwise from the positive x-axis to the element axis (node 1 $\to$ node 2).

4.2 Worked Example: Three-Element Planar Truss

Example 4.1 (Three-element planar truss under vertical load)**.** A planar truss consists of three elements and three nodes, as shown in Figure 4.2. All members have cross-sectional area $A = 225$ mm^2 and Young's modulus $E = 210{,}000$ N/mm^2. A vertical load $F = 10$ kN acts downward at the free node 2.

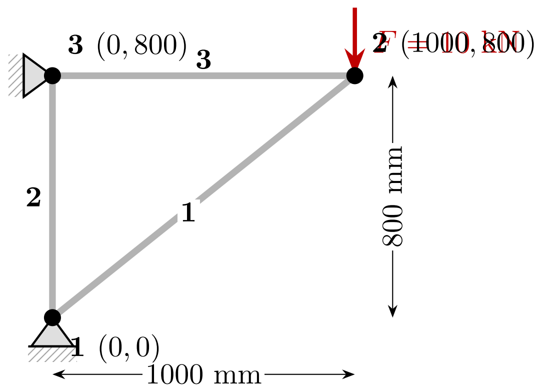

Figure 4.2: Three-node, three-element planar truss. Node coordinates in mm. Pin supports at nodes 1 and 3; 10 kN downward load at node 2.

Step 1: Element properties.

The DOF ordering is $[U_1, V_1, U_2, V_2, U_3, V_3]$ = DOFs $[1, 2, 3, 4, 5, 6]$. Table 4.1 lists the geometric and stiffness properties for each element.

Step 2: Element global stiffness matrices.

Using Equation (4.3) with the values from Table 4.1:

Table 4.1: Element properties for the three-element truss.

Elem	Nodes	L (mm)	θ (°)	c	s	EA/L (N/mm)	DOFs
1	$1 \to 2$	1280.6	38.66	0.7809	0.6247	36 896	1,2,3,4
2	$1 \to 3$	800.0	90.00	0	1	59 063	1,2,5,6
3	$3 \to 2$	1000.0	0.00	1	0	47 250	5,6,3,4

$$\boldsymbol{k}^{(1)} = \begin{bmatrix} 22\,498 & 17\,998 & -22\,498 & -17\,998 \\ 17\,998 & 14\,398 & -17\,998 & -14\,398 \\ -22\,498 & -17\,998 & 22\,498 & 17\,998 \\ -17\,998 & -14\,398 & 17\,998 & 14\,398 \end{bmatrix} \text{ N/mm}, \quad \text{DOFs } [1, 2, 3, 4]$$

$$\boldsymbol{k}^{(2)} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 59\,063 & 0 & -59\,063 \\ 0 & 0 & 0 & 0 \\ 0 & -59\,063 & 0 & 59\,063 \end{bmatrix} \text{ N/mm}, \quad \text{DOFs } [1, 2, 5, 6]$$

$$\boldsymbol{k}^{(3)} = \begin{bmatrix} 47\,250 & 0 & -47\,250 & 0 \\ 0 & 0 & 0 & 0 \\ -47\,250 & 0 & 47\,250 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \text{ N/mm}, \quad \text{DOFs } [5, 6, 3, 4]$$

Step 3: Global assembly.

Assembling the three element matrices into the 6×6 global stiffness:

$$\boldsymbol{K} = \begin{bmatrix} 22\,498 & 17\,998 & -22\,498 & -17\,998 & 0 & 0 \\ 17\,998 & 73\,461 & -17\,998 & -14\,398 & 0 & -59\,063 \\ -22\,498 & -17\,998 & 69\,748 & 17\,998 & -47\,250 & 0 \\ -17\,998 & -14\,398 & 17\,998 & 14\,398 & 0 & 0 \\ 0 & 0 & -47\,250 & 0 & 47\,250 & 0 \\ 0 & -59\,063 & 0 & 0 & 0 & 59\,063 \end{bmatrix} \text{ N/mm}$$

Figure 4.3 shows the assembly schematically. Each element contributes a 4×4 block at its DOFs: element 1 (blue) at DOFs 1–4, element 2 (red) at the scattered DOFs 1, 2, 5, 6, and element 3 (green) at DOFs 3–6. The three 2×2 violet patches in the global matrix mark cells where two elements contribute simultaneously.

The three violet overlaps explain the entries that grow on the diagonal: $K_{22} = 14\,398 + 59\,063 = 73\,461$ from elements 1 and 2, and $K_{33} = 22\,498 + 47\,250 = 69\,748$ from elements 1 and 3.

Step 4: Boundary conditions and reduced system.

Nodes 1 and 3 are both pin supports: $U_1 = V_1 = U_3 = V_3 = 0$ (DOFs 1, 2, 5, 6 fixed). The only free DOFs are U_2 and V_2 (DOFs 3 and 4). The load vector has $F_{y2} = -10\,000$ N at DOF 4 and zero elsewhere. The reduced 2×2 system is:

$$\begin{bmatrix} 69\,748 & 17\,998 \\ 17\,998 & 14\,398 \end{bmatrix} \begin{bmatrix} U_2 \\ V_2 \end{bmatrix} = \begin{bmatrix} 0 \\ -10\,000 \end{bmatrix} \text{ N.} \tag{4.4}$$

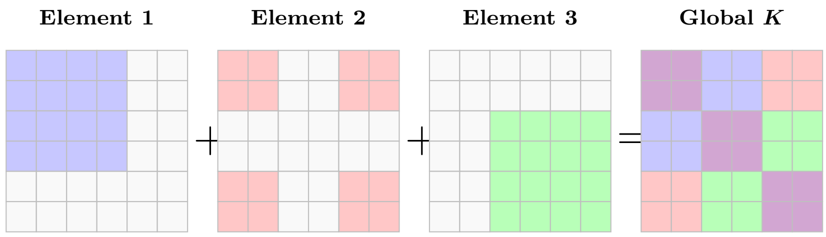

Figure 4.3: Schematic graphical assembly of the global stiffness matrix. Element 1 (blue) fills the 4×4 block at DOFs 1–4; element 2 (red) fills the four corners of a 4×4 pattern at DOFs 1, 2, 5, 6; element 3 (green) fills the 4×4 block at DOFs 3–6. Three 2×2 violet patches mark the overlaps: $e_1 + e_2$ (top-left), $e_1 + e_3$ (centre), and $e_2 + e_3$ (bottom-right). Numerical values appear in the matrix display above.

Step 5: Solution.

Solving Equation (4.4) (see Section A.5 for the MATLAB implementation):

$$U_2 = +0.2646 \text{ mm} \quad \text{(rightward)},$$
$$V_2 = -1.0252 \text{ mm} \quad \text{(downward)}.$$

Step 6: Axial forces.

The axial force in each element is $N^e = (EA/L)\,\delta^e$ where $\delta^e = [-c,\, -s,\, c,\, s]\,\boldsymbol{u}^e$ is the axial elongation:

- **Element 1** (diagonal): $\delta_1 = 0.7809 \times 0.2646 + 0.6247 \times (-1.0252) = -0.4338$ mm,

$$N_1 = 36{,}896 \times (-0.4338) = -\mathbf{16{,}008} \text{ N (compression)}.$$

- **Element 2** (vertical, between two fixed pins): $\delta_2 = s_2(V_3 - V_1) = 1 \times (0 - 0) = 0$,

$$N_2 = 0.$$

  Both endpoints are fixed, so the element undergoes no axial deformation and carries no force under this loading.

- **Element 3** (horizontal): $\delta_3 = U_2 - U_3 = 0.2646$ mm,

$$N_3 = 47{,}250 \times 0.2646 = +\mathbf{12{,}500} \text{ N (tension)}.$$

Step 7: Support reactions.

Multiplying the full 6×6 system $\boldsymbol{K}\,\boldsymbol{u} = \boldsymbol{f}_{\text{ext}}$ and extracting the fixed-DOF rows:

$$R_{1x} = +12{,}500 \text{ N}, \qquad R_{1y} = +10{,}000 \text{ N},$$
$$R_{3x} = -12{,}500 \text{ N}, \qquad R_{3y} = 0 \text{ N}.$$

Equilibrium check:

$$\sum F_x = 12{,}500 + (-12{,}500) = 0 \checkmark, \qquad \sum F_y = 10{,}000 + 0 - 10{,}000 = 0 \checkmark.$$

Chapter 5

Euler-Bernoulli Beam Element

5.1 Direct Formulation of the Beam Element

The Euler-Bernoulli beam element models transverse bending under the assumption that plane cross-sections remain plane and perpendicular to the neutral axis (no shear deformation). Consider a two-node element of length L, bending stiffness EI, with nodes 1 and 2 as shown in Figure 5.1.

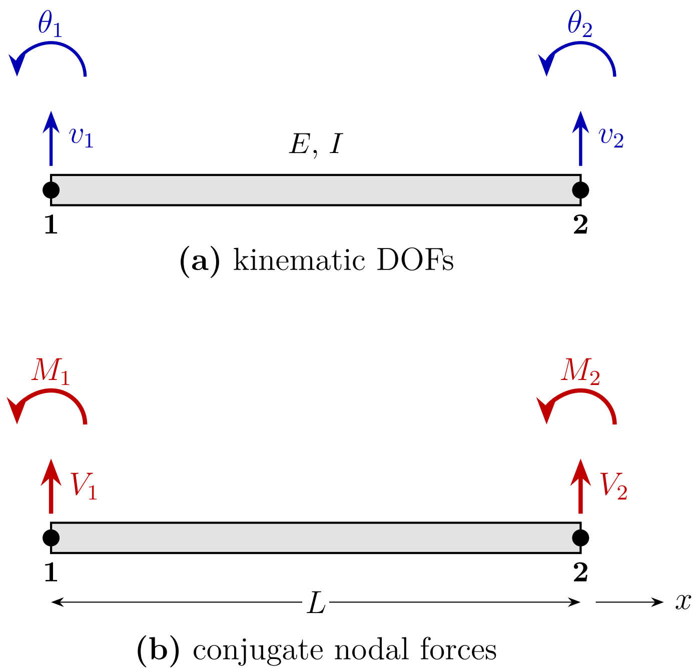

Figure 5.1: Two-node Euler–Bernoulli beam element of length L and bending stiffness EI. (a) The four kinematic DOFs: transverse displacement v_i (upward positive) and rotation θ_i (CCW positive) at each node. (b) Their conjugate nodal forces: shear V_i acts in the same upward sense as v_i, and bending moment M_i in the same CCW sense as θ_i.

5.1.1 Element Kinematics

The Euler-Bernoulli hypothesis states that the transverse displacement $v(x)$ completely describes the element state. The curvature and bending moment are:

$$\kappa(x) = \frac{\mathrm{d}^2 v}{\mathrm{d}x^2}, \qquad M(x) = EI\,\kappa(x) = EI\,\frac{\mathrm{d}^2 v}{\mathrm{d}x^2}, \tag{5.1}$$

and the shear force is $V(x) = \mathrm{d}M/\mathrm{d}x = EI\,v'''(x)$. The governing strong-form equation for a distributed transverse load $q(x)$ (positive upward) is:

$$\frac{\mathrm{d}^2}{\mathrm{d}x^2}\left(EI\,\frac{\mathrm{d}^2 v}{\mathrm{d}x^2}\right) = q(x), \qquad 0 < x < L. \tag{5.2}$$

5.1.2 Stiffness Matrix from the Unit Displacement Method

The element has four DOFs: $\boldsymbol{u}^e = [v_1, \theta_1, v_2, \theta_2]^\mathsf{T}$, where $v_i = v(x_i)$ and $\theta_i = v'(x_i)$. The j-th column of $\boldsymbol{k}_b^e$ gives the nodal forces produced when DOF j is set to unity and all others are zero, while the element is in a state of pure bending (no distributed load, $q = 0$).

For unit displacement $v_1 = 1$ ($\theta_1 = v_2 = \theta_2 = 0$), the displacement field is the unique cubic satisfying these four boundary conditions:

$$v(x) = 1 - 3\xi^2 + 2\xi^3, \qquad \xi = x/L.$$

The moment and shear distributions follow from $M = EIv''$ and $V = EIv'''$, and the four nodal force components are evaluated at $x = 0$ and $x = L$. Repeating this procedure for each of the four unit DOFs and assembling the results gives the 4×4 element stiffness:

$$\boldsymbol{k}_b^e = \frac{EI}{L^3}\begin{bmatrix} 12 & 6L & -12 & 6L \\ 6L & 4L^2 & -6L & 2L^2 \\ -12 & -6L & 12 & -6L \\ 6L & 2L^2 & -6L & 4L^2 \end{bmatrix} \tag{5.3}$$

Key Result

$$\boldsymbol{k}_b^e = \frac{EI}{L^3}\begin{bmatrix} 12 & 6L & -12 & 6L \\ 6L & 4L^2 & -6L & 2L^2 \\ -12 & -6L & 12 & -6L \\ 6L & 2L^2 & -6L & 4L^2 \end{bmatrix}$$

5.1.3 Unit Displacement Interpretation

The columns of $\boldsymbol{k}_b^e$ carry physical meaning analogous to the bar. Setting $v_1 = 1$ with $\theta_1 = v_2 = \theta_2 = 0$ requires forces $V_1 = 12EI/L^3$ (upward at node 1), $M_1 = 6EI/L^2$ (CCW), $V_2 = -12EI/L^3$ (downward at node 2), $M_2 = 6EI/L^2$ (CCW) — exactly the first column of $\boldsymbol{k}_b^e$.

Remark 5.1. The beam stiffness matrix is symmetric (Maxwell's reciprocal theorem) and singular: a uniform translation $v_1 = v_2 = c$, $\theta_1 = \theta_2 = 0$ or a rigid rotation produce no curvature and hence no forces. The matrix has a two-dimensional null space corresponding to the two rigid-body modes of the unconstrained beam element.

5.2 Displacement Field Approximation

5.2.1 Hermite Shape Functions

A unique cubic polynomial is determined by four conditions: values and slopes at both endpoints. Setting $v(0) = 1$, $v'(0) = 0$, $v(L) = 0$, $v'(L) = 0$ gives H_1; similarly for H_2, H_3, H_4. With $\xi = x/L$:

$$H_1(\xi) = 1 - 3\xi^2 + 2\xi^3, \qquad H_2(\xi) = L\Big(\xi - 2\xi^2 + \xi^3\Big),$$
$$H_3(\xi) = 3\xi^2 - 2\xi^3, \qquad H_4(\xi) = L\Big(-\xi^2 + \xi^3\Big). \tag{5.4}$$

The displacement field within the element is:

$$v(x) = H_1 v_1 + H_2 \theta_1 + H_3 v_2 + H_4 \theta_2 = \boldsymbol{H}(x)\, \boldsymbol{u}^e. \tag{5.5}$$

Remark 5.2. The Hermite functions satisfy the Kronecker delta property at the nodes ($H_i(x_j) = \delta_{ij}$ for displacement DOFs, analogously for rotations) and ensure C^1 continuity: both v and $v' = \theta$ are continuous across element boundaries. This is essential for the Euler-Bernoulli beam, whose bending energy requires one derivative of v to be square-integrable.

5.2.2 Curvature-Displacement Matrix

Differentiating Equation (5.5) twice,

$$\kappa(x) = \frac{\mathrm{d}^2 v}{\mathrm{d}x^2} = \boldsymbol{B}_b(x)\, \boldsymbol{u}^e, \qquad \boldsymbol{B}_b = \frac{1}{L^2}\left[\frac{12x}{L} - 6,\quad 6\frac{x}{L} - 4,\quad 6 - \frac{12x}{L},\quad 6\frac{x}{L} - 2\right]. \tag{5.6}$$

The bending moment follows as $M(x) = EI\, \boldsymbol{B}_b\, \boldsymbol{u}^e$.

5.2.3 Enrichment by a Particular Solution

The Hermite cubic Equation (5.5) satisfies the homogeneous beam equation ($q = 0$) exactly but represents only a linear moment distribution. Under a distributed load $q(x) \neq 0$, the exact deflection field includes higher polynomial terms that the cubic cannot capture.

Analogous to the bar element enrichment, augment the field with a particular solution $v_p(x)$:

$$v(x) = \boldsymbol{H}(x)\, \boldsymbol{u}^e + v_p(x), \tag{5.7}$$

where $v_p(x)$ satisfies the inhomogeneous equation with zero endpoint displacements and slopes:

$$EI\, v_p'''' = q(x), \qquad v_p(0) = v_p'(0) = v_p(L) = v_p'(L) = 0. \tag{5.8}$$

For uniform upward load $q(x) = q_0$, integrating four times and applying the four boundary conditions gives:

$$v_p(x) = \frac{q_0}{24EI}\, x^2 (L - x)^2. \tag{5.9}$$

One can verify that $EIv_p'''' = q_0$ and all four endpoint conditions are satisfied. The enriched field is nodally exact: the FEM solution at the nodes is unaffected (since v_p vanishes there), but the interior fields correctly represent the quartic deflection under uniform load.

Remark 5.3. For downward load of intensity $\bar{q} > 0$ (upward-positive convention: $q_0 = -\bar{q}$), the particular solution becomes $v_p = -\bar{q}\,x^2(L-x)^2/(24EI) < 0$ (downward deflection).

5.3 Element Load Vector

For a distributed transverse load $q(x)$ (upward positive), the consistent element load vector is

$$\boldsymbol{f}^e = \int_0^L \boldsymbol{H}(x)^\mathsf{T} q(x)\,\mathrm{d}x. \tag{5.10}$$

Evaluating for uniform $q(x) = q_0$:

$$\begin{aligned} f_1^e &= \int_0^L H_1\,q_0\,\mathrm{d}x = \frac{q_0 L}{2}, & f_2^e &= \int_0^L H_2\,q_0\,\mathrm{d}x = \frac{q_0 L^2}{12}, \\ f_3^e &= \int_0^L H_3\,q_0\,\mathrm{d}x = \frac{q_0 L}{2}, & f_4^e &= \int_0^L H_4\,q_0\,\mathrm{d}x = -\frac{q_0 L^2}{12}, \end{aligned} \tag{5.11}$$

so that

$$\boldsymbol{f}^e = \frac{q_0 L}{12}\begin{bmatrix} 6 \\ L \\ 6 \\ -L \end{bmatrix}. \tag{5.12}$$

For downward load $\bar{q} > 0$, set $q_0 = -\bar{q}$ to obtain equivalent downward nodal forces $\mp\bar{q}L/2$ and moments.

5.4 Global Assembly

A beam mesh with n nodes has $2n$ DOFs: $[v_1, \theta_1, v_2, \theta_2, \ldots, v_n, \theta_n]$. The global stiffness matrix is assembled exactly as for bar elements: each element's 4×4 $\boldsymbol{k}_b^e$ is added into the global matrix at the four DOFs corresponding to nodes i and $j = i + 1$. For element e connecting nodes i and $i + 1$, the DOF mapping is

$$\text{local DOFs } [1, 2, 3, 4] \;\longleftrightarrow\; \text{global DOFs } [2i-1,\ 2i,\ 2(i+1)-1,\ 2(i+1)]. \tag{5.13}$$

Overlapping entries (shared between adjacent elements) are added, yielding the banded symmetric global stiffness $\boldsymbol{K}$.

5.5 Boundary Conditions

Essential (kinematic) boundary conditions prescribe specific DOF values:

Simply supported / roller: $v_i = 0$, θ_i free. No moment resistance; the rotation adjusts freely to equilibrium.

Clamped (fixed): $v_i = 0$ and $\theta_i = 0$. Both displacement and rotation are constrained; full moment resistance.

Free end: no essential BC imposed. Shear and moment at the free end are natural BCs, automatically satisfied by the variational formulation.

Essential BCs are applied by striking the rows and columns corresponding to prescribed DOFs from the global system, or equivalently by the penalty method: add a large stiffness $\alpha \gg \|\boldsymbol{K}\|$ to K_{ii} and set $F_i = \alpha\, \bar{u}_i$ for prescribed value $\bar{u}_i$.

5.6 Worked Example: Simply Supported Beam

Example 5.1 (Two-element simply supported beam)**.** A beam with total length $L = 2$ m, bending stiffness $EI = 10^4$ N m^2, simply supported at both ends (nodes 1 and 3), with a central point load $P = 1000$ N downward at node 2. Discretize with two equal elements of length $L_e = 1$ m.

DOF ordering: $[v_1, \theta_1, v_2, \theta_2, v_3, \theta_3]$ = DOFs $[1, 2, 3, 4, 5, 6]$.

Step 1: Element stiffness matrix.

Both elements are identical ($L_e = 1$ m). With $K_0 = EI/L_e^3 = 10^4$ N/m:

$$\boldsymbol{k}_b^e = K_0 \begin{bmatrix} 12 & 6 & -12 & 6 \\ 6 & 4 & -6 & 2 \\ -12 & -6 & 12 & -6 \\ 6 & 2 & -6 & 4 \end{bmatrix} \begin{bmatrix} \text{N/m} \\ \text{N} \\ \text{N/m} \\ \text{N} \end{bmatrix} \text{ (and N m for moment rows/cols).}$$

Step 2: Global assembly.

Element 1 occupies DOFs $[1, 2, 3, 4]$; element 2 occupies $[3, 4, 5, 6]$. The graphical assembly is shown in Figure 5.2; the resulting 6×6 global stiffness, divided by K_0, is

$$\frac{\boldsymbol{K}}{K_0} = \begin{bmatrix} 12 & 6 & -12 & 6 & 0 & 0 \\ 6 & 4 & -6 & 2 & 0 & 0 \\ -12 & -6 & 24 & 0 & -12 & 6 \\ 6 & 2 & 0 & 8 & -6 & 2 \\ 0 & 0 & -12 & -6 & 12 & -6 \\ 0 & 0 & 6 & 2 & -6 & 4 \end{bmatrix}.$$

The 2×2 overlap at DOFs 3–4 receives contributions from both elements, giving $K_{33} = 12 + 12 = 24$, $K_{34} = -6 + 6 = 0$, and $K_{44} = 4 + 4 = 8$ (in units of K_0).

$$\underbrace{\begin{bmatrix} 12 & 6 & -12 & 6 & 0 & 0 \\ 6 & 4 & -6 & 2 & 0 & 0 \\ -12 & -6 & 12 & -6 & 0 & 0 \\ 6 & 2 & -6 & 4 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}}_{\textbf{Element 1}} + \underbrace{\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 12 & 6 & -12 & 6 \\ 0 & 0 & 6 & 4 & -6 & 2 \\ 0 & 0 & -12 & -6 & 12 & -6 \\ 0 & 0 & 6 & 2 & -6 & 4 \end{bmatrix}}_{\textbf{Element 2}} = \underbrace{\begin{bmatrix} 12 & 6 & -12 & 6 & 0 & 0 \\ 6 & 4 & -6 & 2 & 0 & 0 \\ -12 & -6 & 24 & 0 & -12 & 6 \\ 6 & 2 & 0 & 8 & -6 & 2 \\ 0 & 0 & -12 & -6 & 12 & -6 \\ 0 & 0 & 6 & 2 & -6 & 4 \end{bmatrix}}_{\textbf{Global } \boldsymbol{K}/K_0}$$

Figure 5.2: Graphical assembly of the global stiffness matrix (in units of $K_0 = EI/L_e^3$). Element 1 (blue) fills DOFs 1–4, element 2 (red) fills DOFs 3–6. The 2×2 violet block at DOFs 3–4 receives contributions from both elements; its entries $24, 0, 0, 8$ are the sums of the two overlapping 4×4 blocks.

Step 3: Load vector and boundary conditions.

Applied loads: $F_3 = -P = -1000$ N (downward at DOF 3), all other loads zero. Essential BCs: $v_1 = 0$ (DOF 1) and $v_3 = 0$ (DOF 5). Free DOFs: $[\theta_1, v_2, \theta_2, \theta_3]$ = DOFs $[2, 3, 4, 6]$.

The reduced 4×4 system:

$$K_0 \begin{bmatrix} 4 & -6 & 2 & 0 \\ -6 & 24 & 0 & 6 \\ 2 & 0 & 8 & 2 \\ 0 & 6 & 2 & 4 \end{bmatrix} \begin{bmatrix} \theta_1 \\ v_2 \\ \theta_2 \\ \theta_3 \end{bmatrix} = \begin{bmatrix} 0 \\ -P \\ 0 \\ 0 \end{bmatrix}. \tag{5.14}$$

Step 4: Solution (MATLAB-verified, Section A.6).

$$\begin{aligned} \theta_1 &= -0.025 \text{ rad}, & v_2 &= -16.67 \text{ mm} \quad \text{(downward)}, \\ \theta_2 &= 0 \text{ rad}, & \theta_3 &= +0.025 \text{ rad}. \end{aligned}$$

By symmetry, $\theta_2 = 0$ exactly. The FEM result is nodally exact because the loading is a concentrated nodal force:

$$|v_2| = \frac{PL^3}{48EI} = \frac{1000 \times 2^3}{48 \times 10^4} = 0.01667 \text{ m} = \mathbf{16.67} \text{ mm}.$$

Step 5: Reactions and moment diagram.

The support reactions follow from $\boldsymbol{K}\,\boldsymbol{u} = \boldsymbol{F}_{\text{ext}}$:

$$R_1 = R_3 = 500 \text{ N} \quad (\textstyle\sum F = 500 + 500 - 1000 = 0 \checkmark).$$

The maximum bending moment occurs at midspan (node 2):

$$M_{\max} = \frac{PL}{4} = \frac{1000 \times 2}{4} = 500 \text{ N m} \quad \text{(sagging)}.$$

The MATLAB script Section A.6 plots the deflection and moment diagram (Figure 5.3).

5.7 Worked Example: Three-Span Continuous Beam

Example 5.2 (Three-element continuous beam with mixed loading)**.** A continuous beam with $E = 210{,}000$ N/mm^2 and I-section moment of inertia $I = 5{,}481{,}140$ mm^4 ($z_{\max} = 70$ mm) is discretized with three elements, as shown in Figure 5.4.

Problem data:

$$\begin{gathered} EI = 210{,}000 \times 5{,}481{,}140 = 1.1510 \times 10^{12} \text{ N mm}^2 \\ L_1 = 3000 \text{ mm}, \quad L_2 = 2000 \text{ mm}, \quad L_3 = 1000 \text{ mm} \end{gathered}$$

DOF map: $[w_1, \beta_1, w_2, \beta_2, w_3, \beta_3, w_4, \beta_4]$ = DOFs $[1 \ldots 8]$.
Fixed DOFs: $\{1, 3, 7\}$ ($w_1 = w_2 = w_4 = 0$); free DOFs: $\{2, 4, 5, 6, 8\}$.

Step 1: Element stiffness constants.

$$K_1 = \frac{EI}{L_1^3} = 42.63 \,\frac{\text{N}}{\text{mm}}, \quad K_2 = \frac{EI}{L_2^3} = 143.88 \,\frac{\text{N}}{\text{mm}}, \quad K_3 = \frac{EI}{L_3^3} = 1151.0 \,\frac{\text{N}}{\text{mm}}.$$

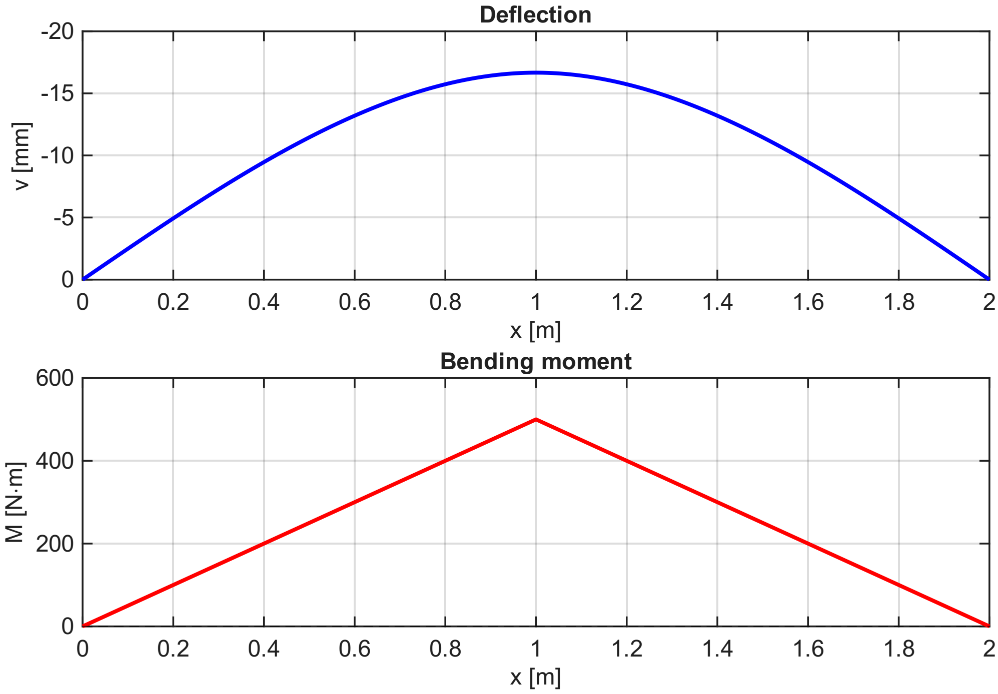

Figure 5.3: Two-element simply supported beam: deflection (top, downward displacement plotted positive) and bending moment diagram (bottom). Maximum deflection $|v_2| = 16.67$ mm; $M_{\max} = 500$ N m.

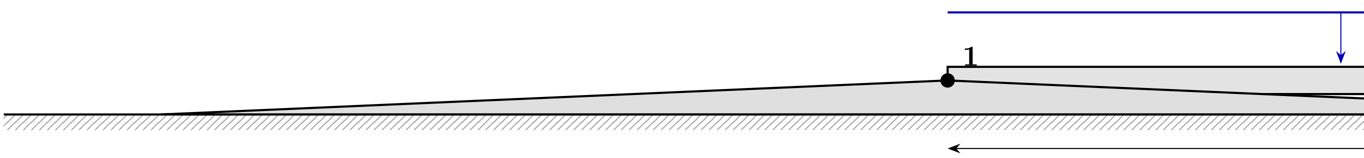

Figure 5.4: Three-element continuous beam. Distributed load $q = 10$ N/mm on element 1; concentrated load $P = 30$ kN at node 3. Pin support at node 1; rollers at nodes 2 and 4.

Each element stiffness matrix uses the pattern of Equation (5.3) with its own K_i and L_i. The dominant off-diagonal entries for element 1 are:

$$k_{12}^{(1)} = K_1{\cdot}6L_1 = 42.63{\times}18{,}000 = 767{,}340\ \text{N}, \qquad k_{22}^{(1)} = K_1{\cdot}4L_1^2 = 42.63{\times}36{\times}10^6 = 1.535{\times}10^9\ \text{N mm}.$$

Step 2: Load vectors.

Distributed load $\bar{q} = 10$ N/mm (downward) on element 1 only ($q_0 = -\bar{q} = -10$ N/mm):

$$\boldsymbol{f}_q^{(1)} = \frac{q_0 L_1}{12}\begin{bmatrix}6\\L_1\\6\\-L_1\end{bmatrix} = \frac{-10\times 3000}{12}\begin{bmatrix}6\\3000\\6\\-3000\end{bmatrix} = \begin{bmatrix}-15{,}000\\-7.5\times10^6\\-15{,}000\\+7.5\times10^6\end{bmatrix}\begin{bmatrix}\text{N}\\\text{N mm}\\\text{N}\\\text{N mm}\end{bmatrix} \quad \text{(DOFs 1,2,3,4)}.$$

Elements 2 and 3 carry no distributed load. The concentrated load $P = 30{,}000$ N (downward) acts at DOF 5:

$$F_5 = -30{,}000\ \text{N}.$$

The global load vector (non-zero entries only):

$$\begin{aligned} F_1 &= -15{,}000\ \text{N}, & F_2 &= -7.5\times10^6\ \text{N mm},\\ F_3 &= -15{,}000\ \text{N}, & F_4 &= +7.5\times10^6\ \text{N mm},\\ F_5 &= -30{,}000\ \text{N}. \end{aligned}$$

Step 3: Reduced system.

Retaining only free DOFs $\{2,4,5,6,8\}$, the 5×5 reduced stiffness matrix is (all entries $\times10^{-9}$ N^{-1}mm^{-1} for rotation DOFs, or as listed by MATLAB):

$$10^9\times\boldsymbol{K}_{ff} = \begin{bmatrix}1534.7 & 767.4 & 0 & 0 & 0\\ 767.4 & 3836.8 & -1.7 & 1151.0 & 0\\ 0 & -1.7 & 0.0052 & 5.2 & 6.9\\ 0 & 1151.0 & 5.2 & 6906.2 & 2302.1\\ 0 & 0 & 6.9 & 2302.1 & 4604.2\end{bmatrix}\ \text{N/mm or N or N mm} \tag{5.15}$$

with load vector $\boldsymbol{F}_{ff} = [-7.5\times10^6,\ 7.5\times10^6,\ -30{,}000,\ 0,\ 0]^T$.

Step 4: Solution (MATLAB-verified, Section A.7).

$$\begin{aligned} \beta_1 &= -4.434\times10^{-3}\ \text{rad}, & \beta_2 &= -9.050\times10^{-4}\ \text{rad},\\ w_3 &= -6.838\ \text{mm}\quad\text{(downward)}, & \beta_3 &= +2.232\times10^{-3}\ \text{rad},\\ \beta_4 &= +9.140\times10^{-3}\ \text{rad}. \end{aligned}$$

All other DOFs are zero (prescribed supports).

Step 5: Support reactions.

$$R_1 = +10{,}900\ \text{N}, \qquad R_2 = +33{,}190\ \text{N}, \qquad R_4 = +15{,}900\ \text{N}.$$

Equilibrium check: $\sum F = 10{,}900 + 33{,}190 + 15{,}900 - 10\times3000 - 30{,}000 = 60{,}000 - 60{,}000 = 0$ ✓.

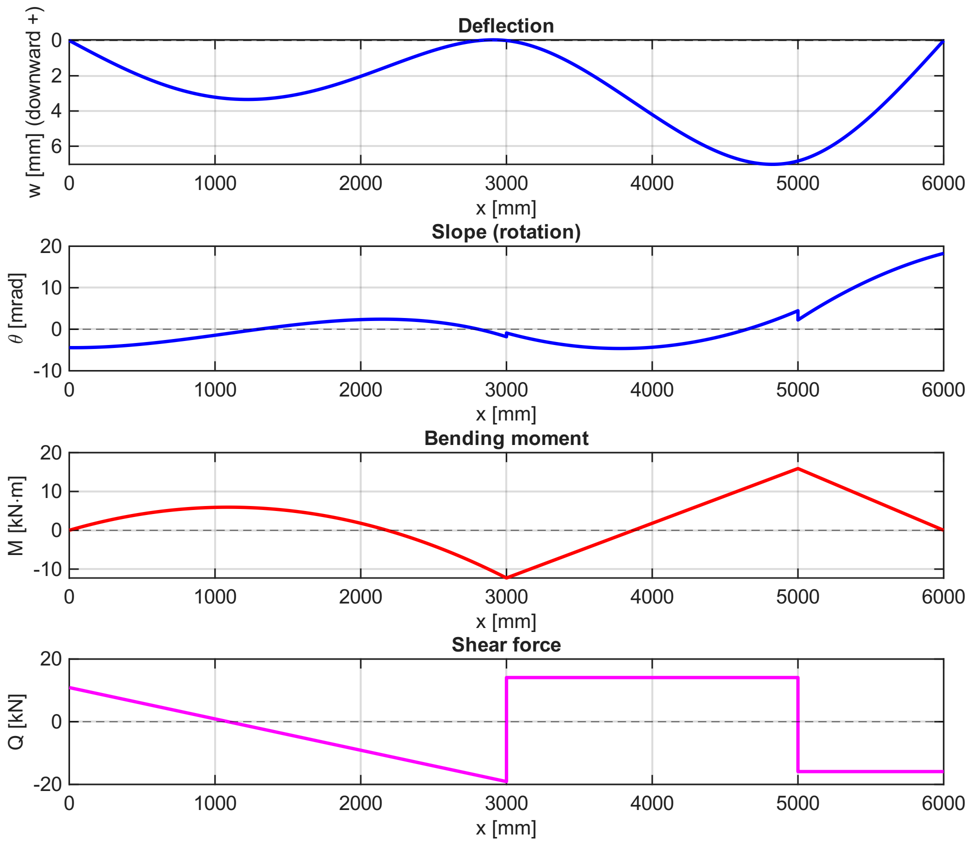

Figure 5.5: Continuous beam internal fields. Top to bottom: deflection (downward positive), slope, bending moment (sagging positive), shear force. Key values: $w_3 = 6.84$ mm (downward), $M_{\max} = +15.90$ kN m (sagging at node 3), $M_{\min} = -12.29$ kN m (hogging at node 2).

Step 6: Internal fields and cross-section stress.

Using Hermite interpolation with particular solution enrichment on element 1, the displacement, slope, bending moment ($M = EIv''$, sagging positive), and shear force ($V = EIv'''$) are computed element-by-element and plotted in Figure 5.5.

The maximum bending moment magnitude is $|M|_{\max} = 15.90 \times 10^6$ N mm at $x = 5000$ mm. The maximum normal bending stress:

$$\sigma_{\max} = \frac{|M|_{\max}\, z_{\max}}{I_y} = \frac{15.90 \times 10^6 \times 70}{5{,}481{,}140} = \mathbf{203.1} \text{ MPa}. \tag{5.16}$$

5.8 Principle of Virtual Work for the Beam

5.8.1 Statement of the Principle

Key Result

Principle of Virtual Work for the Beam: For a beam in static equilibrium, and for any *kinematically admissible* virtual displacement field $\delta v(x)$ (continuous with continuous slope, and $\delta v = \delta v' = 0$ wherever v or v' is prescribed):

$$\delta W_{\text{int}} = \delta W_{\text{ext}}, \tag{5.17}$$

with

$$\delta W_{\text{int}} = \int_0^L M(x)\,\delta\kappa(x)\;\mathrm{d}x = \int_0^L EI\,\frac{\mathrm{d}^2 v}{\mathrm{d}x^2}\,\frac{\mathrm{d}^2 \delta v}{\mathrm{d}x^2}\;\mathrm{d}x,$$

$$\delta W_{\text{ext}} = \int_0^L q(x)\,\delta v\;\mathrm{d}x + \sum_i F_i\,\delta v(x_i) + \sum_j M_j\,\delta v'(x_j).$$

5.8.2 Derivation from Equilibrium

Starting from the strong form Equation (5.2), multiply by an arbitrary virtual displacement $\delta v(x)$ and integrate:

$$\int_0^L \left(EI\,v''\right)''\delta v\;\mathrm{d}x = \int_0^L q\,\delta v\;\mathrm{d}x.$$

Integration by parts once (on the left):

$$\int_0^L \left(EI\,v''\right)'\delta v\;\mathrm{d}x = \Big[(EI\,v'')'\;\delta v\Big]_0^L - \int_0^L \left(EI\,v''\right)'\,\delta v'\;\mathrm{d}x. \tag{5.18}$$

Integration by parts a second time (on the remaining integral):

$$\int_0^L \left(EI\,v''\right)'\,\delta v'\;\mathrm{d}x = \Big[EI\,v''\,\delta v'\Big]_0^L - \int_0^L EI\,v''\,\delta v''\;\mathrm{d}x. \tag{5.19}$$

Combining and rearranging:

$$\int_0^L EI\,v''\,\delta v''\;\mathrm{d}x = \int_0^L q\,\delta v\;\mathrm{d}x + \Big[(EI\,v'')'\,\delta v\Big]_0^L - \Big[EI\,v''\,\delta v'\Big]_0^L. \tag{5.20}$$

The boundary terms yield the natural boundary conditions: $V = (EIv'')'$ (shear) at a transverse-free end contributes to δW_{ext} via $V\,\delta v$, and $M = EIv''$ (moment) at a rotationally-free end via $M\,\delta v'$. At nodes where displacements or rotations are prescribed, admissibility forces $\delta v = 0$ or $\delta v' = 0$, respectively, so those boundary terms vanish. This recovers Equation (5.17) exactly.

Remark 5.4. Equation Equation (5.20) is the *weak form* of the beam equation. The strong form Equation (5.2) requires v to be four-times differentiable; the weak form requires only that v'' and $\delta v''$ are square-integrable. Hermite cubic shape functions satisfy this requirement with their C^1 continuity, making them the natural choice for beam finite elements.

5.8.3 Recovery of the Beam Stiffness Matrix

Substitute the Hermite interpolation $v = \boldsymbol{H}\,\boldsymbol{u}^e$, $\delta v = \boldsymbol{H}\,\delta\boldsymbol{u}^e$, $v'' = \boldsymbol{B}_b\,\boldsymbol{u}^e$ into the left-hand side of Equation (5.20):

$$\delta\boldsymbol{u}^{e^\mathsf{T}}\left[\int_0^L EI\,\boldsymbol{B}_b^\mathsf{T}\boldsymbol{B}_b\,\mathrm{d}x\right]\boldsymbol{u}^e = \delta\boldsymbol{u}^{e^\mathsf{T}}\,\boldsymbol{k}_b^e\,\boldsymbol{u}^e,$$

giving:

$$\boldsymbol{k}_b^e = \int_0^L EI\,\boldsymbol{B}_b^\mathsf{T}\boldsymbol{B}_b\,\mathrm{d}x = \frac{EI}{L^3}\begin{bmatrix} 12 & 6L & -12 & 6L \\ 6L & 4L^2 & -6L & 2L^2 \\ -12 & -6L & 12 & -6L \\ 6L & 2L^2 & -6L & 4L^2 \end{bmatrix}, \tag{5.21}$$

identical to Equation (5.3). The right-hand side of Equation (5.20) produces $\boldsymbol{f}^e = \int_0^L \boldsymbol{H}^\mathsf{T} q\,\mathrm{d}x$, recovering Equation (5.12).

5.9 Galerkin Finite Element Formulation of the Beam

Remark 5.5. Chapter 2 applied the Galerkin method with a single global polynomial to the axially loaded bar. The formulation here uses the Hermite shape functions H_1–H_4 as both trial and test functions on each element — the standard Bubnov–Galerkin approach for the beam.

5.9.1 Weighted Residual Statement

Substitute the Hermite trial $v^h = \boldsymbol{H}\,\boldsymbol{u}^e$ into the strong form residual:

$$R(x) = \left(EI\,v^{h''}\right)'' - q(x). \tag{5.22}$$

Since the Hermite cubics have zero fourth derivative, $v^{h''''} = 0$, so $R = -q(x)$ — the same situation as for the bar. The Galerkin statement requires the residual to be orthogonal to each test (shape) function:

$$\int_0^L H_i(x)\,R(x)\,\mathrm{d}x = 0, \qquad i = 1, 2, 3, 4. \tag{5.23}$$

5.9.2 Integration by Parts to the Weak Form

Substituting $R = (EIv^{h\prime\prime})'' - q$ and integrating by parts twice (identical to Equations (5.18) and (5.19)):

$$\int_0^L EI\, H_i''\, v^{h\prime\prime}\, \mathrm{d}x = \int_0^L H_i\, q\, \mathrm{d}x + F_i^{\text{natural}}, \tag{5.24}$$

where F_i^{natural} collects the shear/moment boundary terms. Writing all four rows simultaneously:

$$\underbrace{\left[\int_0^L EI\, \boldsymbol{B}_b^{\mathsf{T}} \boldsymbol{B}_b\, \mathrm{d}x\right]}_{\boldsymbol{k}_b^e} \boldsymbol{u}^e = \underbrace{\int_0^L \boldsymbol{H}^{\mathsf{T}} q\, \mathrm{d}x}_{\boldsymbol{f}^e}. \tag{5.25}$$

Key Result

The Galerkin FE formulation yields the same element matrices as direct derivation:

$$\boldsymbol{k}_b^e = \int_0^L EI\, \boldsymbol{B}_b^{\mathsf{T}} \boldsymbol{B}_b\, \mathrm{d}x = \frac{EI}{L^3} \begin{bmatrix} 12 & 6L & -12 & 6L \\ 6L & 4L^2 & -6L & 2L^2 \\ -12 & -6L & 12 & -6L \\ 6L & 2L^2 & -6L & 4L^2 \end{bmatrix}, \quad \boldsymbol{f}^e = \int_0^L \boldsymbol{H}^{\mathsf{T}} q\, \mathrm{d}x = \frac{q_0 L}{12} \begin{bmatrix} 6 \\ L \\ 6 \\ -L \end{bmatrix}.$$

5.10 Equivalence of Galerkin and Virtual Work Formulations

The Galerkin weak form Equation (5.24) and the PVW equation Equation (5.20) are identical statements. Setting $w_i = \delta v$:

$$\begin{aligned} \text{PVW:} \quad & \int_0^L EI\, \frac{\mathrm{d}^2 v}{\mathrm{d}x^2} \frac{\mathrm{d}^2 \delta v}{\mathrm{d}x^2}\, \mathrm{d}x = \int_0^L q\, \delta v\, \mathrm{d}x + \sum_i F_i\, \delta v(x_i), \\ \text{Galerkin:} \quad & \int_0^L EI\, v^{h\prime\prime}\, w_i''\, \mathrm{d}x = \int_0^L q\, w_i\, \mathrm{d}x + F_i^{\text{natural}}. \end{aligned} \tag{5.26}$$

Under $w_i = \delta v$ the two are identical: both enforce the same integral condition over every kinematically admissible test field.

Remark 5.6. An important distinction for beams: the test functions $w_i = H_i$ and the virtual displacements δv must be C^1 (continuous with continuous slope), because the weak form involves w_i''. This is more demanding than for bar elements (C^0 suffices) and is why Hermite cubics — rather than linear Lagrange polynomials — are required. The Galerkin method's mandate to choose test functions from the same space as the trial directly prescribes this continuity requirement.

5.11 Principle of Minimum Potential Energy

5.11.1 Total Potential Energy of the Beam

Definition 5.1. The **bending strain energy** of a linearly elastic beam is

$$U = \frac{1}{2}\int_0^L EI\left(\frac{\mathrm{d}^2 v}{\mathrm{d}x^2}\right)^2 \mathrm{d}x. \tag{5.27}$$

The **total potential energy** is

$$\Pi[v] = U - W_{\text{ext}}, \qquad W_{\text{ext}} = \int_0^L q\,v\;\mathrm{d}x + \sum_i F_i\,v(x_i) + \sum_j M_j\,v'(x_j). \tag{5.28}$$

5.11.2 Stationarity Condition and the Weak Form

Key Result

Principle of Minimum Potential Energy (PMPE): Of all kinematically admissible displacement fields, the one satisfying equilibrium makes Π stationary:

$$\delta\Pi = 0 \tag{5.29}$$

for every admissible variation δv. For a stable elastic equilibrium, Π is a global minimum.

Taking the first variation of Equation (5.28):

$$\delta\Pi = \int_0^L EI\,v''\,\delta v''\;\mathrm{d}x - \int_0^L q\,\delta v\;\mathrm{d}x - \sum_i F_i\,\delta v(x_i) - \sum_j M_j\,\delta v'(x_j) = 0. \tag{5.30}$$

Identifying $\delta W_{\text{int}} = \int EIv''\delta v''\;\mathrm{d}x$ and δW_{ext} from the remaining terms, this is precisely the PVW statement Equation (5.17):

$$\delta\Pi = 0 \iff \delta W_{\text{int}} = \delta W_{\text{ext}}. \tag{5.31}$$

Remark 5.7. PMPE is a special case of PVW for conservative (elastic) systems. Both principles are equivalent for the Euler-Bernoulli beam with fixed applied loads. PMPE provides the variational foundation that guarantees PVW holds; PVW generalizes to non-conservative systems (e.g. aerodynamic follower loads) where no potential energy functional exists.

5.11.3 Element Stiffness Matrix and Load Vector from PMPE

Substitute $v = \boldsymbol{H}\,\boldsymbol{u}^e$, $v'' = \boldsymbol{B}_b\,\boldsymbol{u}^e$ into the element bending energy:

$$U^e = \frac{1}{2}\int_0^L EI\,\boldsymbol{u}^{e^\mathsf{T}}\boldsymbol{B}_b^\mathsf{T}\boldsymbol{B}_b\,\boldsymbol{u}^e\;\mathrm{d}x = \frac{1}{2}\,\boldsymbol{u}^{e^\mathsf{T}}\underbrace{\left[\int_0^L EI\,\boldsymbol{B}_b^\mathsf{T}\boldsymbol{B}_b\;\mathrm{d}x\right]}_{\boldsymbol{k}_b^e}\boldsymbol{u}^e.$$

The external work term gives $W_{\text{ext}}^e = \boldsymbol{u}^{e^\mathsf{T}}\boldsymbol{f}^e$, so the element potential energy is the quadratic:

$$\Pi^e = \frac{1}{2}\,\boldsymbol{u}^{e^\mathsf{T}}\boldsymbol{k}_b^e\,\boldsymbol{u}^e - \boldsymbol{u}^{e^\mathsf{T}}\boldsymbol{f}^e. \tag{5.32}$$

Stationarity $\partial\Pi^e/\partial\boldsymbol{u}^e = \boldsymbol{0}$ gives $\boldsymbol{k}_b^e\,\boldsymbol{u}^e = \boldsymbol{f}^e$, recovering Equation (5.3) and Equation (5.12).

Key Result

Minimizing the element potential energy yields the stiffness matrix:

$$\boldsymbol{k}_b^e = \int_0^L EI\,\boldsymbol{B}_b^\mathsf{T}\boldsymbol{B}_b\,\mathrm{d}x = \frac{EI}{L^3}\begin{bmatrix} 12 & 6L & -12 & 6L \\ 6L & 4L^2 & -6L & 2L^2 \\ -12 & -6L & 12 & -6L \\ 6L & 2L^2 & -6L & 4L^2 \end{bmatrix}.$$

The stiffness matrix is the Hessian of the bending strain energy with respect to the nodal DOFs.

5.11.4 Summary: Four Equivalent Paths to the Beam Element Matrices

Table 5.1: Four derivation paths for the Euler-Bernoulli beam element matrices.

Method	**Starting point**	**Path to $\boldsymbol{k}_b^e$**	**Load vector $\boldsymbol{f}^e$**
Direct Formulation	$M = EIv''$; unit displacements	Four unit BVPs; nodal forces from M and V	Particular solution enrichment (Section 5.2)
PVW	$\delta W_{\mathrm{int}} = \delta W_{\mathrm{ext}}$ (Equation (5.17))	Sub. $v = \boldsymbol{H}\boldsymbol{u}^e$, $\delta v = \boldsymbol{H}\delta\boldsymbol{u}^e$ into weak form	RHS of weak form (Section 5.8)
Galerkin FE	$\int H_i R\,\mathrm{d}x = 0$, $w_i = H_i$ (Equation (5.23))	IBP twice + $\boldsymbol{B}_b$-matrix (Section 5.9)	RHS of weak form (Equation (5.25))
PMPE	$\delta\Pi = 0$, $\Pi = U - W_{\mathrm{ext}}$ (Equation (5.28))	Sub. FE trial into U^e; stationarity	$\partial W_{\mathrm{ext}}^e/\partial\boldsymbol{u}^e$ (Section 5.11.3)

All four paths produce Equation (5.3) and Equation (5.12). PVW and Galerkin extend unchanged to frame, plate, and shell elements; PMPE extends to any conservative elastic system. The four-path convergence for the bar element is documented in Table 3.1.

5.12 Natural Coordinates for the Beam Element

For generality and numerical integration, map $x \in [0, L]$ to $\xi \in [0, 1]$ via $x = \xi L$, $\mathrm{d}x = L\,\mathrm{d}\xi$. The Hermite functions in natural coordinates are:

$$\tilde{H}_1(\xi) = 1 - 3\xi^2 + 2\xi^3, \qquad \tilde{H}_2(\xi) = \xi - 2\xi^2 + \xi^3,$$
$$\tilde{H}_3(\xi) = 3\xi^2 - 2\xi^3, \qquad \tilde{H}_4(\xi) = -\xi^2 + \xi^3, \tag{5.33}$$

so that $H_1(x) = \tilde{H}_1(\xi)$ and $H_2(x) = L\tilde{H}_2(\xi)$, etc.

The second derivative transforms via the chain rule: $\mathrm{d}^2v/\,\mathrm{d}x^2 = (1/L^2)\,\mathrm{d}^2v/\,\mathrm{d}\xi^2$. The stiffness integral in natural coordinates becomes:

$$\boldsymbol{k}_b^e = \int_0^1 EI\,\tilde{\boldsymbol{B}}_b(\xi)^\mathsf{T}\tilde{\boldsymbol{B}}_b(\xi)\,\frac{\mathrm{d}\xi}{L^3}, \tag{5.34}$$

where $\tilde{\boldsymbol{B}}_b = (1/L^2)[d^2\tilde{H}_i/d\xi^2]$ collects the curvature shape derivatives. Since $\tilde{H}_i$ are cubic polynomials, $\tilde{\boldsymbol{B}}_b^\mathsf{T}\tilde{\boldsymbol{B}}_b$ is a polynomial of degree 2 in ξ; a 2-point Gauss–Legendre quadrature (mapped to $[0,1]$) integrates it exactly. This is the foundation for numerical integration of beam and frame elements with variable $EI(x)$.

Chapter 6

2D Frame Element

6.1 Frame Element

A *frame element* (beam-column) combines the axial stiffness of a bar element with the bending stiffness of an Euler-Bernoulli beam element into a single structural member. Unlike truss elements, which transmit axial force only through pinned joints, frame elements transmit *axial force, shear force, and bending moment* at rigid joints. Frames are the appropriate model whenever joint rigidity induces bending in the members: portal frames, multi-storey building frames, and any structure in which loads are not applied along member axes.

In the *local coordinate system* $(\bar{x}, \bar{y})$ aligned with the element axis, each node carries three DOFs — axial displacement $\bar{u}$, transverse displacement $\bar{v}$, and rotation θ (positive counter-clockwise). The six-component local DOF vector and its conjugate nodal force vector are:

$$\bar{\boldsymbol{u}}^e = [\bar{u}_1,\ \bar{v}_1,\ \theta_1,\ \bar{u}_2,\ \bar{v}_2,\ \theta_2]^\mathsf{T}, \qquad \bar{\boldsymbol{f}}^e = [\bar{N}_1,\ \bar{V}_1,\ \bar{M}_1,\ \bar{N}_2,\ \bar{V}_2,\ \bar{M}_2]^\mathsf{T}, \tag{6.1}$$

where $\bar{N}_i$ is the axial force, $\bar{V}_i$ the shear force, and $\bar{M}_i$ the bending moment at node i (see Figure 6.1).

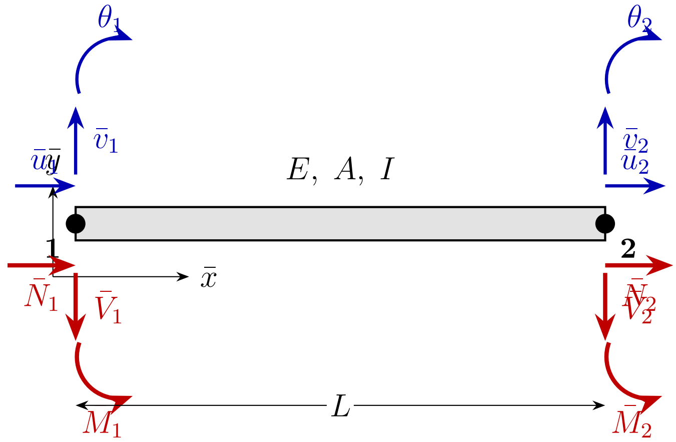

Figure 6.1: Two-node frame element in the local frame $(\bar{x}, \bar{y})$. Displacements $\bar{u}_i, \bar{v}_i$ (blue) and rotations θ_i (blue, CCW positive) are the kinematic DOFs. Conjugate nodal forces $\bar{N}_i$ (axial), $\bar{V}_i$ (shear), and moments $\bar{M}_i$ (red) are the static conjugates.

6.2 Local Stiffness Matrix

In the local frame, axial deformation (bar action) and transverse bending (beam action) are decoupled: an axial displacement $\bar{u}$ produces no bending forces, and a transverse displacement $\bar{v}$ produces no axial force. The local stiffness matrix $\bar{\boldsymbol{k}}^e$ is therefore block-diagonal, assembling the bar stiffness (Equation (3.7)) and the Euler-Bernoulli beam stiffness (Equation (5.3)) into a single 6×6 matrix:

$$\bar{\boldsymbol{k}}^e = \begin{bmatrix} \frac{EA}{L} & 0 & 0 & -\frac{EA}{L} & 0 & 0 \\ 0 & \frac{12EI}{L^3} & \frac{6EI}{L^2} & 0 & -\frac{12EI}{L^3} & \frac{6EI}{L^2} \\ 0 & \frac{6EI}{L^2} & \frac{4EI}{L} & 0 & -\frac{6EI}{L^2} & \frac{2EI}{L} \\ -\frac{EA}{L} & 0 & 0 & \frac{EA}{L} & 0 & 0 \\ 0 & -\frac{12EI}{L^3} & -\frac{6EI}{L^2} & 0 & \frac{12EI}{L^3} & -\frac{6EI}{L^2} \\ 0 & \frac{6EI}{L^2} & \frac{2EI}{L} & 0 & -\frac{6EI}{L^2} & \frac{4EI}{L} \end{bmatrix} \tag{6.2}$$

Key Result

The frame local stiffness is block-diagonal: rows and columns $\{1, 4\}$ carry the bar block (EA/L); rows and columns $\{2, 3, 5, 6\}$ carry the beam block (EI/L^3 scaling). Axial and bending deformations are *uncoupled* in the local frame.

Remark 6.1. The unconstrained frame element has a three-dimensional null space: rigid translation in $\bar{x}$, rigid translation in $\bar{y}$, and rigid rotation. At least three linearly independent kinematic constraints are needed to make the assembled stiffness non-singular.

6.3 Element Load Vector

When distributed loads act along the element, the consistent nodal load vector $\bar{\boldsymbol{f}}^e$ is formed by superposing the bar and beam contributions derived in Chapters 3 and 5. For a uniform *axial* distributed load p_0 (positive in $+\bar{x}$) and a uniform *transverse* distributed load q_0 (positive in $+\bar{y}$):

$$\bar{\boldsymbol{f}}^e = \underbrace{\frac{p_0 L}{2}\begin{bmatrix}1\\0\\0\\1\\0\\0\end{bmatrix}}_{\text{axial } (\textit{Equation } (3.20))} + \underbrace{\frac{q_0 L}{12}\begin{bmatrix}0\\6\\L\\0\\6\\-L\end{bmatrix}}_{\text{transverse } (\textit{Equation } (5.12))} \tag{6.3}$$

The two contributions act on separate DOF subsets: the axial terms go to DOFs 1 and 4 ($\bar{u}$ displacements); the transverse terms go to DOFs 2, 3, 5, and 6 ($\bar{v}$ displacements and θ rotations). In the worked example of Section 6.5, the beam (element 2) carries a downward transverse load, so $q_0 = -q < 0$ and $p_0 = 0$; the two columns carry no distributed load.

6.4 Coordinate Transformation

Frame members are generally inclined in the global (x, y) plane. The 6×6 transformation matrix $\boldsymbol{T}_6$ maps from the global frame to the local frame. With $c = \cos\alpha$ and $s = \sin\alpha$, where α is the element inclination angle:

$$\boldsymbol{T}_6 = \begin{bmatrix} c & s & 0 & 0 & 0 & 0 \\ -s & c & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & c & s & 0 \\ 0 & 0 & 0 & -s & c & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \tag{6.4}$$

$\boldsymbol{T}_6$ consists of two identical 3×3 rotation blocks (one per node), each rotating the displacement pair (u_i, v_i) from global to local while leaving the rotation θ_i unchanged. The structure mirrors the truss transformation (Equation (4.2)), extended by the scalar rotation DOF.

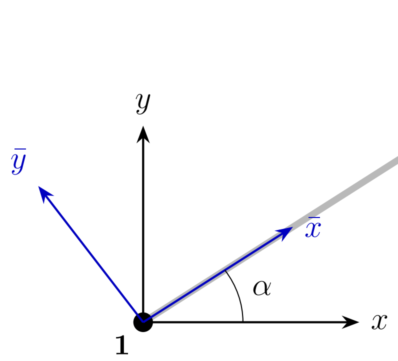

Figure 6.2: Frame element inclined at angle α (CCW from the positive x-axis). Global axes (x, y); local axes $(\bar{x}, \bar{y})$ in blue. $\boldsymbol{T}_6$ rotates the global DOFs (u_i, v_i, θ_i) to local $(\bar{u}_i, \bar{v}_i, \theta_i)$.

Key Result

Angle convention: α is measured counter-clockwise from the positive x-axis to the element axis directed from node 1 to node 2.

Horizontal beam, rightward:	$\alpha = 0°$,	$c = 1,\ s = 0$
Vertical column, upward:	$\alpha = 90°$,	$c = 0,\ s = 1$
Vertical column, downward:	$\alpha = -90°$,	$c = 0,\ s = -1$

The element stiffness and load vector in global coordinates are:

$$\boldsymbol{k}^e = \boldsymbol{T}_6^\top \bar{\boldsymbol{k}}^e \boldsymbol{T}_6, \qquad \boldsymbol{f}^e = \boldsymbol{T}_6^\top \bar{\boldsymbol{f}}^e. \tag{6.5}$$

For vertical columns ($c = 0$, $|s| = 1$), $\boldsymbol{T}_6$ exchanges the axial and transverse DOF blocks: the column's axial stiffness EA/L appears in the vertical (v-v) entries of the global stiffness, and its bending stiffness $12EI/L^3$ appears in the horizontal (u-u) entries.

Remark 6.2. Global assembly, boundary condition enforcement, and solution proceed identically to the truss (Chapter 4) and beam (Chapter 5) chapters. The PVW and PMPE derivations for the frame element are direct extensions of Sections 5.8 and 5.11: replace the 4×4 beam integrals with the 6×6 frame counterparts after applying $\boldsymbol{T}_6$.

6.5 Worked Example: Portal Frame under Combined Loading

Example 6.1 (Portal frame — lateral force and distributed beam load)**.** A symmetric portal frame has column height $H = 3000$ mm and beam span $B = 4000$ mm. All members share the same cross-section: $E = 210{,}000$ N/mm^2, $A = 6{,}000$ mm^2, $I = 10{,}000{,}000$ mm^4. Both column bases are clamped (fixed). The frame carries:

- a rightward horizontal force $F = 10$ kN at the top of the left column (node 2), and
- a uniform downward load $q = 5$ N/mm on the beam (element 2).

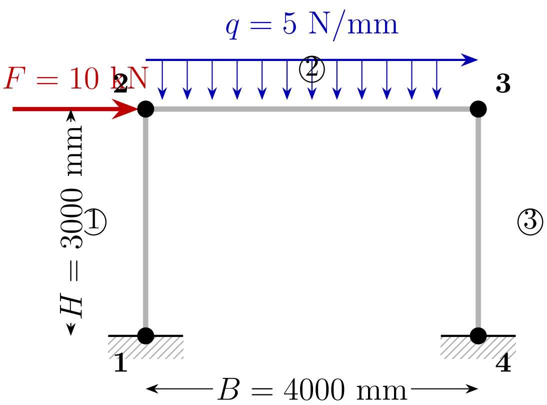

Figure 6.3: Portal frame: four nodes, three elements, clamped bases at nodes 1 and 4. Combined loading: horizontal force F at node 2 and uniform vertical load q on the beam.

DOF ordering: $[u_1, v_1, \theta_1,\ u_2, v_2, \theta_2,\ u_3, v_3, \theta_3,\ u_4, v_4, \theta_4] =$ DOFs $[1 \ldots 12]$.
Clamped DOFs: $\{1, 2, 3, 10, 11, 12\} = \mathbf{0}$. Free DOFs: $\{4, 5, 6, 7, 8, 9\}$.

Step 1: Element data.

Table 6.1: Element data for the portal frame example.

Elem	Nodes	L (mm)	α (°)	c	s	Global DOFs
1	$1 \to 2$	3000	90	0	+1	1,2,3,4,5,6
2	$2 \to 3$	4000	0	1	0	4,5,6,7,8,9
3	$3 \to 4$	3000	−90	0	−1	7,8,9,10,11,12

Step 2: Local stiffness constants.

Let $a = EA/L$, $b = 12EI/L^3$, $d = 6EI/L^2$, $e_4 = 4EI/L$, $e_2 = 2EI/L$.

For the columns (elements 1 and 3, $L_c = 3000$ mm):

$$a_c = \frac{210{,}000 \times 6{,}000}{3{,}000} = 420{,}000\ \frac{\text{N}}{\text{mm}}, \qquad b_c = \frac{12 \times 210{,}000 \times 10^7}{3{,}000^3} = 933.3\ \frac{\text{N}}{\text{mm}},$$

$$d_c = \frac{6 \times 210{,}000 \times 10^7}{3{,}000^2} = 1{,}400{,}000\ \text{N}, \quad e_{4c} = \frac{4 \times 210{,}000 \times 10^7}{3{,}000} = 2.800 \times 10^9\ \text{N mm},$$

$$e_{2c} = \tfrac{1}{2}\, e_{4c} = 1.400 \times 10^9\ \text{N mm}.$$

For the beam (element 2, $L_b = 4000$ mm):

$$a_b = \frac{210{,}000 \times 6{,}000}{4{,}000} = 315{,}000 \ \tfrac{\mathrm{N}}{\mathrm{mm}}, \qquad b_b = \frac{12 \times 210{,}000 \times 10^7}{4{,}000^3} = 393.75 \ \tfrac{\mathrm{N}}{\mathrm{mm}},$$

$$d_b = \frac{6 \times 210{,}000 \times 10^7}{4{,}000^2} = 787{,}500 \ \mathrm{N}, \qquad e_{4b} = \frac{4 \times 210{,}000 \times 10^7}{4{,}000} = 2.100 \times 10^9 \ \mathrm{N\,mm},$$

$$e_{2b} = 1.050 \times 10^9 \ \mathrm{N\,mm}.$$

Step 3: Transformation matrices.

For elements 1 and 3, the column geometry gives $c = 0$, $|s| = 1$:

$$\boldsymbol{T}_6^{(1)} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \quad \boldsymbol{T}_6^{(3)} = \begin{bmatrix} 0 & -1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

For the beam ($\alpha = 0°$): $\boldsymbol{T}_6^{(2)} = \boldsymbol{I}_6$, so the beam's local and global stiffness matrices are identical.

Step 4: Global element stiffnesses.

Applying $\boldsymbol{k}^{(e)} = (\boldsymbol{T}_6^{(e)})^\top \bar{\boldsymbol{k}}^{(e)} \boldsymbol{T}_6^{(e)}$ to the columns with $c = 0$, $s = \pm 1$ exchanges the axial and transverse blocks in the global frame: the column's axial stiffness a_c appears in the v-v coupling (vertical DOFs), and the bending stiffness b_c appears in the u-u coupling (horizontal DOFs). For element 1 (DOFs 1,2,3,4,5,6):

$$\boldsymbol{k}^{(1)} = \begin{bmatrix} b_c & 0 & d_c & -b_c & 0 & d_c \\ 0 & a_c & 0 & 0 & -a_c & 0 \\ d_c & 0 & e_{4c} & -d_c & 0 & e_{2c} \\ -b_c & 0 & -d_c & b_c & 0 & -d_c \\ 0 & -a_c & 0 & 0 & a_c & 0 \\ d_c & 0 & e_{2c} & -d_c & 0 & e_{4c} \end{bmatrix}$$

and symmetrically for element 3 (DOFs 7,8,9,10,11,12) with $s = -1$ (sign changes on the d_c off-diagonal terms involving the v–θ coupling). Element 2 is unchanged by the identity transformation.

Step 5: Global assembly.

The 12×12 global stiffness $\boldsymbol{K}$ is assembled by adding each $\boldsymbol{k}^{(e)}$ into the positions given by its DOF list (Table 6.1). Node 2 (DOFs 4,5,6) receives contributions from elements 1 and 2; node 3 (DOFs 7,8,9) receives contributions from elements 2 and 3. The full 12×12 matrix is computed by Listing A.9.

Step 6: Load vector and boundary conditions.

The global load vector has two contributions:

- **Lateral force** $F = 10{,}000$ N (rightward) at DOF 4 (u_2): $F_4 = +10{,}000$ N.

- **Distributed load** $q = 5$ N/mm (downward, $q_0 = -5$ N/mm) on element 2 ($L_b = 4000$ mm, $\boldsymbol{T}_6^{(2)} = \boldsymbol{I}$):

$$\boldsymbol{f}_q^{(2)} = \frac{q_0 L_b}{12}\begin{bmatrix} 0 \\ 6 \\ L_b \\ 0 \\ 6 \\ -L_b \end{bmatrix} = \frac{-5 \times 4000}{12}\begin{bmatrix} 0 \\ 6 \\ 4000 \\ 0 \\ 6 \\ -4000 \end{bmatrix} = \begin{bmatrix} 0 \\ -10{,}000 \\ -6.667 \times 10^6 \\ 0 \\ -10{,}000 \\ +6.667 \times 10^6 \end{bmatrix}\begin{bmatrix} \\ \mathrm{N} \\ \mathrm{N\,mm} \\ \\ \mathrm{N} \\ \mathrm{N\,mm} \end{bmatrix} \qquad \text{(DOFs 4–9)}.$$

Applying the clamped BCs (DOFs $\{1, 2, 3, 10, 11, 12\} = 0$) and retaining only the six free DOFs gives the reduced 6×6 system:

$$\boldsymbol{K}_{ff}\,\boldsymbol{u}_f = \boldsymbol{F}_f, \qquad \boldsymbol{u}_f = [u_2, v_2, \theta_2, u_3, v_3, \theta_3]^{\mathsf{T}},$$

$$\boldsymbol{F}_f = [+10{,}000,\ -10{,}000,\ -6.667 \times 10^6,\ 0,\ -10{,}000,\ +6.667 \times 10^6]^{\mathsf{T}} \text{ (N and N mm)}.$$

Step 7: Solution (MATLAB-verified, Listing A.9).

$$\begin{aligned} u_2 &= +8.296 \text{ mm}, & v_2 &= -0.017 \text{ mm}, & \theta_2 &= -3.687 \times 10^{-3} \text{ rad}, \\ u_3 &= +8.272 \text{ mm}, & v_3 &= -0.031 \text{ mm}, & \theta_3 &= -2.15 \times 10^{-4} \text{ rad}. \end{aligned}$$

The dominant response is the lateral sway ($u_2 \approx u_3 \approx 8.28$ mm) induced by the horizontal force F. The small difference $u_2 - u_3 = 0.024$ mm reflects the beam's axial flexibility. Both joints deflect slightly downward ($v_2, v_3 < 0$) under the gravity load.

Step 8: Support reactions.

$$\begin{aligned} R_{1x} &= -2{,}580.8 \text{ N}, & R_{1y} &= +6{,}932.9 \text{ N}, & M_1 &= +6{,}451{,}977 \text{ N mm}, \\ R_{4x} &= -7{,}419.2 \text{ N}, & R_{4y} &= +13{,}067.1 \text{ N}, & M_4 &= +11{,}279{,}478 \text{ N mm}. \end{aligned}$$

Equilibrium check:

$$\begin{aligned} \textstyle\sum F_x &= -2{,}580.8 + (-7{,}419.2) + 10{,}000 = 0 \checkmark \\ \textstyle\sum F_y &= 6{,}932.9 + 13{,}067.1 - 5 \times 4000 = 0 \checkmark \end{aligned}$$

Step 9: Element internal forces.

Local end forces are recovered via $\bar{\boldsymbol{f}}_{\text{int}}^{(e)} = \bar{\boldsymbol{k}}^{(e)}\left(\boldsymbol{T}_6^{(e)}\boldsymbol{u}^{(e)}\right) - \bar{\boldsymbol{f}}_{\text{load}}^{(e)}$ (see Section 6.6). The values at each end are:

The MATLAB script (Listing A.9) plots the deflected shape and the bending moment, shear force, and axial force diagrams (Figure 6.4).

6.6 Internal Force Recovery

Once the global displacement vector $\boldsymbol{u}$ is known, the internal forces at any point along a frame element are recovered in four steps:

Table 6.2: Element end forces in local coordinates (N and N mm). Tension positive for $\bar{N}$; sagging positive for $\bar{M}$.

Element	$\bar{N}_i$ (N)	$\bar{V}_i$ (N)	$\bar{M}_i$ (N mm)	$\bar{N}_j$ (N)	$\bar{V}_j$ (N)	$\bar{M}_j$ (N mm)
1 – left col	+6,932.9	+2,580.8	+6,451,977	−6,932.9	−2,580.8	+1,290,284
2 – beam	+7,419.2	+6,932.9	−1,290,284	−7,419.2	+13,067.1	−10,978,261
3 – right col	+13,067.1	+7,419.2	+10,978,261	−13,067.1	−7,419.2	+11,279,478

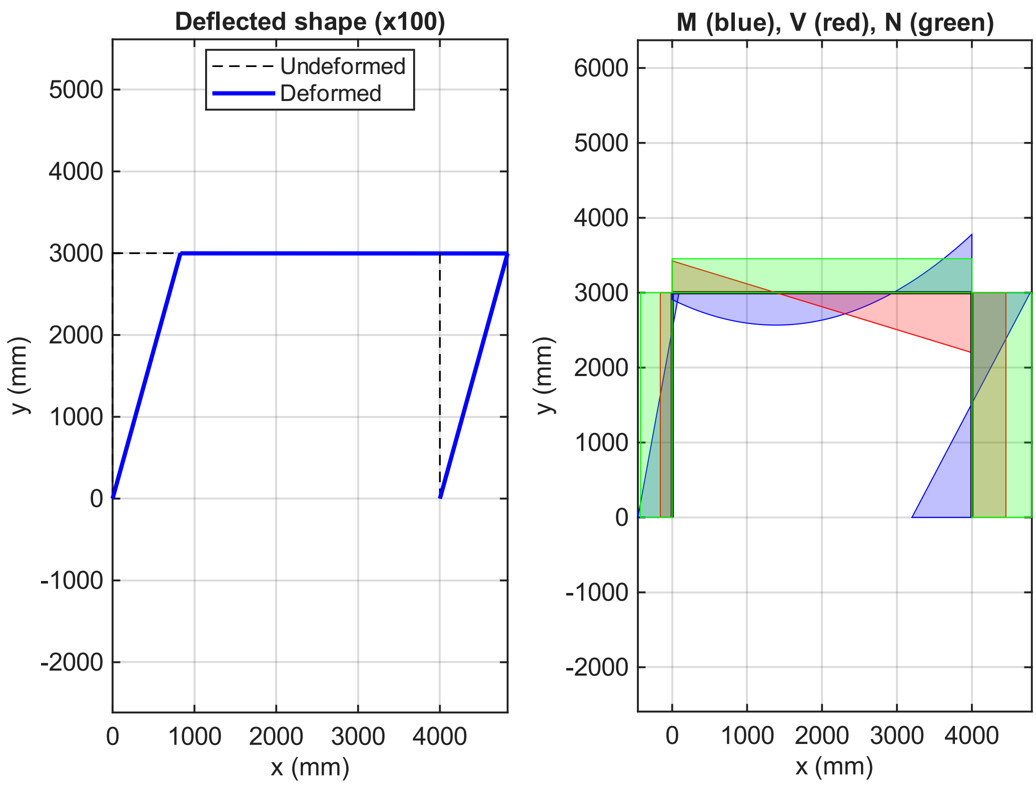

Figure 6.4: Portal frame results: deflected shape (left, displacements magnified ×100) and internal force diagrams (right) for bending moment (blue), shear (red), and axial force (green).

1. **Extract global nodal DOFs.** For element e connecting nodes i and j, assemble $\boldsymbol{u}^e = [u_i,\, v_i,\, \theta_i,\, u_j,\, v_j,\, \theta_j]^{\mathsf{T}}$.

2. **Transform to local coordinates.**

$$\bar{\boldsymbol{u}}^e = \boldsymbol{T}_6\, \boldsymbol{u}^e.$$

3. **Compute local end forces.**

$$\bar{\boldsymbol{f}}^e_{\text{int}} = \bar{\boldsymbol{k}}^e\, \bar{\boldsymbol{u}}^e - \bar{\boldsymbol{f}}^e_{\text{load}}, \tag{6.6}$$

where $\bar{\boldsymbol{f}}^e_{\text{load}}$ is the element load vector from Equation (6.3) (zero for elements carrying no distributed load). The six components are the end axial forces, shear forces, and bending moments in local coordinates.

4. **Distribute along the element.** Between the ends, the internal forces vary with position $\bar{x} \in [0, L]$:

$$N(\bar{x}) = \bar{N}_i + \frac{EA}{L}(\bar{u}_j - \bar{u}_i)\frac{\bar{x}}{L} \quad \text{(linear under axial load; constant if none)}, \tag{6.7}$$

$$V(\bar{x}) = \bar{V}_i + q_0\, \bar{x} \quad \text{(linear under uniform } q_0\text{; constant if } q_0 = 0\text{)}, \tag{6.8}$$

$$M(\bar{x}) = \bar{M}_i - \bar{V}_i\, \bar{x} - \tfrac{1}{2} q_0\, \bar{x}^2 \quad \text{(quadratic under } q_0\text{; linear if } q_0 = 0\text{)}. \tag{6.9}$$

Remark 6.3. For a vertical column ($\alpha = 90°$), the local $\bar{x}$ axis points upward and the local $\bar{y}$ axis points leftward. The local axial force $\bar{N}$ is therefore the vertical internal force (gravity direction) in the column, and the local shear $\bar{V}$ is horizontal. In the portal frame example, the column axial forces carry the vertical gravity load from the beam, while the column shear forces are in equilibrium with the applied lateral force F.

Chapter 7

Two-Dimensional Continuum Elements

7.1 Plane Elasticity

Two-dimensional continuum elements model thin plates or long prismatic bodies [1, 3]. Two assumptions are common:

Plane stress Thin plates loaded in their plane: $\sigma_z = \tau_{xz} = \tau_{yz} = 0$.

Plane strain Long bodies with constrained out-of-plane strain: $\varepsilon_z = \gamma_{xz} = \gamma_{yz} = 0$.

7.1.1 Stress and Strain Vectors

For both cases, the in-plane state is described by:

$$\boldsymbol{\sigma} = \begin{bmatrix} \sigma_x \\ \sigma_y \\ \tau_{xy} \end{bmatrix}, \qquad \boldsymbol{\varepsilon} = \begin{bmatrix} \varepsilon_x \\ \varepsilon_y \\ \gamma_{xy} \end{bmatrix} \tag{7.1}$$

7.1.2 Constitutive Matrix

Plane stress (ν = Poisson's ratio, E = Young's modulus):

$$\boldsymbol{D}_{\mathrm{ps}} = \frac{E}{1-\nu^2} \begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \frac{1-\nu}{2} \end{bmatrix} \tag{7.2}$$

Plane strain:

$$\boldsymbol{D}_{\mathrm{pe}} = \frac{E}{(1+\nu)(1-2\nu)} \begin{bmatrix} 1-\nu & \nu & 0 \\ \nu & 1-\nu & 0 \\ 0 & 0 & \frac{1-2\nu}{2} \end{bmatrix} \tag{7.3}$$

7.2 Constant Strain Triangle (CST)

The CST (or T3) is the simplest 2D element: 3 nodes, 2 DOFs per node (6 total), with linear displacement fields and constant strains throughout.

7.2.1 Shape Functions

For a triangle with nodes (x_1, y_1), (x_2, y_2), (x_3, y_3) and element area A_e, the shape functions are the *area coordinates*:

$$N_i(x, y) = \frac{1}{2A_e}(a_i + b_i x + c_i y), \quad i = 1, 2, 3 \tag{7.4}$$

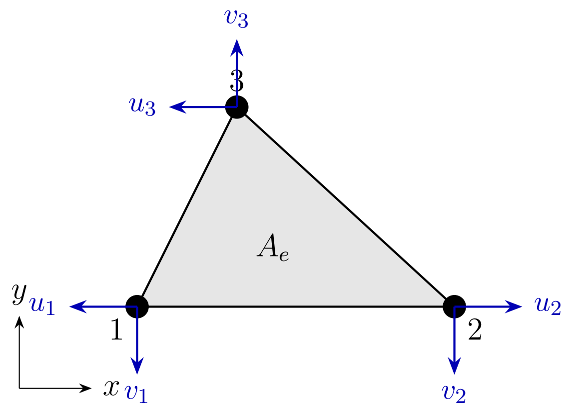

Figure 7.1: CST element: 3 nodes, 2 DOFs per node (u_i, v_i), element area A_e.

where:

$$a_1 = x_2 y_3 - x_3 y_2, \quad b_1 = y_2 - y_3, \quad c_1 = x_3 - x_2$$

$$2A_e = \det \begin{bmatrix} 1 & x_1 & y_1 \\ 1 & x_2 & y_2 \\ 1 & x_3 & y_3 \end{bmatrix}$$

(and cyclic permutations for $i = 2, 3$).

7.2.2 Strain-Displacement Matrix

$$\boldsymbol{\varepsilon} = \boldsymbol{B}\,\boldsymbol{u}^e, \qquad \boldsymbol{B} = \frac{1}{2A_e}\begin{bmatrix} b_1 & 0 & b_2 & 0 & b_3 & 0 \\ 0 & c_1 & 0 & c_2 & 0 & c_3 \\ c_1 & b_1 & c_2 & b_2 & c_3 & b_3 \end{bmatrix} \tag{7.5}$$

Note that $\boldsymbol{B}$ is constant $\Rightarrow$ strain is constant over the CST element.

7.2.3 Element Stiffness Matrix

$$\boldsymbol{k}^e = \int_{A_e} \boldsymbol{B}^\mathsf{T}\boldsymbol{D}\boldsymbol{B}\, t\, \mathrm{d}A = t\, A_e\, \boldsymbol{B}^\mathsf{T}\boldsymbol{D}\boldsymbol{B} \tag{7.6}$$

where t is the element thickness.

Key Result

The CST stiffness is evaluated in *closed form* (no numerical integration required) because $\boldsymbol{B}$ and $\boldsymbol{D}$ are both constant over the element:

$$\boldsymbol{k}^e = t\, A_e\, \boldsymbol{B}^\mathsf{T}\boldsymbol{D}\boldsymbol{B} \in \mathbb{R}^{6\times 6}$$

7.2.4 Worked Example: Two-Element CST Mesh in Uniaxial Tension

Example 7.1 (Two-element CST plate under uniaxial tension)**.** A 1 m $\times$ 1 m square plate of thickness $t = 0.01$ m is modelled with two CST elements formed by splitting the square along its diagonal, as shown in Figure 7.2. Material properties: $E = 200$ GPa, $\nu = 0.3$, plane stress. A uniform tensile stress $\sigma_x = 100$ MPa is applied to the right edge, equivalent to a total force $F = \sigma_x \cdot t \cdot 1$ m $= 1$ MN split equally between nodes 2 and 3 as $F/2 = 500$ kN each. The left-edge nodes are restrained: pin at node 1 ($u_1 = v_1 = 0$) and horizontal roller at node 4 ($u_4 = 0$).

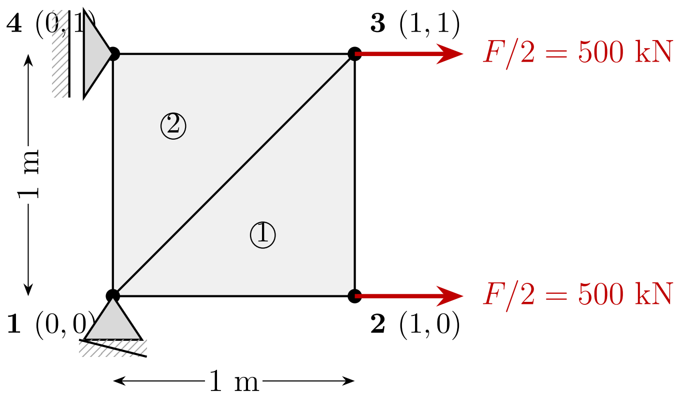

Figure 7.2: Two-element CST mesh: 4 nodes, 2 triangular elements formed by the diagonal from node 1 to node 3. Pin support at node 1, horizontal roller at node 4, and 500 kN horizontal force at nodes 2 and 3.

DOF ordering: $[u_1, v_1, u_2, v_2, u_3, v_3, u_4, v_4]$ = DOFs $[1 \ldots 8]$.

Table 7.1: Node coordinates and element connectivity.

Elem	Nodes	Node coords (m)	A_e (m^2)	Global DOFs
1	$1 \to 2 \to 3$	$(0,0), (1,0), (1,1)$	0.5	1,2,3,4,5,6
2	$1 \to 3 \to 4$	$(0,0), (1,1), (0,1)$	0.5	1,2,5,6,7,8

Step 1: Element geometry and B-matrices.

For a CST with nodes at (x_1, y_1), (x_2, y_2), (x_3, y_3), the coefficients are $b_i = y_j - y_k$ and $c_i = x_k - x_j$ (cyclic, Equation (7.4)), and $\boldsymbol{B}$ is constant (Equation (7.5)).

Element 1 (nodes $1 \to 2 \to 3$, coordinates $(0,0), (1,0), (1,1)$):

$$b_1 = -1, \quad b_2 = 1, \quad b_3 = 0, \qquad c_1 = 0, \quad c_2 = -1, \quad c_3 = 1, \qquad 2A_e = 1 \text{ m}^2$$

$$\boldsymbol{B}^{(1)} = \begin{bmatrix} -1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 1 \\ 0 & -1 & -1 & 1 & 1 & 0 \end{bmatrix} \text{ m}^{-1} \tag{7.7}$$

Element 2 (nodes $1 \to 3 \to 4$, coordinates $(0,0), (1,1), (0,1)$):

$$b_1 = 0, \quad b_2 = 1, \quad b_3 = -1, \qquad c_1 = -1, \quad c_2 = 0, \quad c_3 = 1, \qquad 2A_e = 1 \text{ m}^2$$

$$\boldsymbol{B}^{(2)} = \begin{bmatrix} 0 & 0 & 1 & 0 & -1 & 0 \\ 0 & -1 & 0 & 0 & 0 & 1 \\ -1 & 0 & 0 & 1 & 1 & -1 \end{bmatrix} \text{ m}^{-1} \tag{7.8}$$

The two B-matrices differ because the diagonal splits the square asymmetrically: element 1 has its right-angle corner at node 2, element 2 at node 4.

Step 2: Constitutive matrix.

Plane stress with $E = 200$ GPa, $\nu = 0.3$:

$$\boldsymbol{D} = \frac{200 \times 10^9}{1 - 0.09} \begin{bmatrix} 1 & 0.3 & 0 \\ 0.3 & 1 & 0 \\ 0 & 0 & 0.35 \end{bmatrix} = \begin{bmatrix} 2.198 \times 10^{11} & 6.593 \times 10^{10} & 0 \\ 6.593 \times 10^{10} & 2.198 \times 10^{11} & 0 \\ 0 & 0 & 7.692 \times 10^{10} \end{bmatrix} \text{ Pa} \tag{7.9}$$

Step 3: Element stiffness matrices.

Using $\boldsymbol{k}^e = t\,A_e\,\boldsymbol{B}^\top \boldsymbol{D}\boldsymbol{B}$ (Equation (7.6)) with $t\,A_e = 0.01 \times 0.5 = 0.005$ m^3:

$$\boldsymbol{k}^{(1)} = 10^8 \times \begin{bmatrix} 10.99 & 0 & -10.99 & 3.30 & 0 & -3.30 \\ 0 & 3.85 & 3.85 & -3.85 & -3.85 & 0 \\ -10.99 & 3.85 & 14.84 & -7.14 & -3.85 & 3.30 \\ 3.30 & -3.85 & -7.14 & 14.84 & 3.85 & -10.99 \\ 0 & -3.85 & -3.85 & 3.85 & 3.85 & 0 \\ -3.30 & 0 & 3.30 & -10.99 & 0 & 10.99 \end{bmatrix} \frac{\text{N}}{\text{m}} \quad (\text{DOFs } u_1, v_1, u_2, v_2, u_3, v_3) \tag{7.10}$$

$$\boldsymbol{k}^{(2)} = 10^8 \times \begin{bmatrix} 3.85 & 0 & 0 & -3.85 & -3.85 & 3.85 \\ 0 & 10.99 & -3.30 & 0 & 3.30 & -10.99 \\ 0 & -3.30 & 10.99 & 0 & -10.99 & 3.30 \\ -3.85 & 0 & 0 & 3.85 & 3.85 & -3.85 \\ -3.85 & 3.30 & -10.99 & 3.85 & 14.84 & -7.14 \\ 3.85 & -10.99 & 3.30 & -3.85 & -7.14 & 14.84 \end{bmatrix} \frac{\text{N}}{\text{m}} \quad (\text{DOFs } u_1, v_1, u_3, v_3, u_4, v_4) \tag{7.11}$$

Remark 7.1. $\boldsymbol{k}^{(1)} \neq \boldsymbol{k}^{(2)}$ even though both elements have the same area and material. The two triangles are geometrically distinct (different orientations of the hypotenuse) so their B-matrices differ, leading to different stiffness matrices.

Step 4: Global assembly.

The 8×8 global stiffness $\boldsymbol{K}$ is assembled from the two element matrices. Element 1 contributes to DOFs $\{1, 2, 3, 4, 5, 6\}$; element 2 to DOFs $\{1, 2, 5, 6, 7, 8\}$. DOFs $\{1, 2, 5, 6\}$ (nodes 1 and 3) receive contributions from both elements:

$$\boldsymbol{K} = 10^8 \times \begin{bmatrix} 14.84 & 0 & -10.99 & 3.30 & 0 & -7.14 & -3.85 & 3.85 \\ 0 & 14.84 & 3.85 & -3.85 & -7.14 & 0 & 3.30 & -10.99 \\ -10.99 & 3.85 & 14.84 & -7.14 & -3.85 & 3.30 & 0 & 0 \\ 3.30 & -3.85 & -7.14 & 14.84 & 3.85 & -10.99 & 0 & 0 \\ 0 & -7.14 & -3.85 & 3.85 & 14.84 & 0 & -10.99 & 3.30 \\ -7.14 & 0 & 3.30 & -10.99 & 0 & 14.84 & 3.85 & -3.85 \\ -3.85 & 3.30 & 0 & 0 & -10.99 & 3.85 & 14.84 & -7.14 \\ 3.85 & -10.99 & 0 & 0 & 3.30 & -3.85 & -7.14 & 14.84 \end{bmatrix} \frac{\text{N}}{\text{m}} \tag{7.12}$$

Note that all diagonal entries equal 14.84×10^8 N/m — a consequence of the mesh symmetry.

Step 5: Load vector and boundary conditions.

The global load vector has two non-zero entries: $F_3 = F_5 = 500\,000$ N (forces at DOFs 3 and 5, i.e. u_2 and u_3).

Essential BCs: $u_1 = 0$ (DOF 1), $v_1 = 0$ (DOF 2), $u_4 = 0$ (DOF 7). The five free DOFs are $\{3, 4, 5, 6, 8\}$ $(u_2, v_2, u_3, v_3, v_4)$. The reduced 5×5 system is:

$$10^8 \times \begin{bmatrix} 14.84 & -7.14 & -3.85 & 3.30 & 0 \\ -7.14 & 14.84 & 3.85 & -10.99 & 0 \\ -3.85 & 3.85 & 14.84 & 0 & 3.30 \\ 3.30 & -10.99 & 0 & 14.84 & -3.85 \\ 0 & 0 & 3.30 & -3.85 & 14.84 \end{bmatrix} \begin{bmatrix} u_2 \\ v_2 \\ u_3 \\ v_3 \\ v_4 \end{bmatrix} = \begin{bmatrix} 500\,000 \\ 0 \\ 500\,000 \\ 0 \\ 0 \end{bmatrix} \text{ N} \tag{7.13}$$

Step 6: Solution (MATLAB-verified, Section A.8).

$$\begin{aligned} u_2 &= +5.000 \times 10^{-4} \text{ m}, & v_2 &= 0 \text{ m}, \\ u_3 &= +5.000 \times 10^{-4} \text{ m}, & v_3 &= -1.500 \times 10^{-4} \text{ m}, \\ u_4 &= 0 \text{ m}, & v_4 &= -1.500 \times 10^{-4} \text{ m}. \end{aligned}$$

The right edge moves uniformly rightward ($u_2 = u_3 = 0.5$ mm) with no shear distortion. The top-right and top-left nodes contract vertically by 0.15 mm due to the Poisson effect.

Step 7: Stress recovery.

For each element, the constant strain and stress vectors are:

$$\boldsymbol{\varepsilon}^e = \boldsymbol{B}^e\,\boldsymbol{u}^e, \qquad \boldsymbol{\sigma}^e = \boldsymbol{D}\,\boldsymbol{\varepsilon}^e.$$

Element 1 ($\boldsymbol{u}^{(1)} = [0, 0, 5\times10^{-4}, 0, 5\times10^{-4}, -1.5\times10^{-4}]^\mathsf{T}$ m):

$$\boldsymbol{\varepsilon}^{(1)} = \boldsymbol{B}^{(1)}\boldsymbol{u}^{(1)} = \begin{bmatrix} 5 \times 10^{-4} \\ -1.5 \times 10^{-4} \\ 0 \end{bmatrix} \begin{bmatrix} \text{m/m} \\ \text{m/m} \\ \text{rad} \end{bmatrix}$$

$$\boldsymbol{\sigma}^{(1)} = \boldsymbol{D}\,\boldsymbol{\varepsilon}^{(1)} = \begin{bmatrix} 100 \\ 0 \\ 0 \end{bmatrix} \text{ MPa}$$

Element 2 ($\boldsymbol{u}^{(2)} = [0, 0, 5\times10^{-4}, -1.5\times10^{-4}, 0, -1.5\times10^{-4}]^\mathsf{T}$ m):

$$\boldsymbol{\varepsilon}^{(2)} = \boldsymbol{\varepsilon}^{(1)}, \qquad \boldsymbol{\sigma}^{(2)} = \boldsymbol{\sigma}^{(1)} = \begin{bmatrix} 100 \\ 0 \\ 0 \end{bmatrix} \text{ MPa}$$

Both elements give identical stresses: the CST mesh is in a state of uniform $\sigma_x = 100$ MPa, $\sigma_y = 0$, $\tau_{xy} = 0$.

Step 8: Analytical verification.

The exact solution for a uniformly loaded elastic plate is:

$$\begin{aligned}
\sigma_x &= \frac{F}{t \cdot L} = \frac{1 \times 10^6}{0.01 \times 1} = 100 \text{ MPa} \checkmark \\
\sigma_y &= 0 \checkmark, \qquad \tau_{xy} = 0 \checkmark \\
u(x, y) &= \frac{\sigma_x}{E} x = 5 \times 10^{-4} x \text{ m} \;\Rightarrow\; u_2 = u_3 = 5 \times 10^{-4} \text{ m} \checkmark \\
v(x, y) &= -\frac{\nu \sigma_x}{E} y = -1.5 \times 10^{-4} y \text{ m} \;\Rightarrow\; v_3 = v_4 = -1.5 \times 10^{-4} \text{ m} \checkmark
\end{aligned}$$

Key Result

The CST recovers the *exact* analytical solution for uniform stress states. This is because the true displacement field (u linear in x, v linear in y) lies within the CST polynomial space — a polynomial that the element can represent exactly. The FEM error is zero for any problem whose exact solution is linear.

The deformed mesh, magnified $\times 800$, is shown in Figure 7.3.

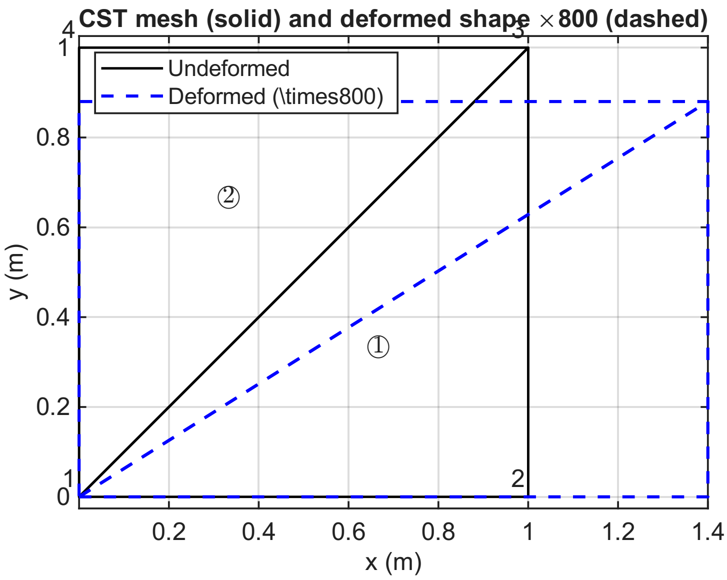

Figure 7.3: CST mesh (solid) and deformed shape (dashed, magnified $\times 800$). The right edge moves uniformly rightward; the top nodes contract downward by the Poisson effect. No shear distortion occurs.

7.3 Isoparametric Quadrilateral Element (Q4)

The 4-node quadrilateral (Q4) uses the same polynomials to interpolate both geometry and displacement — this is the *isoparametric* concept.

7.3.1 Shape Functions in Natural Coordinates

In the reference square $(\xi, \eta) \in [-1, 1]^2$:

$$N_i(\xi, \eta) = \frac{1}{4}(1 + \xi_i \xi)(1 + \eta_i \eta), \quad i = 1, 2, 3, 4 \tag{7.14}$$

where (ξ_i, η_i) are the corner coordinates: $(-1,-1), (1,-1), (1,1), (-1,1)$.

7.3.2 Jacobian Mapping

The physical coordinates are mapped as:

$$x = \sum_{i=1}^{4} N_i\, x_i, \qquad y = \sum_{i=1}^{4} N_i\, y_i \tag{7.15}$$

The Jacobian matrix $\boldsymbol{J}$ relates derivatives:

$$\boldsymbol{J} = \begin{bmatrix} \frac{\partial x}{\partial \xi} & \frac{\partial y}{\partial \xi} \\ \frac{\partial x}{\partial \eta} & \frac{\partial y}{\partial \eta} \end{bmatrix} = \sum_{i=1}^{4} \begin{bmatrix} \frac{\partial N_i}{\partial \xi} \\ \frac{\partial N_i}{\partial \eta} \end{bmatrix} \begin{bmatrix} x_i & y_i \end{bmatrix} \tag{7.16}$$

Physical derivatives are recovered as:

$$\begin{bmatrix} \frac{\partial N_i}{\partial x} \\ \frac{\partial N_i}{\partial y} \end{bmatrix} = \boldsymbol{J}^{-1} \begin{bmatrix} \frac{\partial N_i}{\partial \xi} \\ \frac{\partial N_i}{\partial \eta} \end{bmatrix} \tag{7.17}$$

7.3.3 Numerical Integration (Gauss Quadrature)

The element stiffness integral is evaluated by 2×2 Gauss quadrature (sufficient for Q4):

$$\boldsymbol{k}^e = t \int_{-1}^{1}\int_{-1}^{1} \boldsymbol{B}(\xi,\eta)^\mathsf{T} \boldsymbol{D}\, \boldsymbol{B}(\xi,\eta)\, |\boldsymbol{J}|\, \mathrm{d}\xi\, \mathrm{d}\eta \approx t \sum_{i=1}^{n_g}\sum_{j=1}^{n_g} w_i w_j\, \boldsymbol{B}(\xi_i,\eta_j)^\mathsf{T} \boldsymbol{D}\, \boldsymbol{B}(\xi_i,\eta_j)\, |\boldsymbol{J}(\xi_i,\eta_j)| \tag{7.18}$$

For 2×2 Gauss: $n_g = 2$, $\xi_i = \pm 1/\sqrt{3}$, $w_i = 1$.

Order	Points per direction	Exact for polynomials up to
1	1	degree 1
2	2	degree 3
3	3	degree 5

7.3.4 Worked Example: Single Q4 Element (Cook Membrane)

Example 7.2 (Cook membrane — single Q4 element under in-plane shear)**.** The Cook membrane is a tapered quadrilateral plate used as a benchmark for 2D elements. A single Q4 element discretises the plate, as shown in Figure 7.4. Material properties: $E = 70{,}000$ N/mm^2, $\nu = 0.3$, $t = 1$ mm, plane stress. The left edge is clamped (nodes 1 and 4). A total vertical force $P = 100$ N acts uniformly on the right edge; the consistent nodal forces are $F/2 = 50$ N upward at each of nodes 2 and 3.

DOF ordering: $[u_1, v_1, u_2, v_2, u_3, v_3, u_4, v_4]$ = DOFs $[1 \ldots 8]$.
Fixed: $\{1, 2, 7, 8\}$. Free: $\{3, 4, 5, 6\}$ (u_2, v_2, u_3, v_3).

Step 1: Shape function natural derivatives.
From Equation (7.14), the derivatives are:

Step 2: Symbolic Jacobian.

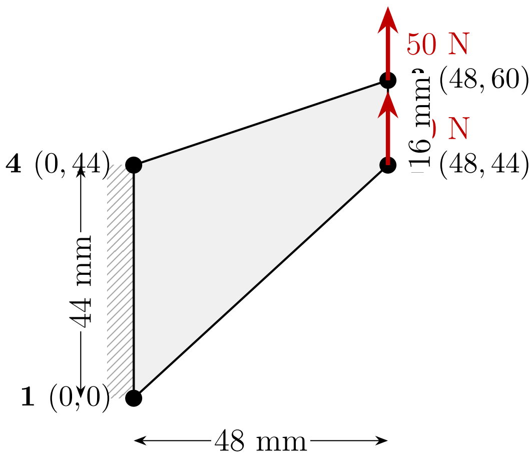

Figure 7.4: Single Q4 element for the Cook membrane benchmark. Node coordinates in mm. Left edge clamped; 50 N vertical nodal forces at nodes 2 and 3.

Table 7.2: Natural derivatives of Q4 shape functions.

Node i	$\partial N_i/\partial\xi$	$\partial N_i/\partial\eta$
1	$-(1-\eta)/4$	$-(1-\xi)/4$
2	$+(1-\eta)/4$	$-(1+\xi)/4$
3	$+(1+\eta)/4$	$+(1+\xi)/4$
4	$-(1+\eta)/4$	$+(1-\xi)/4$

Substituting the node coordinates $(x_1, y_1) = (0, 0)$, $(x_2, y_2) = (48, 44)$, $(x_3, y_3) = (48, 60)$, $(x_4, y_4) = (0, 44)$ into Equation (7.16):

$$J_{11} = \frac{\partial x}{\partial \xi} = \sum_i \frac{\partial N_i}{\partial \xi} x_i = \frac{-(1-\eta)}{4}\cdot 0 + \frac{(1-\eta)}{4}\cdot 48 + \frac{(1+\eta)}{4}\cdot 48 + \frac{-(1+\eta)}{4}\cdot 0 = 24$$

$$J_{12} = \frac{\partial y}{\partial \xi} = \frac{(1-\eta)}{4}\cdot 44 + \frac{(1+\eta)}{4}\cdot 60 - \frac{(1+\eta)}{4}\cdot 44 = 15 - 7\eta$$

$$J_{21} = \frac{\partial x}{\partial \eta} = \frac{-(1+\xi)}{4}\cdot 48 + \frac{(1+\xi)}{4}\cdot 48 = 0$$

$$J_{22} = \frac{\partial y}{\partial \eta} = \frac{-(1+\xi)}{4}\cdot 44 + \frac{(1+\xi)}{4}\cdot 60 + \frac{(1-\xi)}{4}\cdot 44 = 15 - 7\xi$$

$$\boldsymbol{J}(\xi,\eta) = \begin{bmatrix} 24 & 15-7\eta \\ 0 & 15-7\xi \end{bmatrix}, \qquad \det|\boldsymbol{J}| = 24(15-7\xi) \tag{7.19}$$

$$\boldsymbol{J}^{-1} = \frac{1}{24(15-7\xi)} \begin{bmatrix} 15-7\xi & -(15-7\eta) \\ 0 & 24 \end{bmatrix} = \begin{bmatrix} \dfrac{1}{24} & \dfrac{-(15-7\eta)}{24(15-7\xi)} \\ 0 & \dfrac{1}{15-7\xi} \end{bmatrix} \tag{7.20}$$

Remark 7.2. $J_{21} = 0$ because the left and right edges are both vertical ($x_1 = x_4 = 0$, $x_2 = x_3 = 48$ mm), so x depends only on ξ, not η. Consequently $\det|\boldsymbol{J}| = 24(15-7\xi)$ varies with ξ only — Gauss points at the same ξ share the same determinant regardless of η.

Step 3: Gauss-point table.

For 2×2 quadrature with $g = 1/\sqrt{3}$, $w_i = 1$:

Table 7.3: Jacobian data at the four Gauss points ($g = 1/\sqrt{3} \approx 0.5774$). $J_{11} = 24$ and $J_{21} = 0$ at all points; $J_{11}^{-1} = 1/24 \approx 0.04167$ and $J_{21}^{-1} = 0$ at all points.

GP	(ξ, η)	$J_{12} = 15{-}7\eta$	$J_{22} = 15{-}7\xi$	$\det\lvert\boldsymbol{J}\rvert$	J_{12}^{-1}	J_{22}^{-1}
1	$(-g, -g)$	19.04	19.04	457.0	$-1/24$	0.05252
2	$(+g, -g)$	19.04	10.96	263.0	-0.07240	0.09125
3	$(+g, +g)$	10.96	10.96	263.0	$-1/24$	0.09125
4	$(-g, +g)$	10.96	19.04	457.0	-0.02398	0.05252

Step 4: Strain-displacement matrix at GP1.

The physical derivatives $\partial N_i/\partial x$ and $\partial N_i/\partial y$ follow from $[\partial N/\partial x,\ \partial N/\partial y]^\mathsf{T} = \boldsymbol{J}^{-1}[\partial N/\partial \xi,\ \partial N/\partial \eta]^\mathsf{T}$. At GP1 ($\xi = \eta = -g$), substituting $J_{12}^{-1} = -1/24$ and $J_{22}^{-1} = 1/19.04$ (Equation (7.20)):

$$\frac{\partial N_i}{\partial x} = \frac{1}{24}\frac{\partial N_i}{\partial \xi} - \frac{1}{24}\frac{\partial N_i}{\partial \eta},$$

$$\frac{\partial N_i}{\partial y} = \frac{1}{19.04}\frac{\partial N_i}{\partial \eta}.$$

Evaluating for all four nodes yields (in 10^{-3} mm^{-1}):

$$\frac{\partial \boldsymbol{N}}{\partial x} = [0,\ 20.83,\ 0,\ -20.83], \qquad \frac{\partial \boldsymbol{N}}{\partial y} = [-20.71,\ -5.549,\ 5.549,\ 20.71]$$

Assembling $\boldsymbol{B} = \boldsymbol{B}(\xi, \eta)$ at GP1:

$$\boldsymbol{B}(\text{GP1}) = 10^{-3}\times\begin{bmatrix} 0 & 0 & 20.83 & 0 & 0 & 0 & -20.83 & 0 \\ 0 & -20.71 & 0 & -5.549 & 0 & 5.549 & 0 & 20.71 \\ -20.71 & 0 & -5.549 & 20.83 & 5.549 & 0 & 20.71 & -20.83 \end{bmatrix} \text{mm}^{-1} \tag{7.21}$$

Remark 7.3. Unlike the CST where $\boldsymbol{B}$ is constant, the Q4 $\boldsymbol{B}(\xi, \eta)$ changes at every Gauss point because the physical derivatives depend on $\boldsymbol{J}^{-1}(\xi, \eta)$, which varies over the element. Gauss quadrature is therefore *essential* — no closed-form integral exists.

Step 5: Constitutive matrix.

$$\boldsymbol{D} = \frac{70{,}000}{1 - 0.09}\begin{bmatrix} 1 & 0.3 & 0 \\ 0.3 & 1 & 0 \\ 0 & 0 & 0.35 \end{bmatrix} = \begin{bmatrix} 76{,}923 & 23{,}077 & 0 \\ 23{,}077 & 76{,}923 & 0 \\ 0 & 0 & 26{,}923 \end{bmatrix} \text{N/mm}^2 \tag{7.22}$$

Step 6: Element stiffness matrix.

The 8×8 element stiffness is accumulated over the four Gauss points (Equation (7.18)), with all weights $w_i = 1$:

$$\boldsymbol{k}^e = t\sum_{p=1}^{4} \boldsymbol{B}_p^\mathsf{T}\, \boldsymbol{D}\, \boldsymbol{B}_p\, \det\lvert\boldsymbol{J}_p\rvert$$

Since $\boldsymbol{B}$ varies at each Gauss point, this sum must be computed numerically (Section A.9). The MATLAB-verified result is:

$$\boldsymbol{k}^e = 10^3 \times \begin{bmatrix} 14.33 & -1.28 & 0.98 & -9.95 & -0.98 & -3.51 & -14.33 & 14.74 \\ -1.28 & 34.78 & -8.02 & 19.75 & -3.51 & -19.75 & 12.82 & -34.78 \\ 0.98 & -8.02 & 85.76 & -34.66 & -37.68 & 21.20 & -49.06 & 21.49 \\ -9.95 & 19.75 & -34.66 & 77.89 & 23.13 & -61.06 & 21.49 & -36.58 \\ -0.98 & -3.51 & -37.68 & 23.13 & 37.68 & -9.66 & 0.98 & -9.95 \\ -3.51 & -19.75 & 21.20 & -61.06 & -9.66 & 61.06 & -8.02 & 19.75 \\ -14.33 & 12.82 & -49.06 & 21.49 & 0.98 & -8.02 & 62.40 & -26.28 \\ 14.74 & -34.78 & 21.49 & -36.58 & -9.95 & 19.75 & -26.28 & 51.60 \end{bmatrix} \frac{\text{N}}{\text{mm}} \tag{7.23}$$

Step 7: Boundary conditions and reduced system.

Clamping the left edge sets DOFs $\{1, 2, 7, 8\} = 0$. The four free DOFs $\{3, 4, 5, 6\}$ (u_2, v_2, u_3, v_3) satisfy the 4×4 reduced system:

$$10^3 \times \begin{bmatrix} 85.76 & -34.66 & -37.68 & 21.20 \\ -34.66 & 77.89 & 23.13 & -61.06 \\ -37.68 & 23.13 & 37.68 & -9.66 \\ 21.20 & -61.06 & -9.66 & 61.06 \end{bmatrix} \begin{bmatrix} u_2 \\ v_2 \\ u_3 \\ v_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 50 \\ 0 \\ 50 \end{bmatrix} \text{N} \tag{7.24}$$

Step 8: Solution (MATLAB-verified, Section A.9).

$$u_2 = -9.3 \times 10^{-5} \text{ mm}, \qquad v_2 = +8.210 \times 10^{-3} \text{ mm},$$
$$u_3 = -2.926 \times 10^{-3} \text{ mm}, \qquad v_3 = +8.597 \times 10^{-3} \text{ mm}.$$

The dominant response is vertical (in the direction of loading): both right-edge nodes deflect upward by approximately 8.4 μm. The small horizontal displacements ($\sim$3 μm) reflect the shear-induced distortion of the tapered geometry. The deformed shape is shown in Figure 7.5.

Step 9: Stress recovery at Gauss points.

The stresses $\boldsymbol{\sigma}^e = \boldsymbol{D}\,\boldsymbol{B}(\xi_p, \eta_p)\,\boldsymbol{u}^e$ are evaluated at each Gauss point:

Table 7.4: Stresses at the four Gauss points (N/mm^2).

GP	(ξ, η)	σ_x	σ_y	τ_{xy}
1	$(-g, -g)$	-0.100	$+0.121$	$+4.181$
2	$(+g, -g)$	$+5.436$	$+2.608$	$+1.608$
3	$(+g, +g)$	$+0.173$	$+1.029$	$+1.860$
4	$(-g, +g)$	-3.128	-0.788	$+4.327$

Remark 7.4. The stresses vary significantly across the element — a key difference from the CST, which yields constant stress. The large τ_{xy} values at GPs 1 and 4 (left side of the element) reflect the shear dominance near the clamped edge, where bending is most constrained. This stress variation within the element is what the Q4 brings over the CST: it can represent linearly varying strains, which better captures the true behaviour of the Cook membrane under in-plane shear.

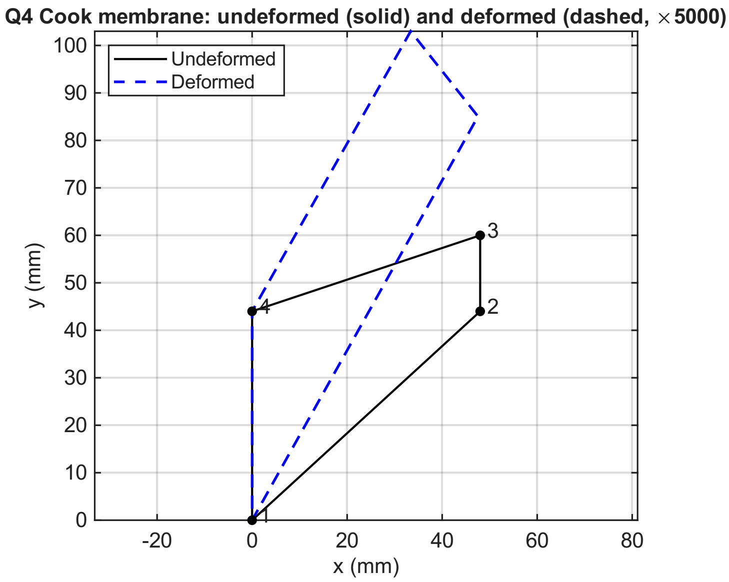

Figure 7.5: Cook membrane: undeformed (solid) and deformed shape (dashed, magnified $\times 5000$). Both right-edge nodes deflect vertically; the tapered geometry induces small lateral movement.

Appendix A

MATLAB Code Listings

This appendix provides MATLAB implementations for the key algorithms presented in each chapter. All scripts are self-contained and use only core MATLAB (no additional toolboxes required).

A.1 1D Bar Element Assembly and Solution

```matlab
% fem_bar_1d.m -- 3-element 1D bar, tip load
% Example from Chapter 2

clear; clc;

%% Parameters
nElem = 3;              % number of elements
nNode = nElem + 1;      % number of nodes
L     = 0.9;            % total length [m]
EA    = 70e6;           % axial stiffness [N]
F_tip = 10000;          % tip force [N]
Le    = L / nElem;      % element length [m]

%% Element stiffness matrix
ke = (EA/Le) * [1 -1; -1 1];

%% Global assembly
K = zeros(nNode, nNode);
for e = 1:nElem
    dofs = [e, e+1];            % global DOFs for element e
    K(dofs, dofs) = K(dofs, dofs) + ke;
end

%% Force vector
f = zeros(nNode, 1);
f(nNode) = F_tip;               % tip load at last node

%% Apply BC: node 1 fixed (u1 = 0)
freeDoFs = 2:nNode;
K_red = K(freeDoFs, freeDoFs);
f_red = f(freeDoFs);
```

```
%% Solve
u_free = K_red \ f_red;
u = [0; u_free];                        % full displacement vector

%% Post-processing
fprintf('Nodal displacements [m]:\n');
for i = 1:nNode
    fprintf('  u(%d) = %.6e m\n', i, u(i));
end
```

Listing A.1: Three-element bar under tip load (Chapter 3)

A.2 Stepped Bar with Distributed and Concentrated Loads

This listing reproduces the worked example of Example 3.2: a two-element cantilever bar with a step in cross-section, a distributed axial load on the first element, and concentrated forces at nodes 2 and 3. The script assembles the global stiffness and force vectors element by element, applies the fixed boundary condition, solves the reduced system, recovers the reaction, and plots the displacement and internal-force fields using the enriched trial Equation (3.12).

```
% stepped_bar_2elem.m -- two-element stepped bar (Chapter 3)
% Cantilever bar with two segments:
%   E1: L1 = 500 mm, area  A,  distributed load p = -4 N/mm
%   E2: L2 = 400 mm, area 2A, no distributed load
% Concentrated forces: +6000 N at node 2, -15000 N at node 3.

clear; clc; close all;

%% Material and geometry
E   = 210000;          % Young's modulus [MPa = N/mm^2]
A   = 225;             % element-1 cross-section area [mm^2]
p_e1 = -4;             % distributed load on element 1 [N/mm]
p_e2 =  0;             % distributed load on element 2 [N/mm]

%% Concentrated forces
P_node2 =   6000;     % +x at node 2 [N]
P_node3 = -15000;     % -x at node 3 [N]

%% Discretization
nodes_x   = [0, 500, 900];       % node coordinates [mm]
nElem     = 2;
nNode     = numel(nodes_x);
nDof      = nNode;               % 1D bar: 1 DOF per node
elem_conn = [1 2; 2 3];          % element connectivity
elem_area = [A, 2*A];
elem_p    = [p_e1, p_e2];
elem_L    = diff(nodes_x);
```

```
%% Initialize globals
KG     = zeros(nDof, nDof);       % global stiffness
FG_p   = zeros(nDof, 1);          % distributed-load contribution
FG_c   = zeros(nDof, 1);          % concentrated forces

%% Element-by-element assembly
for e = 1:nElem
    L    = elem_L(e);
    Ae   = elem_area(e);
    pe   = elem_p(e);
    dofs = elem_conn(e,:);
    ke   = (E*Ae/L) * [1 -1; -1 1];
    fe_p = (pe*L/2) * [1; 1];
    KG(dofs, dofs) = KG(dofs, dofs) + ke;
    FG_p(dofs)     = FG_p(dofs)     + fe_p;
end

%% Add concentrated forces and form total
FG_c(2) = P_node2;
FG_c(3) = P_node3;
FG = FG_p + FG_c;

%% Boundary conditions: node 1 fixed
fixed_dofs = 1;
free_dofs  = setdiff(1:nDof, fixed_dofs);

UG              = zeros(nDof, 1);
UG(free_dofs)   = KG(free_dofs, free_dofs) \ FG(free_dofs);
RG              = KG*UG - FG;

fprintf('UG = '); disp(UG');
fprintf('RG = '); disp(RG');

%% Element-level postprocessing using the enriched trial:
%   u(x)   = N1*u1 + N2*u2 + p*x*(L-x)/(2EA)
%   N(x)   = EA*(u2-u1)/L + p*(L-2x)/2
xg = []; ug = []; Ng = [];
for e = 1:nElem
    L    = elem_L(e);
    Ae   = elem_area(e);
    pe   = elem_p(e);
    dofs = elem_conn(e,:);
    u1   = UG(dofs(1));  u2 = UG(dofs(2));
    x    = linspace(0, L, 200);
    xi   = x / L;
    u_e  = (1-xi)*u1 + xi*u2 + pe*x.*(L - x)/(2*E*Ae);
    N_e  = E*Ae*(u2-u1)/L + pe*(L - 2*x)/2;
    xg = [xg, nodes_x(dofs(1)) + x];
    ug = [ug, u_e];
    Ng = [Ng, N_e];
```

```
end

%% Plots
figure('Color','w'); plot(xg, ug, 'LineWidth', 1.5); grid on;
xlabel('x [mm]'); ylabel('u(x) [mm]');
title('Displacement along the stepped bar');

figure('Color','w'); plot(xg, Ng, 'LineWidth', 1.5); grid on;
xlabel('x [mm]'); ylabel('N(x) [N]');
title('Internal axial force along the stepped bar');
```

Listing A.2: Stepped bar with distributed and concentrated loads (Chapter 3)

A.3 Verification of the Principle of Virtual Work

This listing supports Example 3.3: it takes the solved stepped bar (the same u_1, u_2, u_3) and, for three different kinematically admissible virtual displacements, evaluates the internal and external virtual work separately. All three cases satisfy $\delta W_{\text{int}} = \delta W_{\text{ext}}$ to numerical precision, confirming the principle of virtual work.

```
% pvw_verification_stepped_bar.m
% Numerically verify the principle of virtual work for the
    SOLVED
% stepped bar (the worked example of Section "Stepped Bar").
% For three different kinematically admissible virtual
    displacements,
% the script computes the internal and external virtual work
    separately
% and shows they are equal.

clear; clc;

%% Stepped bar parameters
E   = 210000;
A   = 225;
EA1 = E*A;          % element 1 axial stiffness [N]
EA2 = E*2*A;        % element 2 axial stiffness [N]
L1  = 500;          % element 1 length [mm]
L2  = 400;          % element 2 length [mm]
p1  = -4;           % element 1 distributed load [N/mm]
F2  =  6000;        % concentrated force at node 2 [N]
F3  = -15000;       % concentrated force at node 3 [N]

%% Exact (FEM) nodal displacements from the Stepped Bar example
u1 = 0;              % fixed
u2 = -20/189;        % mm   (= -0.10582)
u3 = -32/189;        % mm   (= -0.16931)

%% Internal axial force in each element using the enriched
    trial:
%   N(x) = EA*(u_b - u_a)/L   +  p*(L - 2x_local)/2
```

```
N1 = @(x) EA1*(u2-u1)/L1 + p1*(L1 - 2*x)/2;          % x measured from node 1
N2 = @(x) EA2*(u3-u2)/L2 + 0*x;                      % x measured from node 2

%% Three admissible virtual displacements (each must satisfy du(x=0)=0)
%   Each entry stores du(x), du'(x), and the nodal values du(node_k)
testCases = struct( ...
  'name',     {}, ...
  'du',       {}, ...
  'du_dx',    {}, ...
  'du_n2',    {}, ...
  'du_n3',    {});

% Case A -- tent: du_2 = 1, du_3 = 0
testCases(end+1) = struct( ...
  'name',  'Case A:  (du_1, du_2, du_3) = (0, 1, 0)', ...
  'du',    @(x_local, elem) (elem==1).*(x_local/L1) + (elem==2).*(1 - x_local/L2), ...
  'du_dx', @(x_local, elem) (elem==1)*(1/L1)         + (elem==2)*(-1/L2), ...
  'du_n2', 1, ...
  'du_n3', 0);

% Case B -- ramp + plateau: du_2 = 1, du_3 = 1
testCases(end+1) = struct( ...
  'name',  'Case B:  (du_1, du_2, du_3) = (0, 1, 1)', ...
  'du',    @(x_local, elem) (elem==1).*(x_local/L1) + (elem==2).*1, ...
  'du_dx', @(x_local, elem) (elem==1)*(1/L1)         + (elem==2)*0, ...
  'du_n2', 1, ...
  'du_n3', 1);

% Case C -- global quadratic du(x_glob) = (x_glob / Ltot)^2
Ltot = L1 + L2;
testCases(end+1) = struct( ...
  'name',  'Case C:  du(x) = (x / 900)^2  (smooth, non-piecewise-linear)', ...
  'du',    @(x_local, elem) ((elem==1).*x_local + (elem==2).*(L1 + x_local)).^2 / Ltot^2, ...
  'du_dx', @(x_local, elem) 2*((elem==1).*x_local + (elem==2).*(L1 + x_local)) / Ltot^2, ...
  'du_n2', (L1/Ltot)^2, ...
  'du_n3', 1);

%% Verify PVW for each case
fprintf('Principle of Virtual Work verification -- stepped bar\n');
```

```
fprintf('Solved nodal displacements:  u1 = 0,  u2 = %.5f,  u3 =
    %.5f mm\n\n', u2, u3);
fprintf('%-58s %14s %14s %14s\n', 'Admissible virtual
   displacement', ...
        'dW_int [Nmm]', 'dW_ext [Nmm]', '|diff|');
fprintf('%s\n', repmat('-', 1, 102));

for k = 1:length(testCases)
    tc = testCases(k);

    % Internal virtual work -- sum element contributions
    dWint_e1 = integral(@(x) N1(x).*tc.du_dx(x,1), 0, L1);
    dWint_e2 = integral(@(x) N2(x).*tc.du_dx(x,2), 0, L2);
    dWint    = dWint_e1 + dWint_e2;

    % External virtual work -- distributed load (element 1 only
       ) and
    % concentrated forces; reaction at node 1 contributes 0
       because du(0)=0.
    dWext_d  = integral(@(x) p1.*tc.du(x,1), 0, L1);
    dWext_c  = F2*tc.du_n2 + F3*tc.du_n3;
    dWext    = dWext_d + dWext_c;

    fprintf('%-58s %14.5f %14.5f %14.2e\n', tc.name, dWint,
       dWext, abs(dWint-dWext));
end

fprintf(['\nFor every kinematically admissible virtual
   displacement,\n', ...
         'dW_int = dW_ext to numerical precision: PVW holds.\n'
            ]);
```

Listing A.3: PVW verification for the stepped bar (Chapter 3)

A.4 Three-Node Quadratic Rod Element

```
% Chapter 3 -- Three-node quadratic rod element
% Worked example (sec:ex_bar_quad): cantilever bar of length L,
    axial
% stiffness EA, uniform distributed load p0, single 3-node
   element.
% Demonstrates the 2-point Gauss-Legendre loop that produces ke
    and fe.

clear; clc; close all;

%% Parameters
L  = 1.0;        % length [m]
EA = 1.0;        % axial stiffness [N]
p0 = 1.0;        % uniform load per unit length [N/m]
```

```
%% Quadratic Lagrange shape functions on the parent domain xi in [-1, 1]
% Each handle returns a 3-by-1 column vector [N1; N2; N3] or its derivative
N    = @(xi) [0.5*xi*(xi - 1); 1 - xi*xi; 0.5*xi*(xi + 1)];
dNdx = @(xi) [xi - 0.5;         -2*xi;      xi + 0.5       ];

%% Two-point Gauss-Legendre rule on [-1, 1]
g  = 1/sqrt(3);
gp = [-g, +g];
w  = [1,  1 ];

%% Element matrices via Gauss quadrature
% Three nodes are equally spaced at x = 0, L/2, L, so the Jacobian
% J = dx/dxi = L/2 is constant; the loop is written generally so that the
% same code applies to a distorted interior node where J(xi) varies.
ke = zeros(3);
fe = zeros(3, 1);
for q = 1:2
    xi   = gp(q);
    Nv   = N(xi);
    dNv  = dNdx(xi);
    Jq   = dNv' * [0; L/2; L];       % geometry interpolation
    Binv = 1/Jq;
    B    = Binv * dNv';              % 1-by-3 strain-displacement row
    ke   = ke + w(q) * EA * (B' * B) * Jq;
    fe   = fe + w(q) * Nv * p0 * Jq;
end

fprintf('=== ELEMENT STIFFNESS (multiplied by 3L/EA) ===\n');
disp(ke * 3*L/EA);
fprintf('=== ELEMENT LOAD (multiplied by 6/(p0 L)) ===\n');
disp(fe * 6/(p0*L));

%% Apply boundary condition u1 = 0 and solve
free    = [2, 3];
u       = zeros(3, 1);
u(free) = ke(free, free) \ fe(free);

fprintf('\n=== NODAL DISPLACEMENTS (in units p0 L^2 / EA) ===\n');
fprintf('u1 = %+.4f   (exact:  0.0000)\n', u(1) * EA / (p0*L^2));
fprintf('u2 = %+.4f   (exact:  0.3750 = 3/8)\n', u(2) * EA / (p0*L^2));
```

```
fprintf('u3 = %+.4f   (exact:  0.5000 = 1/2)\n', u(3) * EA / (
   p0*L^2));

%% Strain at the three nodes (xi = -1, 0, +1)
xi_nodes = [-1, 0, +1];
strain   = zeros(3, 1);
for k = 1:3
    Bk = (2/L) * dNdx(xi_nodes(k))';
    strain(k) = Bk * u;
end
fprintf('\n=== NODAL STRAIN (in units p0 L / EA) ===\n');
fprintf('eps1 = %+.4f   (exact:  1.0000)\n', strain(1) * EA / (
   p0*L));
fprintf('eps2 = %+.4f   (exact:  0.5000)\n', strain(2) * EA / (
   p0*L));
fprintf('eps3 = %+.4f   (exact:  0.0000)\n', strain(3) * EA / (
   p0*L));

%% Reaction at the fixed support
R1 = ke(1, :) * u - fe(1);
fprintf('\n=== REACTION AT FIXED END ===\n');
fprintf('R1 = %+.4f p0 L   (exact: -1.0000)\n', R1 / (p0*L));

%% Plot FE vs analytical displacement field
nx        = 201;
xx        = linspace(0, L, nx);
xi_grid   = 2*xx/L - 1;
u_fe      = zeros(1, nx);
for k = 1:nx
    u_fe(k) = N(xi_grid(k))' * u;
end
u_exact = (p0/EA) * (L*xx - 0.5*xx.^2);

fig = figure('Units','centimeters','Position',[2, 2, 14, 9]);
plot(xx/L, u_exact * EA/(p0*L^2), 'k-',  'LineWidth', 1.8);
   hold on;
plot(xx/L, u_fe    * EA/(p0*L^2), 'b--', 'LineWidth', 1.8);
plot([0, 0.5, 1.0], u * EA/(p0*L^2), 'ro', ...
     'MarkerSize', 8, 'MarkerFaceColor', 'r');
grid on; box on;
xlabel('x / L');
ylabel('u EA / (p_0 L^2)');
legend({'Exact','FE field','FE nodes'}, 'Location','northwest')
   ;
title('Cantilever bar -- single 3-node quadratic element');

book_style(fig);
if ~exist('../figures', 'dir'), mkdir('../figures'); end
exportgraphics(fig, '../figures/bar_3node_results.pdf', '
   ContentType','vector');
```

```
exportgraphics(fig, '../figures/bar_3node_results.png', '
    Resolution', 300);
fprintf('\nFigure saved to ../figures/bar_3node_results.pdf and
    .png\n');
```

Listing A.4: Cantilever bar under uniform load, single 3-node quadratic element with 2-point Gauss quadrature (Example 3.4).

A.5 2D Truss Analysis

```
% fem_truss_2d.m -- Chapter 4: Worked Example, 3-element 2D
    truss
% Nodes: 1=(0,0), 2=(1000,800), 3=(0,800) [mm]
% BCs:   node 1 pinned (U1=V1=0), node 3 pinned (U3=V3=0)
% Load:  Fy = -10000 N (downward) at node 2
clear; clc; close all;

coords = [   0     0;
           1000   800;
              0   800];
conn = [1 2; 1 3; 3 2];
E = 210000;  A = 225;
nEl = 3;  nDOF = 6;

for e = 1:nEl
    n1 = conn(e,1); n2 = conn(e,2);
    dx = coords(n2,1) - coords(n1,1);
    dy = coords(n2,2) - coords(n1,2);
    Le(e)  = sqrt(dx^2 + dy^2);
    ce(e)  = dx / Le(e);
    se(e)  = dy / Le(e);
    EAL(e) = E * A / Le(e);
end

K = zeros(nDOF);
for e = 1:nEl
    c = ce(e); s = se(e); k = EAL(e);
    ke = k * [ c^2   c*s  -c^2  -c*s;
               c*s   s^2  -c*s  -s^2;
              -c^2  -c*s   c^2   c*s;
              -c*s  -s^2   c*s   s^2];
    n1 = conn(e,1); n2 = conn(e,2);
    dofs = [2*n1-1, 2*n1, 2*n2-1, 2*n2];
    K(dofs, dofs) = K(dofs, dofs) + ke;
end

F = zeros(nDOF, 1);
F(4) = -10000;
fixedDOFs = [1 2 5 6];
freeDOFs  = [3 4];
```

```

u_free = K(freeDOFs, freeDOFs) \ F(freeDOFs);
u = zeros(nDOF, 1);
u(freeDOFs) = u_free;

for e = 1:nEl
    c = ce(e); s = se(e);
    n1 = conn(e,1); n2 = conn(e,2);
    dofs = [2*n1-1, 2*n1, 2*n2-1, 2*n2];
    N(e) = EAL(e) * ([-c -s c s] * u(dofs));
end

R = K * u - F;

fprintf('U2 = %.4f mm,  V2 = %.4f mm\n', u(3), u(4));
fprintf('N1 = %.1f N,  N2 = %.1f N,  N3 = %.1f N\n', N(1), N(2)
    , N(3));
fprintf('R1x = %.1f N,  R1y = %.1f N\n', R(1), R(2));
fprintf('R3x = %.1f N,  R3y = %.1f N\n', R(5), R(6));

scale = 500;
figure; hold on; axis equal;
for e = 1:nEl
    n1 = conn(e,1); n2 = conn(e,2);
    xo = [coords(n1,1) coords(n2,1)];
    yo = [coords(n1,2) coords(n2,2)];
    plot(xo, yo, 'k--', 'LineWidth', 1.2);
    plot(xo + scale*[u(2*n1-1) u(2*n2-1)], ...
         yo + scale*[u(2*n1)   u(2*n2)  ], 'b-', 'LineWidth',
            2.5);
end
xlabel('x [mm]'); ylabel('y [mm]');
title(sprintf('Deformed shape (x%d)', scale));
exportgraphics(gcf,'../figures/truss2d_deformed.pdf','
    ContentType','vector');
exportgraphics(gcf,'../figures/truss2d_deformed.png','
    Resolution',300);
```

Listing A.5: Three-element planar truss (Chapter 4)

A.6 Euler-Bernoulli Beam: Simply Supported

```
% fem_beam_simple.m -- Chapter 5: 2-element simply supported
    beam
% L = 2 m, EI = 1e4 N*m^2, P = 1000 N downward at midspan
% Sign convention: v upward positive, theta CCW positive
clear; clc; close all;

L = 2; Le = L/2; EI = 1e4; P = 1000;
nNodes = 3; nDOF = 6;
```

```
K0 = EI / Le^3;  % = 1e4 N/m

Ce = [12    6*Le    -12    6*Le;
       6*Le  4*Le^2  -6*Le  2*Le^2;
     -12   -6*Le     12   -6*Le;
       6*Le  2*Le^2  -6*Le  4*Le^2];
ke = K0 * Ce;

K = zeros(nDOF);
for e = 1:2
    dofs = [2*e-1, 2*e, 2*e+1, 2*e+2];
    K(dofs, dofs) = K(dofs, dofs) + ke;
end

F = zeros(nDOF,1);
F(3) = -P;                      % downward load at DOF 3 (v2)
fixedDOFs = [1, 5];             % v1=0, v3=0
freeDOFs  = [2, 3, 4, 6];

u_free = K(freeDOFs,freeDOFs) \ F(freeDOFs);
u = zeros(nDOF,1);  u(freeDOFs) = u_free;

R = K*u - F;
fprintf('th1=%.4f rad, v2=%.4f mm, th2=%.4f, th3=%.4f rad\n',
   ...
        u(2), u(3)*1000, u(4), u(6));
fprintf('R1=%.1f N, R3=%.1f N\n', R(1), R(5));
fprintf('v2_analytic = %.4f mm\n', P*L^3/(48*EI)*1000);
```

Listing A.6: Two-element simply supported beam (Chapter 5)

A.7 Euler-Bernoulli Beam: Continuous Beam

```
% fem_beam_3elem.m -- Chapter 5: 3-element continuous beam
% Nodes: 1=0, 2=3000, 3=5000, 4=6000 mm
% BCs: w1=0 (pin), w2=0 (roller), w4=0 (roller)
% Load: q=10 N/mm on element 1; P=30000 N at node 3
clear; clc; close all;

E=210000; Iy=5481140; EI=E*Iy; z_max=70;
xn=[0;3000;5000;6000]; Le=diff(xn);
nEl=3; nDOF=8;

K=zeros(nDOF);
for e=1:nEl
    L=Le(e); EIL3=EI/L^3;
    ke=EIL3*[12,6*L,-12,6*L; 6*L,4*L^2,-6*L,2*L^2;
             -12,-6*L,12,-6*L; 6*L,2*L^2,-6*L,4*L^2];
    dofs=[2*e-1,2*e,2*e+1,2*e+2];
    K(dofs,dofs)=K(dofs,dofs)+ke;
```

```
end

F=zeros(nDOF,1);
q=10; L1=Le(1);
F([1,2,3,4])=F([1,2,3,4])+(-q*L1/12)*[6;L1;6;-L1];
F(5)=F(5)-30000;

fixedDOFs=[1,3,7]; freeDOFs=[2,4,5,6,8];
u=zeros(nDOF,1);
u(freeDOFs)=K(freeDOFs,freeDOFs)\F(freeDOFs);
R=K*u-F;

fprintf('th1=%.4e, th2=%.4e, w3=%.4f mm\n',u(2),u(4),u(5));
fprintf('th3=%.4e, th4=%.4e\n',u(6),u(8));
fprintf('R1=%.1f N, R2=%.1f N, R4=%.1f N\n',R(1),R(3),R(7));

My_max=0;
for e=1:nEl
    L=Le(e); dofs=[2*e-1,2*e,2*e+1,2*e+2]; ue=u(dofs);
    xi=linspace(0,L,201);
    H1pp=12*xi/L^3-6/L^2; H2pp=6*xi/L^2-4/L;
    H3pp=-12*xi/L^3+6/L^2; H4pp=6*xi/L^2-2/L;
    v2nd=H1pp*ue(1)+H2pp*ue(2)+H3pp*ue(3)+H4pp*ue(4);
    if e==1
        v2nd=v2nd+(-q/(24*EI))*(2*L^2-12*L*xi+12*xi.^2);
    end
    My_el=EI*v2nd;
    My_max=max(My_max,max(abs(My_el)));
end
sig_max=My_max*z_max/Iy;
fprintf('|M|_max=%.4g Nmm, sigma_max=%.4g MPa\n',My_max,sig_max
    );
```

Listing A.7: Three-element continuous beam (Chapter 5)

A.8 Chapter 7 — Two-Element CST Mesh

```
% Chapter 7 -- Two-element CST mesh under uniaxial tension
% Plate: 1 m x 1 m x 0.01 m, plane stress, E=200 GPa, nu=0.3
% Nodes: 1(0,0) 2(1,0) 3(1,1) 4(0,1)
% Elements: 1=[1,2,3], 2=[1,3,4]
% BCs: pin at node 1 (u1=v1=0), roller at node 4 (u4=0)
% Load: F_x=500 kN at nodes 2 and 3
clear; clc; close all;

%% Parameters
E  = 200e9;    % Pa
nu = 0.3;
t  = 0.01;     % m

```

```
%% Nodes and connectivity
xy   = [0,0; 1,0; 1,1; 0,1];
conn = [1,2,3; 1,3,4];

%% Constitutive matrix (plane stress)
D = E/(1-nu^2) * [1, nu, 0; nu, 1, 0; 0, 0, (1-nu)/2];

%% Assemble global K
n_dof = 8;
K = zeros(n_dof);
B_all = cell(2,1);

for e = 1:2
    nd = conn(e,:);
    x  = xy(nd,1);  y = xy(nd,2);
    Ae = 0.5*abs((x(2)-x(1))*(y(3)-y(1)) - (x(3)-x(1))*(y(2)-y(1)));
    b  = [y(2)-y(3); y(3)-y(1); y(1)-y(2)];
    c  = [x(3)-x(2); x(1)-x(3); x(2)-x(1)];
    B  = 1/(2*Ae) * [b(1),0,b(2),0,b(3),0; ...
                     0,c(1),0,c(2),0,c(3); ...
                     c(1),b(1),c(2),b(2),c(3),b(3)];
    ke = t * Ae * B' * D * B;
    B_all{e} = B;
    dofs = [2*nd(1)-1, 2*nd(1), 2*nd(2)-1, 2*nd(2), 2*nd(3)-1, 2*nd(3)];
    K(dofs,dofs) = K(dofs,dofs) + ke;
end

%% Load vector and BCs
F = zeros(n_dof,1);
F(3) = 500e3;   % u2
F(5) = 500e3;   % u3

fixed = [1, 2, 7];   % u1, v1, u4
free  = setdiff(1:n_dof, fixed);
u = zeros(n_dof,1);
u(free) = K(free,free) \ F(free);

%% Print results
fprintf('=== NODAL DISPLACEMENTS ===\n')
for n = 1:4
    fprintf('Node %d: u=%.4e m  v=%.4e m\n', n, u(2*n-1), u(2*n))
end

fprintf('\n=== ELEMENT STRESSES ===\n')
for e = 1:2
    nd   = conn(e,:);
    dofs = [2*nd(1)-1,2*nd(1),2*nd(2)-1,2*nd(2),2*nd(3)-1,2*nd(3)];
```

```
    sig  = D * B_all{e} * u(dofs);
    fprintf('Elem %d: sx=%.4e Pa  sy=%.4e Pa  txy=%.4e Pa\n', ...
            e, sig(1), sig(2), sig(3))
end

%% Figure: mesh + deformed shape
scale = 800;
fig = figure('Units','centimeters','Position',[2,2,14,10]);
hold on; axis equal; grid on;

tri_x = xy([1,2,3,1],1); tri_y = xy([1,2,3,1],2);
plot(tri_x, tri_y, 'k-', 'LineWidth', 1.2)
tri_x = xy([1,3,4,1],1); tri_y = xy([1,3,4,1],2);
plot(tri_x, tri_y, 'k-', 'LineWidth', 1.2)

% Deformed nodes
xy_def = xy + scale * reshape(u,2,[])';
tri_x = xy_def([1,2,3,1],1); tri_y = xy_def([1,2,3,1],2);
plot(tri_x, tri_y, 'b--', 'LineWidth', 1.5)
tri_x = xy_def([1,3,4,1],1); tri_y = xy_def([1,3,4,1],2);
plot(tri_x, tri_y, 'b--', 'LineWidth', 1.5)

% Node labels
for n = 1:4
    text(xy(n,1)-0.07, xy(n,2)+0.04, sprintf('  %d',n), 'FontSize',10)
end

% Element labels
c1 = mean(xy([1,2,3],:)); text(c1(1),c1(2),'\textcircled{1}','Interpreter','latex','FontSize',12,'HorizontalAlignment','center')
c2 = mean(xy([1,3,4],:)); text(c2(1),c2(2),'\textcircled{2}','Interpreter','latex','FontSize',12,'HorizontalAlignment','center')

xlabel('x (m)'); ylabel('y (m)')
title(sprintf('CST mesh (solid) and deformed shape \\times%d (dashed)', scale))
legend({'Undeformed','','Deformed (\\times800)',''}, 'Location','northwest')

book_style(fig);
exportgraphics(fig,'../figures/cst_example_results.pdf','ContentType','vector')
exportgraphics(fig,'../figures/cst_example_results.png','Resolution',300)
fprintf('\nFigures saved to ../figures/\n')
```

Listing A.8: Two-element CST plate in uniaxial tension (Example 7.1).

Chapter 6 — 2D Frame Element

```
% Chapter 6 -- Portal Frame under Combined Loading
% Geometry: H=3000 mm (columns), B=4000 mm (beam)
% Nodes:  1(0,0)  2(0,3000)  3(4000,3000)  4(4000,0)
% Elements: 1=left col (1->2), 2=beam (2->3), 3=right col
  (3->4)
% BCs: nodes 1 and 4 fully clamped
% Loading: F_x=10000 N at node 2; q=5 N/mm downward on beam
clear; clc; close all;

%% Parameters
E      = 210000;   % N/mm^2
A      = 6000;     % mm^2
Iz     = 10e6;     % mm^4
H      = 3000;     % mm  column height
B      = 4000;     % mm  beam span
F_lat  = 10000;    % N   horizontal force at node 2 (global x)
q_beam = 5;        % N/mm uniform downward on beam (element 2)

%% Node coordinates [x, y]
coords = [0, 0; 0, H; B, H; B, 0];
n_dof  = 12;   % 4 nodes x 3 DOFs

%% Element connectivity and inclination angles (deg, CCW from +
  x)
conn   = [1 2; 2 3; 3 4];
alpha_deg = [90; 0; -90];
n_elem = 3;

%% Assemble global K and F
K = zeros(n_dof);
F = zeros(n_dof, 1);

for e = 1:n_elem
    ni = conn(e,1); nj = conn(e,2);
    L  = norm(coords(nj,:) - coords(ni,:));
    c  = cosd(alpha_deg(e));
    s  = sind(alpha_deg(e));
    T6 = frame_T6(c, s);
    kbar = frame_local_k(E, A, Iz, L);
    ke   = T6' * kbar * T6;
    dofs = elem_dofs(ni, nj);
    K(dofs, dofs) = K(dofs, dofs) + ke;

    if e == 2  % uniform downward load on beam
        q0   = -q_beam;  % upward-positive convention
        fbar = [0; q0*L/2; q0*L^2/12; 0; q0*L/2; -q0*L^2/12];
        F(dofs) = F(dofs) + T6' * fbar;
    end
end
```

```

% Lateral point force at node 2, global x-direction (DOF 4)
F(4) = F(4) + F_lat;

%% Apply BCs: fix nodes 1 (DOFs 1-3) and 4 (DOFs 10-12)
fixed = [1 2 3 10 11 12];
free  = setdiff(1:n_dof, fixed);
u = zeros(n_dof, 1);
u(free) = K(free, free) \ F(free);

%% Reactions
R = K * u - F;

%% Print verified results
fprintf('=== FREE-DOF DISPLACEMENTS ===\n');
fprintf('Node 2: u2=%+.5f mm  v2=%+.5f mm  th2=%+.6f rad\n', u
    (4),u(5),u(6));
fprintf('Node 3: u3=%+.5f mm  v3=%+.5f mm  th3=%+.6f rad\n', u
    (7),u(8),u(9));
fprintf('\n=== SUPPORT REACTIONS ===\n');
fprintf('Node 1: Rx1=%+.1f N  Ry1=%+.1f N  M1=%+.1f N*mm\n', R
    (1),R(2),R(3));
fprintf('Node 4: Rx4=%+.1f N  Ry4=%+.1f N  M4=%+.1f N*mm\n',R
    (10),R(11),R(12));
fprintf('\n=== EQUILIBRIUM CHECK ===\n');
fprintf('Sum Fx (reactions+applied) = %.1f N\n', R(1)+R(10)+
    F_lat);
fprintf('Sum Fy (reactions+applied) = %.1f N\n', R(2)+R(11)-
    q_beam*B);

%% Element internal forces (local coordinates)
fprintf('\n=== ELEMENT END FORCES (local) ===\n');
fprintf('%-10s %10s %10s %12s %10s %10s %12s\n', ...
        'Element','N_i(N)','V_i(N)','M_i(N*mm)','N_j(N)','V_j(N
            )','M_j(N*mm)');
for e = 1:n_elem
    ni = conn(e,1); nj = conn(e,2);
    L  = norm(coords(nj,:) - coords(ni,:));
    c  = cosd(alpha_deg(e)); s = sind(alpha_deg(e));
    dofs  = elem_dofs(ni, nj);
    T6    = frame_T6(c, s);
    uloc  = T6 * u(dofs);
    kbar  = frame_local_k(E, A, Iz, L);
    if e == 2
        q0   = -q_beam;
        fbar_load = [0;q0*L/2;q0*L^2/12;0;q0*L/2;-q0*L^2/12];
    else
        fbar_load = zeros(6,1);
    end
    fint = kbar * uloc - fbar_load;
```

```
    fprintf('%-10d %10.1f %10.1f %12.0f %10.1f %10.1f %12.0f\n', ...
            e, fint(1),fint(2),fint(3),fint(4),fint(5),fint(6));
end

%% Figures
n_pts = 80;
xi_v  = linspace(0, 1, n_pts);

X_el = cell(n_elem,1); Y_el = X_el;
Xd_el = X_el; Yd_el = X_el;
N_el = X_el; V_el = X_el; M_el = X_el;

scale_disp = 100;  % displacement magnification for deflected shape

for e = 1:n_elem
    ni = conn(e,1); nj = conn(e,2);
    xi = coords(ni,1); yi = coords(ni,2);
    xj = coords(nj,1); yj = coords(nj,2);
    L  = norm([xj-xi, yj-yi]);
    c  = cosd(alpha_deg(e)); s = sind(alpha_deg(e));
    dofs = elem_dofs(ni, nj);
    T6   = frame_T6(c, s);
    uloc = T6 * u(dofs);
    kbar = frame_local_k(E, A, Iz, L);
    if e == 2
        q0 = -q_beam;
        fbar_load = [0;q0*L/2;q0*L^2/12;0;q0*L/2;-q0*L^2/12];
    else
        q0 = 0;
        fbar_load = zeros(6,1);
    end
    fint = kbar * uloc - fbar_load;

    xv = xi_v * L;
    N_dist = fint(1) * ones(1, n_pts);
    V_dist = fint(2) + q0 * xv;
    M_dist = fint(3) - fint(2)*xv - q0*xv.^2/2;

    X_gl = xi + xi_v*(xj - xi);
    Y_gl = yi + xi_v*(yj - yi);

    Xd_gl = X_gl + scale_disp*((1-xi_v)*u(dofs(1)) + xi_v*u(dofs(4)));
    Yd_gl = Y_gl + scale_disp*((1-xi_v)*u(dofs(2)) + xi_v*u(dofs(5)));

    X_el{e}=X_gl; Y_el{e}=Y_gl;
    Xd_el{e}=Xd_gl; Yd_el{e}=Yd_gl;
```

```
    N_el{e}=N_dist; V_el{e}=V_dist; M_el{e}=M_dist;
end

M_max = max(abs([M_el{:}])); if M_max<1, M_max=1; end
V_max = max(abs([V_el{:}])); if V_max<1, V_max=1; end
N_max = max(abs([N_el{:}])); if N_max<1, N_max=1; end
diag_scale = 800;

fig = figure('Units','centimeters','Position',[2 2 20 14]);

subplot(1,2,1); hold on; axis equal; grid on;
for e = 1:n_elem
    plot(X_el{e}, Y_el{e}, 'k--', 'LineWidth', 0.8);
    plot(Xd_el{e}, Yd_el{e}, 'b-', 'LineWidth', 2);
end
xlabel('x (mm)'); ylabel('y (mm)');
title(sprintf('Deflected shape (x%d)', scale_disp));
legend({'Undeformed','Deformed'}, 'Location','best');

subplot(1,2,2); hold on; axis equal; grid on;
colors = {'b','r','g'};
data   = {M_el, V_el, N_el};
scales = [M_max, V_max, N_max];

for e = 1:n_elem
    c_ = cosd(alpha_deg(e)); s_ = sind(alpha_deg(e));
    nx = -s_; ny = c_;
    plot(X_el{e}, Y_el{e}, 'k-', 'LineWidth', 1.5);
    for d = 1:3
        vals = data{d}{e} / scales(d) * diag_scale;
        Xoff = X_el{e} + nx*vals;
        Yoff = Y_el{e} + ny*vals;
        fill([X_el{e} fliplr(Xoff)], [Y_el{e} fliplr(Yoff)],
            ...
             colors{d}, 'FaceAlpha', 0.25, 'EdgeColor', colors{
                d});
    end
end
xlabel('x (mm)'); ylabel('y (mm)');
title('M (blue), V (red), N (green)');

book_style(fig);
exportgraphics(fig, '../figures/frame_portal_results.pdf', '
   ContentType', 'vector');
exportgraphics(fig, '../figures/frame_portal_results.png', '
   Resolution', 300);
fprintf('\nFigures saved to ../figures/\n');

%% Local functions
function T6 = frame_T6(c, s)
    T6 = [c  s  0  0  0  0;
```

```
           -s   c   0   0   0   0;
            0   0   1   0   0   0;
            0   0   0   c   s   0;
            0   0   0  -s   c   0;
            0   0   0   0   0   1];
end

function kbar = frame_local_k(E, A, I, L)
    a   = E*A/L;
    b   = 12*E*I/L^3;
    d   = 6*E*I/L^2;
    e4  = 4*E*I/L;
    e2  = 2*E*I/L;
    kbar = [ a    0    0   -a    0    0;
             0    b    d    0   -b    d;
             0    d   e4    0   -d   e2;
            -a    0    0    a    0    0;
             0   -b   -d    0    b   -d;
             0    d   e2    0   -d   e4];
end

function d = elem_dofs(ni, nj)
    d = [3*ni-2, 3*ni-1, 3*ni, 3*nj-2, 3*nj-1, 3*nj];
end
```

Listing A.9: Portal frame under combined loading (Example 6.1).

A.9 Chapter 7 — Q4 Cook Membrane

```
% Chapter 7 -- Q4 Cook membrane under in-plane shear
% One Q4 element: nodes 1(0,0) 2(48,44) 3(48,60) 4(0,44) [mm]
% E=70000 N/mm^2, nu=0.3, t=1 mm, plane stress
% BCs: left edge clamped (nodes 1,4); load: 50 N vertical at
   nodes 2,3
clear; clc; close all;

%% Parameters
E   = 70000;   nu = 0.3;   t = 1;
xy  = [0,0; 48,44; 48,60; 0,44];

%% Constitutive matrix (plane stress)
D = E/(1-nu^2) * [1,nu,0; nu,1,0; 0,0,(1-nu)/2];

%% 2x2 Gauss points and weights
g    = 1/sqrt(3);
gpts = [-g,-g; g,-g; g,g; -g,g];
w    = [1,1,1,1];

%% Assemble element stiffness
ke   = zeros(8);
```

```
B_all = cell(4,1);
J_all = cell(4,1);

for gp = 1:4
    xi  = gpts(gp,1);  eta = gpts(gp,2);

    % Natural derivatives of shape functions
    dN_dxi  = [-(1-eta), (1-eta), (1+eta), -(1+eta)] / 4;
    dN_deta = [-(1-xi), -(1+xi),  (1+xi),  (1-xi) ] / 4;

    % Jacobian and its inverse
    J    = [dN_dxi;  dN_deta] * xy;
    detJ = det(J);
    Jinv = inv(J);

    % Physical derivatives
    dN_dx = Jinv(1,:) * [dN_dxi; dN_deta];
    dN_dy = Jinv(2,:) * [dN_dxi; dN_deta];

    % Strain-displacement matrix B (3x8)
    B = zeros(3,8);
    B(1,1:2:7) = dN_dx;
    B(2,2:2:8) = dN_dy;
    B(3,1:2:7) = dN_dy;
    B(3,2:2:8) = dN_dx;

    ke = ke + t * w(gp) * detJ * B' * D * B;
    B_all{gp} = B;
    J_all{gp} = J;
end

%% Print Gauss-point data
fprintf('=== JACOBIAN AT GAUSS POINTS ===\n')
for gp = 1:4
    J = J_all{gp};
    fprintf('GP%d: J=[%.4f,%.4f;%.4f,%.4f]  detJ=%.4f\n', ...
            gp, J(1,1),J(1,2),J(2,1),J(2,2), det(J_all{gp}))
end
fprintf('\n=== B MATRIX AT GP1 (x1e3) ===\n')
disp(B_all{1}*1e3)
fprintf('\n=== ke (x1e-3 N/mm) ===\n')
disp(ke/1e3)

%% BCs and load
fixed = [1,2,7,8];   % u1,v1,u4,v4
free  = setdiff(1:8, fixed);
F = zeros(8,1);
F(4) = 50;   % v2
F(6) = 50;   % v3

u = zeros(8,1);
```

```
u(free) = ke(free,free) \ F(free);

fprintf('\n=== SOLUTION (mm) ===\n')
fprintf('u2=%+.6f  v2=%+.6f\n', u(3), u(4))
fprintf('u3=%+.6f  v3=%+.6f\n', u(5), u(6))

%% Stresses at Gauss points
fprintf('\n=== STRESSES AT GAUSS POINTS (N/mm^2) ===\n')
for gp = 1:4
    sig = D * B_all{gp} * u;
    fprintf('GP%d: sx=%+.4f  sy=%+.4f  txy=%+.4f\n', gp, sig(1), sig(2), sig(3))
end

%% Figure: mesh and deformed shape
scale = 5000;
fig = figure('Units','centimeters','Position',[2,2,14,12]);
hold on; axis equal; grid on;

% Undeformed
patch('XData', xy([1,2,3,4,1],1), 'YData', xy([1,2,3,4,1],2), ...
      'FaceColor','none', 'EdgeColor','k', 'LineWidth',1.2)

% Deformed
xy_def = xy + scale * reshape(u,2,[])';
patch('XData', xy_def([1,2,3,4,1],1), 'YData', xy_def([1,2,3,4,1],2), ...
      'FaceColor','none', 'EdgeColor','b', 'LineStyle','--', 'LineWidth',1.5)

% Nodes
for n = 1:4
    plot(xy(n,1), xy(n,2), 'ko', 'MarkerSize', 5, 'MarkerFaceColor','k')
    text(xy(n,1)+1.5, xy(n,2)+1, sprintf('%d',n), 'FontSize',10)
end

xlabel('x (mm)'); ylabel('y (mm)')
title(sprintf('Q4 Cook membrane: undeformed (solid) and deformed (dashed, \\times%d)', scale))
legend({'Undeformed','Deformed'}, 'Location','northwest')

book_style(fig);
exportgraphics(fig, '../figures/q4_cook_results.pdf', 'ContentType','vector')
exportgraphics(fig, '../figures/q4_cook_results.png', 'Resolution', 300)
fprintf('\nFigures saved to ../figures/\n')
```

Listing A.10: Single Q4 element for the Cook membrane benchmark (Example 7.2).

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